

# The Metric Blind Spot in Deep Learning-Based Brain MRI Reconstruction: A Systematic Review

Dat Tat Mai<sup>a,\*</sup>, Thai Viet Pham<sup>b</sup>, Thu Nguyen Thi Dang<sup>c</sup>, James Jin Kang<sup>a</sup>

<sup>a</sup>*School of Science, Engineering & Technology, RMIT University Vietnam, Ho Chi Minh City, Vietnam*

<sup>b</sup>*School of Computing Technologies, RMIT University, Melbourne, VIC, 3000, Australia*

<sup>c</sup>*School of Health and Biomedical Science, RMIT University, Melbourne, VIC, 3000, Australia*

---

## Abstract

**Objective:** Deep learning accelerates brain MRI four- to tenfold, but learned models can erase a lesion or invent tissue, and pixel-averaged scores such as PSNR and SSIM miss both. We review whether evaluation practice could detect this blind spot. **Methods:** Following PRISMA 2020 we searched seven databases without date limit, included 263 studies (1995–2026), appraised them with QUADAS-2 and matched instruments, and synthesised narratively. Categories came from titles, abstracts and controlled vocabulary; every prevalence figure is a floor. Duplicate screening achieved substantial agreement (Fleiss’  $\kappa = 0.877$ ); appraisal agreement was substantial (0.788; 0.390 where observable); extraction is unaudited. **Results:** Only 18 of 263 studies (6.8%) record a fidelity metric and a reader assessment on the same data, leaving the field’s central proxy unmeasured. Reader studies mostly measure reader-versus-reader agreement, which is weak: fastMRI 2020 concordance was 0.457 and 0.386 (Kendall’s W), improving only where SSIM had diverged. Erasing a 100 mm<sup>3</sup> lacunar infarct moves global PSNR about 0.03 dB under the stated error model. The corpus grew fivefold; reader assessment fell from 32% to 18%, recovering to 21%. Among the four largest families the most hallucination-associated (generative, 39%) is among the least reader-evaluated (11.3%); self-supervised is higher still (47%) with no readers. One study in twenty both releases code and runs readers; none evaluated a model observer; no named dataset covers acute stroke or

---

\*Corresponding author.

Email addresses: `dat.mai2@rmit.edu.vn` (Dat Tat Mai), `s4229249@student.rmit.edu.au` (Thai Viet Pham), `s4205224@student.rmit.edu.au` (Thu Nguyen Thi Dang), `james.kang@rmit.edu.vn` (James Jin Kang)

25 haemorrhage. **Conclusions:** On these floors, evaluation practice cannot certify  
26 diagnostic safety. We derive five requirements safety-oriented evaluation must  
27 meet.

28 *Keywords:* Deep learning, Brain MRI reconstruction, Reconstruction safety,  
29 Hallucination, Image quality assessment, Systematic review

---

## 30 1. Introduction

31 Brain MRI is widely used because it can clearly distinguish different types  
32 of brain tissue and reveal subtle abnormalities that are difficult to detect with  
33 other imaging modalities. However, this image quality comes at the cost of long  
34 acquisition times. Unlike conventional imaging methods that capture an image  
35 more directly, MRI builds an image from a sequence of raw signal measure-  
36 ments (samples of spatial frequencies, known as  $k$ -space) collected over time.  
37 This prolonged acquisition reduces patient throughput and increases sensitivity  
38 to patient motion, which can introduce artifacts and degrade image quality (Za-  
39 itsev et al., 2015). To address these limitations, acceleration techniques such as  
40 parallel imaging, including Sensitivity Encoding (SENSE) (Pruessmann et al.,  
41 1999), and compressed sensing (Lustig et al., 2007) were developed to reduce  
42 the amount of data that must be collected. More recently, deep learning has  
43 enabled further acceleration by reconstructing high-quality images from incom-  
44 plete MRI measurements, through cascaded convolutional networks (Schlemper  
45 et al., 2017), model-based unrolled networks (Hammernik et al., 2018; Aggar-  
46 wal et al., 2018), generative adversarial networks (Yang et al., 2018), and score-  
47 based diffusion models (Song et al., 2021), at accelerations from fourfold to, in  
48 exploratory work, an order of magnitude (Radmanesh et al., 2022).

49 The reconstructed images look convincing, and that is the problem. Small  
50 changes in the input can produce large errors in the output (Antun et al., 2020;  
51 Gottschling et al., 2025). Four failure types recur across the studies we re-  
52 viewed: strong priors smooth away faint structure; motion causes ghosting the  
53 model never saw in training (Sommer et al., 2020); a change of scanner or sam-

54 pling degrades results in ways that in-distribution testing misses (Güngör et al.,  
 55 2023); and generative models can invent plausible tissue (Zhou et al., 2024).  
 56 The scores used to certify these methods, peak signal-to-noise ratio (PSNR)  
 57 and structural similarity (SSIM) (Wang et al., 2004), agree only partially with  
 58 radiologist quality judgement even on conventional image degradations (Mason  
 59 et al., 2020); and SNR, though reproducible across observers, tracks no-reference  
 60 quality models only moderately (Yu et al., 2018), so an image can score well  
 61 while the error that matters goes unrecorded. We call this the **metric blind**  
 62 **spot**: standard image-quality metrics cannot catch these small but clinically se-  
 63 rious errors, because a lesion of a few cubic millimetres barely moves an average  
 64 taken over millions of voxels.

65 Parts of the field have responded, training quality metrics on radiologist  
 66 rankings (Tang et al., 2025), attaching uncertainty maps to reconstructions  
 67 (Khawaled and Freiman, 2024), or running reader studies that test an acceler-  
 68 ated protocol against a reference one directly (Lang et al., 2024; Radmanesh  
 69 et al., 2022). Causal and counterfactual methods can ask the most direct ques-  
 70 tion, whether a lesion survives a controlled change of acquisition, but have  
 71 mostly been applied to classification rather than reconstruction (Pawlowski  
 72 et al., 2020; Castro et al., 2020). These responses share a root cause. Ac-  
 73 celeration works because a learned prior fills in information the measurement  
 74 does not contain, and that same filled-in information can hide a real finding or  
 75 create a false one, while the field certifies the result with averaged pixel scores  
 76 that cannot see either event.

77 To our knowledge no systematic review has assembled the evidence on this  
 78 problem. Existing surveys cover architectures and benchmark performance  
 79 (Liang et al., 2020; Montalt-Tordera et al., 2021); none asks how often con-  
 80 ventional metrics miss clinically important errors, why they miss them, or what  
 81 evaluation would have to look like to catch them. We address that through  
 82 three questions:

83 **RQ1** *How reliably do conventional fidelity metrics track radiologist diagnostic*

84        *ratings in deep learning-based accelerated brain MRI, and how wide is the*  
85        *evaluation reporting gap in the literature?*

86 **RQ2** *What underlying mathematical mechanisms cause standard metrics to*  
87        *overlook localized pathology erasure and generative hallucinations in deep*  
88        *learning-based reconstruction?*

89 **RQ3** *What can each evaluation paradigm in current use establish about prior-*  
90        *induced diagnostic error in deep learning-based MRI reconstruction, and*  
91        *what requirements for safety-oriented evaluation follow from those limits?*

92        We contribute: (C1) a synthesis of what 263 primary studies report about  
93        agreement between pixel-wise metrics and radiologist judgement; (C2) four  
94        mathematical mechanisms that explain the divergence, with an exact identity  
95        bounding the PSNR cost of erasing a lesion; (C3) an analysis of *where* the  
96        evaluation gap comes from, showing that it co-occurs with a division of evalu-  
97        ation practice between research communities and with a benchmark infrastruc-  
98        ture lacking acute pathology; and (C4) five requirements that safety-oriented  
99        evaluation must meet. The auditing framework these requirements motivate is  
100       developed and evaluated in a companion methods paper, not here.

101       Three boundaries should be stated at the outset, because they determine  
102       what the findings can and cannot be asked to support. First, this is a review of  
103       *evaluation practice*, not of reconstruction performance: we count what studies  
104       measured, not how well their methods performed, and nothing here estimates  
105       how often a deployed reconstruction actually erases a lesion. Second, the mech-  
106       anisms in RQ2 are mathematical properties of the metrics, derived under a  
107       stated error model; they establish that a focal error *can* pass unrecorded, not  
108       that it commonly does. Third, prevalence counts rest on labels extracted from  
109       what studies report, so every one of them is a lower bound on practice rather  
110       than an estimate of it, and we treat the direction of that bias explicitly in Sec-  
111       tion 5.2. The contribution is a diagnosis of what the field’s current instruments  
112       are capable of detecting, which is logically prior to, and considerably cheaper  
113       to establish than, a measurement of how often they fail.

## 114 2. Related Work

115 Four strands bear on this review, and the claim that none of them has  
116 assembled the evidence we assemble here is a claim about each in turn.

117 *Surveys of reconstruction methods.* Reviews of deep learning MRI recon-  
118 struction catalogue architectures, acceleration factors and benchmark accuracy  
119 (Liang et al., 2020; Montalt-Tordera et al., 2021; Knoll et al., 2020a; Kazer-  
120 ouni et al., 2023). They are organised by method, and their outcome variable  
121 is reconstruction fidelity against a fully sampled reference. Safety and failure  
122 modes appear as caveats in discussion sections rather than as an object of syn-  
123 thesis, and none of them tabulates what evaluation each primary study actually  
124 performed, which is the quantity this review extracts.

125 *Task-based image quality assessment.* The tradition that ought to have sup-  
126 plied the answer established, decades before deep learning, that image quality is  
127 meaningful only relative to a diagnostic task. Barrett and colleagues formalised  
128 the observer as part of the measurement: an image is good if an observer per-  
129 forming a specified task on it achieves a high figure of merit, typically the  
130 area under an ROC curve for a detection task, and the mathematical (ideal or  
131 channelised Hotelling) observer supplies a computable surrogate for the human  
132 when the task and image statistics are specified (Barrett et al., 1993; Barrett  
133 and Myers, 2004). ICRU Report 54 made the same argument the international  
134 reference framework for assessing medical imaging quality (International Com-  
135 mission on Radiation Units and Measurements, 1996). That framework already  
136 contains the concept the reconstruction literature lacks: a detectability index  
137 for a specified lesion, which is by construction sensitive to exactly the focal  
138 change a whole-volume  $L_2$  score averages away. It has been carried into the  
139 dose- and protocol-optimisation literatures of other imaging modalities, but it  
140 has not crossed into deep learning MRI reconstruction: no study in this cor-  
141 pus evaluated a model observer, and our first requirement (R1, Section 4.7) is  
142 in substance a restatement of the task-based position for a field that has not  
143 adopted it. The reason it has not is worth naming: model observers require a

144 specified signal and a statistical description of the background, and the null-  
145 space structure that makes learned reconstruction dangerous is precisely what  
146 makes those two ingredients hard to specify.

147 *Failure-mode-specific work.* Artifact detection, motion correction and ro-  
148 bustness under distribution shift are each addressed in depth, but in isolation,  
149 by literatures that do not share an evaluation vocabulary (Esteban et al., 2017;  
150 Sommer et al., 2020; Güngör et al., 2023). Analyses of hallucination in tomo-  
151 graphic reconstruction establish that prior-induced structure lives in the null  
152 space of the forward operator, so it is invisible to any measurement-space con-  
153 sistency check (Bhadra et al., 2021; Gottschling et al., 2025), and causal and  
154 counterfactual methods for medical imaging supply the machinery to intervene  
155 on image content and ask whether a finding survives (Pawlowski et al., 2020;  
156 Castro et al., 2020; Sanchez et al., 2022; Peng et al., 2026). Neither line has  
157 been connected to review-level evidence on how reconstruction is evaluated in  
158 practice.

159 *What is new in the four mechanisms of RQ2, and what is not.* Because  
160 Section 4 states four mechanisms formally, their provenance should be set out  
161 plainly rather than left for a reader to reconstruct. Two are elementary prop-  
162 erties of the objective rather than findings: spatial error averaging follows from  
163 the definition of a mean over the image volume, and loss symmetry from the  
164 fact that squaring discards the sign of the residual. Two are established results  
165 we restate: reference dependence is standard in the image-quality literature,  
166 and the null-space account underlying the realism–reward trade-off is due to  
167 Bhadra et al. (2021) and Gottschling et al. (2025). What this review adds is  
168 not the mechanisms but their quantification and their assembly — the identity  
169 at Equation (2), which converts any lesion volume into the PSNR movement its  
170 erasure would cause, and the setting of all four against what a corpus of 263  
171 studies is recorded as measuring. We claim the assembly and the bound, not  
172 the underlying results.

173 *Benchmark and annotation infrastructure.* The raw  $k$ -space resource the  
174 field benchmarks on is fastMRI (Knoll et al., 2020b), whose brain release fol-

lowed the knee data described in that report, and an annotation layer is emerging on top of it: fastMRI+ adds expert pathology labels to the fastMRI brain and knee data (Zhao et al., 2022), and curated stroke-lesion datasets exist for segmentation (ISLES (Hernandez Petzsche et al., 2022), ATLAS (Liew et al., 2022)). None of these supplies raw  $k$ -space paired with acute vascular pathology, which is the combination reconstruction evaluation needs, and Section 4.5 shows that this is not an incidental gap but the binding constraint on the whole evaluation problem.

What no strand supplies is the join: a systematic account of how often the field measures metric–reader agreement, why the metrics fail when it does not, and what each evaluation paradigm can therefore establish. The four strands are not merely separate; they are held apart by what each can access. The survey and hallucination-analysis literatures work on datasets with raw  $k$ -space and no clinically verified pathology; the task-based tradition needs a specified signal and a labelled task, which those datasets do not carry; and the clinical reader literature has the pathology and the readers but no access to the raw data on which the mechanisms operate. Section 4.5 shows this is not an abstraction but a measurable division in the corpus itself. Supplementary Table S4 sets out each secondary study we identified against the questions asked here.

### 3. Methodology

This systematic review was conducted in accordance with the PRISMA 2020 statement (Page et al., 2021); the completed checklist, search vocabulary, extraction schema, and analysis scripts are available in the supplementary material and on the Open Science Framework (<https://doi.org/10.17605/OSF.IO/832Z6>; timestamped deposit <https://doi.org/10.17605/OSF.IO/EUXA8>).

*Protocol and registration status.* The protocol, covering the research questions, eligibility criteria, search strategy, screening procedure, classification rubric and synthesis approach, was fixed in the project repository before screening began. In the terms of PRISMA 2020 item 24a, **this review was**

204 **not prospectively registered and carries no registration number:**  
 205 PROSPERO’s scope excludes reviews without direct clinical health outcomes,  
 206 and no equivalent prospective registry covers methodological reviews of this  
 207 kind. The Open Science Framework deposit cited above therefore timestamps  
 208 the record but is retrospective, and we make no claim of prospective registration  
 209 on it. Two consequences follow. An external record does not exclude undis-  
 210 closed post-hoc analytic choices; and the two analyses we ourselves identify  
 211 as post-hoc — the acceleration-factor extraction of Table 6 and the venue  
 212 classification of Section 4.5 — are labelled as such wherever they appear.

213 *Search strategy.* Seven bibliographic databases (PubMed/MEDLINE, IEEE  
 214 Xplore, Scopus, Web of Science, SpringerLink, OpenAlex, and Semantic  
 215 Scholar) were searched between 23 and 24 July 2026, with no publication-date  
 216 limit applied; the 1995–2026 span quoted elsewhere in this article is the pub-  
 217 lication range of the studies that were included, not a restriction imposed on  
 218 the search. Queries were executed through each database’s API rather than  
 219 its web interface, so the complete Boolean match set was retrieved for every  
 220 term combination rather than a ranked sample, explaining why the initial  
 221 identification count is large. The strategy combined three required concept  
 222 groups: neuroimaging (Block B1), reconstruction and acceleration (Block B2),  
 223 and safety, failure modes, and evaluation outcomes (Block B3), with MeSH  
 224 headings and wildcard truncation adapted per database (Supplementary Ta-  
 225 ble S5). Table 1 details the executed query structure and hit breakdown across  
 226 databases.

227 The breadth of this search is by design. Evidence of reconstruction failure  
 228 is rarely a paper’s stated subject: artifacts, hallucinations and metric–reader  
 229 disagreements are most often reported in passing inside studies whose declared  
 230 contribution is a new architecture, so a query tuned narrowly to safety vocab-  
 231 ulary would miss much of this evidence. Clinical and engineering literatures  
 232 also use distinct vocabularies — sequences and artifacts against inverse prob-  
 233 lems, unrolled optimization and generative priors — and neither database family  
 234 indexes the other completely. We therefore tuned for sensitivity rather than pre-



Table 1: Query structure executed in each database and record hit breakdown.

| Database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Executed Query Structure                                                                                                                                                 | Controlled Vocabulary                            | Fields Searched       | Hits (N) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------|----------|
| PubMed / MEDLINE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (B1 <sub>ft</sub> [TIAB] OR B1 <sub>mesh</sub> ) AND (B2 <sub>ft</sub> [TIAB] OR B2 <sub>mesh</sub> ) AND (B3 <sub>ft</sub> [TIAB] OR B3 <sub>mesh</sub> ) NOT (X[TIAB]) | Yes: 9 MeSH headings OR-ed into the three blocks | Title/Abstract + MeSH | 24,693   |
| SpringerLink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (B1) AND (B2) AND (B3) NOT (X); quoted phrases, no truncation operators                                                                                                  | No                                               | Full record           | 1,598    |
| IEEE Xplore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (B1) AND (B2) AND (B3) NOT (X); wildcard truncation retained                                                                                                             | No                                               | Metadata              | 778      |
| Scopus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TITLE-ABS-KEY(B1) AND TITLE-ABS-KEY(B2) AND TITLE-ABS-KEY(B3) AND NOT TITLE-ABS-KEY(X)                                                                                   | No                                               | TITLE-ABS-KEY         | 127      |
| Semantic Scholar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (B1) + (B2) + (B3), where   denotes OR and + denotes AND                                                                                                                 | No                                               | Title/Abstract        | 66       |
| OpenAlex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | title_and_-abstract.search:B1, title_and_-abstract.search:B2, title_and_abstract.search:B3                                                                               | No                                               | Title/Abstract        | 34*      |
| Web of Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | TS=(B1) AND TS=(B2) AND TS=(B3) NOT TS=(X)                                                                                                                               | No                                               | Topic (TS)            | 31       |
| Total identified records                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                          |                                                  |                       | 27,327   |
| <p>Concept blocks combine as (B1) AND (B2) AND (B3) NOT (X). <b>B1</b> neuroimaging: "brain MRI", "cerebral MRI", neuroimag*, DWI, FLAIR, T1w, T2w, SWI; MeSH <i>Brain, Magnetic Resonance Imaging, Neuroimaging</i>. <b>B2</b> AI and accelerated reconstruction: "MRI reconstruction", "accelerated MRI", "undersampled reconstruction", "k-space reconstruction", "unrolled network", GAN, "diffusion prior"; MeSH <i>Deep Learning, Machine Learning</i>. <b>B3</b> safety, artifacts and evaluation: hallucinat*, "lesion suppression", reconstruction artifact*, PSNR, SSIM, "reader study"; MeSH <i>Artifacts, Diagnostic Errors, Reproducibility of Results</i>. <b>X</b> exclusion: "animal model", "case report", "in vitro". * denotes truncation. Complete term lists, MeSH headings and per-database syntax are given in Supplementary Table S5. Searches were executed between 2026-07-23 and 2026-07-24, with no publication-date limit. *OpenAlex returned 50 raw hits; 34 passed record-completeness filtering.</p> |                                                                                                                                                                          |                                                  |                       |          |

235 cision and accepted the screening burden. The funnel from 27,327 records to  
236 263 studies is the intended signature of that choice: the burden falls on ini-  
237 tial screening, which reviewers can absorb, rather than on coverage, which no  
238 downstream step could recover.

239 *Eligibility and selection.* Records were managed in the review’s own Python  
240 pipeline rather than a commercial screening platform; the scripts, the per-record  
241 decision files and the per-database query logs are released, so each step below  
242 is re-executable. Deduplication by exact identifier matching (PMID/DOI) and  
243 string/TF-IDF similarity (Jaro-Winkler  $\geq 0.97$ , cosine  $\geq 0.95$ ) removed 403  
244 duplicates (1.47%), leaving 26,924 unique records. That rate is low enough to  
245 suggest a matching failure but is structural: PubMed’s MeSH arm returned  
246 24,693 records and the six engineering arms 2,634 between them, and sets that  
247 barely overlap leave little to deduplicate. This is not a claim of completeness,  
248 and one failure is on record: peer review identified an arXiv preprint and its  
249 journal version that both survived to the included set, carrying different DOIs,  
250 no shared PMID, and titles differing by terminal punctuation — below the  
251 thresholds above. It was removed at revision, taking the corpus from 264 reports  
252 to 263 studies (Section 4). Automated matching will not catch a preprint–  
253 journal pair whose metadata diverges in this way, and the honest inference is  
254 that a residual duplicate rate of the same order cannot be excluded.

255 Three reviewers (D.T.M., T.V.P., T.N.T.D.) screened records against the  
256 pre-specified PICOS criteria (Table 2) at both stages, under a primary-screen-  
257 plus- adjudication design: each record received one primary label of *include*,  
258 *exclude*, *flagged* or *background only*, and every record labelled *flagged* went to  
259 a second reviewer for adjudication. Labelling was deliberately conservative,  
260 so any record with an unclear PICOS dimension, or whose text we could not  
261 obtain, was flagged rather than excluded. Title and abstract screening advanced  
262 1,149 records to full-text assessment, of which 109 were labelled *include* on the  
263 abstract alone and 1,040 *flagged*; both labels carried the record forward, and  
264 the distinction records only whether the abstract was sufficient to decide. A  
265 further 25,573 were excluded and 202 set aside as background. Full texts were  
266 pursued through institutional and title-level library subscriptions, open-access  
267 repositories, preprint servers, PubMed Central and inter-library request until  
268 no route remained: 1,068 of the 1,149 (93.0%) were obtained and assessed, and  
269 the 81 that could not be obtained are broken down by recorded reason with the

270 selection results (Section 4).

271     The three reviewers divided the 1,068 obtained reports between them un-  
272 der the same primary-screen-plus-adjudication design used at title and abstract:  
273 each report received one primary full-text label, and a report was escalated to  
274 a second reviewer wherever the primary label was *flagged*, wherever an inclu-  
275 sion decision rested on a borderline PICOS dimension, or wherever the reviewer  
276 requested a second reading. Escalated reports were resolved by consensus dis-  
277 cussion between the two assessors, with the third reviewer consulted where the  
278 two did not converge. The escalation volumes this produced are reported below,  
279 and the per-report labels, escalation marks and reasons are released in the de-  
280 cision ledger. We state the design in this form because it determines what can  
281 be computed from the ledger: one primary label per report means no chance-  
282 corrected inter-rater coefficient is recoverable from full-text screening, which is  
283 why the separate criterion-reproducibility exercise below was run. In total, 263  
284 primary studies met all inclusion criteria upon full-text verification (two ad-  
285 ditional reports were excluded as duplicate publications). Full-text exclusions  
286 ( $n = 785$ ) were categorized by pre-specified PICOS criteria: wrong intervention  
287 ( $n = 326$ ), out of scope ( $n = 256$ ), wrong outcome ( $n = 98$ ), wrong population  
288 ( $n = 93$ ), wrong study design ( $n = 4$ ), non-English ( $n = 1$ ), and non-primary  
289 articles ( $n = 7$ ). A further 18 reports were retained as background material  
290 rather than as included studies, so the 1,068 assessed reports resolve as 263  
291 included, 785 excluded on PICOS grounds, 2 excluded as duplicate publications  
292 and 18 retained as background. The complete decision ledger is released as  
293 **fulltext\_screening\_decisions.csv**.

294     *Publication year.* Every analysis partitioning the corpus by time uses the  
295 year of the version of record as indexed by the publisher — the issue year for  
296 journal articles, the proceedings year for conference papers — rather than first  
297 online posting or preprint deposit. The rule matters at era boundaries, and we  
298 adopt the indexed year because it is what a reader can verify from the reference  
299 list, Crossref and PubMed without online-first metadata, which is unevenly  
300 available across the 87 venues represented. The same value is recorded in the

301 `year` field of every bibliography entry and in the `publication_year` column of  
 302 `included_characteristics.csv`, so the era counts in Table 9 are recomputable  
 303 from either. One consequence should not be read as an inconsistency: the  
 304 searches ran on 23–24 July 2026, yet 41 of the 263 included studies (15.6%)  
 305 carry a 2026 issue year, being articles published online ahead of print before the  
 306 searches ran but assigned by the publisher to a 2026 issue.

307 *Screening agreement, and what it does and does not cover.* The primary  
 308 screen described above records one label per record, so no chance-corrected co-  
 309 efficient is recomputable from those decisions alone. We therefore ran a separate  
 310 duplicate-screening exercise. A random sample of 214 records already assessed  
 311 at title and abstract was re-screened independently by all three reviewers, blind  
 312 both to the recorded decision and to one another. Reviewers judged the four  
 313 PICOS dimensions from title, journal and abstract, opening the DOI link only  
 314 where the abstract was insufficient, and returned one of three labels per record:  
 315 include, exclude or unsure. The sample was drawn without stratification; the  
 316 drawn records, the three label sets and the agreement computation are released,  
 317 so the sample is auditable record by record. All 642 cells were completed and  
 318 no record was dropped for missingness.

319 Across all 214 records Fleiss’  $\kappa = 0.877$  (95% bootstrap CI 0.823–0.925;  
 320 10,000 resamples, seed 20260821), with observed agreement  $\bar{P} = 0.933$  against  
 321  $\bar{P}_e = 0.455$  expected by chance. The three reviewers agreed unanimously on  
 322 194 of 214 records (90.7%), and pairwise Cohen’s  $\kappa$  ran from 0.845 to 0.931.  
 323 One sensitivity analysis is worth reporting because it is the one a reader is most  
 324 likely to run. Restricting to the 193 records on which no reviewer recorded  
 325 *unsure* raises Fleiss’  $\kappa$  to 0.943 (95% CI 0.900–0.979). That complete-case figure  
 326 discards 21 records, and it discards precisely the records at least one reviewer  
 327 could not decide, so it overstates agreement on the population the coefficient  
 328 is meant to describe. We report 0.877 as the estimate and 0.943 only as the  
 329 value obtained when the hard cases are removed. Two qualifications travel with  
 330 that figure and should not be dropped. First, it measures agreement on the  
 331 screening decision, not on the extraction labels from which every prevalence

count in this article is derived; those remain unaudited for reliability, and the only error rate we can quote for them is the measured 4/61 (6.6%) on the design field (Section 5.2). Second, the conservative labelling policy routed almost the whole full-text workload to adjudication in any case — 1,040 of the 1,149 records advanced at title and abstract were flagged rather than included outright — so a high coefficient is evidence that the criteria were applied consistently, not that the borderline records were easy to classify.

What the primary decision records themselves support, and what we therefore also report, are the escalation volumes that policy produced. At full text, 84 reports were escalated to a second reviewer for adjudication and 29 of those were included; a further 77 reports carry a mandatory human-re-inspection mark in the released files, of which 40 were included after that review. Screening recall is bounded by the search strategy and by the criteria as applied rather than estimated against an independent reference set of known-eligible records, since no such set exists for this question; Section 5.2 treats both points as limitations. The duplicate-screened sample, both label sets and the agreement computation are released as `kappa_human_results.csv`, alongside the full decision ledger in `fulltext_screening_decisions.csv`.

*Data extraction.* Reviewers extracted study characteristics using the pre-specified form (Supplementary Table S6), capturing method family, reported failure modes, evaluation paradigm, field strength, pulse sequence, reference standard and reproducibility indicators, with a verbatim supporting quotation for every assigned value. Method, failure-mode and evaluation categories were assigned from each record’s title, abstract and controlled vocabulary against the vocabulary fixed in advance (Supplementary Table S5); that deliberately conservative scope is why every prevalence count here is a lower bound rather than an exhaustive tally. Field strength and pulse sequence, typically stated only in acquisition subsections, were additionally taken from full text where available. We did not contact investigators: a variable not stated in the source was recorded as *not reported* and never inferred, which is why the 54 studies without a stated field strength divide into 50 whose full text does not state it

Table 2: PICOS elements and their application in this review.

| Element      | Definition                                                                            | Application in this Review                                                                                                                                                                                                                                  |
|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Population   | Target imaging modalities, clinical domains, and scanner physics                      | Human brain MRI acquisitions (acute ischemic stroke, neurovascular lesions, intracranial haemorrhages, tumours, and structural T1w, T2w, FLAIR, DWI scans across 1.5T, 3T, and 7T field strengths).                                                         |
| Intervention | AI-based and accelerated reconstruction methods, and their safety/fidelity evaluation | Deep learning and accelerated reconstruction methods (unrolled physics-based networks, GANs, diffusion priors, self-supervised, denoising, super-resolution), together with methods for evaluating their fidelity, artifacts, robustness, or failure modes. |
| Comparison   | Baseline metrics, observers, and reference standards                                  | Conventional pixel-wise metrics (PSNR, SSIM, NRMSE), perceptual/learned metrics (LPIPS, FID), model observers, fully sampled reference images, or radiologist reader assessments.                                                                           |
| Outcome      | Reconstruction quality, fidelity, and failure characterization                        | Image-quality/fidelity measures, artifact and failure-mode characterization, metric-radiologist agreement, uncertainty estimates, robustness to distribution shift, and diagnostic/observer outcomes.                                                       |
| Study Design | Methodological criteria and empirical validation settings                             | Primary empirical studies, algorithmic benchmarking, counterfactual interventional stress-testing, and human or model observer evaluations.                                                                                                                 |

and 4 assessed on abstract or partial text.

*Quality appraisal, and the reader/algorithmic partition.* Methodological quality was evaluated with three design-matched instruments: QUADAS-2 (Whiting et al., 2011) across its four domains for diagnostic accuracy and radiologist reader studies; a reporting-transparency check against STARD principles (Bossuyt et al., 2015), reported descriptively rather than scored, since STARD yields no risk-of-bias rating; and, for algorithmic benchmarking studies, a reproducibility and robustness appraisal covering code and data availability, use of a named public benchmark, whether the reference standard was a fully sampled acquisition, and whether robustness to acquisition or distribution shift was reported. ROBINS-I (Sterne et al., 2016) was not applied, since few included studies are the non-randomised interventional designs it was built for.

The partition between the two instruments is not the one the extraction labels originally implied, and the correction matters for every reader-share statis-

377 tic here. Extraction recorded 61 studies as observer or reader designs; three-  
 378 reviewer appraisal of their full text found four containing no human reader at  
 379 all, each using the language of observer evaluation for an automated surrogate.  
 380 Those four were reclassified as algorithmic, so **the verified reader subset is**  
 381 **57 and the algorithmic set is 206**, and  $57 + 206 = 263$  closes the corpus  
 382 exactly. All 57 were appraised by all three reviewers across two rounds, 39 in  
 383 the first and 18 in the second, so every QUADAS-2 tally rests on 57 studies and  
 384  $57 \times 7 = 399$  domain judgements. Every reader-derived count in this article  
 385 uses 57, and the released per-study file encodes the same partition with the re-  
 386 classification recorded per study. One further study (S001) carries an appraisal  
 387 note that a human reader could not be *confirmed*, only abstract-level text being  
 388 available; it is retained in the reader subset, since non-ascertainment is not a  
 389 finding of absence, and it is flagged in the deposit.

390 *Appraisal agreement.* The three reviewers rated every applicable domain  
 391 Low, Unclear or High independently, each judgement carrying a verbatim sup-  
 392 porting quotation, before joint review to harmonize thresholds. Five first-round  
 393 studies were later re-appraised on recovered full text (below), so their 35 judge-  
 394 ments no longer rest on three independent ratings and are excluded from the  
 395 coefficient. Over the remaining 52 studies and 364 judgements Fleiss'  $\kappa = 0.788$   
 396 (95% bootstrap CI 0.739–0.833; 10,000 resamples, seed 20260821). That figure  
 397 averages two rounds that behaved differently, and the difference is more infor-  
 398 mative than the average. The 34 first-round studies retaining three indepen-  
 399 dent ratings were unanimous on all 238 of their judgements, so they contribute  
 400  $\kappa = 1.00$  by construction and carry no information about latent reliability. The  
 401 18 of the second round were unanimous on 60 of 126 (47.6%), and across those  
 402 126 Fleiss'  $\kappa = 0.390$  (95% CI 0.293–0.484) — fair on the Landis–Koch bands,  
 403 with an interval reaching into moderate. The procedure was identical across  
 404 rounds; the subsets were not, the second round comprising studies the first pass  
 405 deferred as not straightforward. Two readings remain open — that those studies  
 406 report their methods less completely and are genuinely harder to rate, and that  
 407 the first round was less independent in practice than in design — and these data

do not separate them.

The disagreement has a structure worth recording rather than smoothing. Across the 126 second-round judgements one reviewer rated Low 33.3% of the time and another 61.9%, a difference of threshold rather than of reading, which is why  $\kappa$  sits at 0.390 while raw agreement is 63.5%. Six judgements split three ways with no majority; all six were recorded as *Unclear*, on the ground that a panel unable to converge is itself evidence the report did not support a confident judgement, which is what *Unclear* encodes in QUADAS-2. The rule was applied to those six and no other case, and the per-reviewer labels are released so a reader can apply a different rule and recompute. Section 5.2 states what this does and does not establish. Table 3 summarizes corpus-level quality indicators and the QUADAS-2 domain judgments.

Table 3: Corpus-level provenance and methodological quality indicators across the 263 included studies, with QUADAS-2 domain-wise risk of bias and applicability concerns for the full appraised diagnostic/reader subset ( $n = 57$ , rated across seven domains; independently by all three reviewers except for the five studies re-appraised on recovered full text). The 39 studies of the first appraisal round and the 18 of the second are pooled here; the remaining 206 studies were appraised on reproducibility and robustness instead ( $57 + 206 = 263$ ). Six of the 399 domain judgements split three ways with no majority and are recorded as *Unclear* (Section 3).

| Indicator / Domain                | Value / Risk of Bias Breakdown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Publication period                | 1995–2026; 229/263 (87.1%) published in 2020 or later, and 174/263 (66.2%) in 2023 or later, indicating a predominantly recent evidence base.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Basis of assessment, current      | Full text available for 258/263 (98.1%); partial or truncated full text for 1/263 (0.4%); title, abstract and MeSH only for 4/263 (1.5%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Basis of assessment, at appraisal | The first appraisal round was made on the text obtainable at the time, which for 18/263 studies was abstract and MeSH only (6 of them within the QUADAS-2 subset). A later library sweep recovered full text for 14 of those 18 (including 5 of the 6), and all 14 have since been re-appraised on the recovered full text, so appraisal basis now stands at 257/263 full text, 2/263 partial text and 4/263 abstract and MeSH only — within the QUADAS-2 subset, 56/57 on full text and 1/57 (S001) on abstract-level text. The two rows now differ by a single study (S200), appraised on truncated text whose full version has since become available. |



(Table 3 continued)

|                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Leading venues                                                                          | The eight most frequent of the 87 distinct venue titles represented in the corpus, accounting for 128/263 (48.7%) of included studies: <i>Magnetic Resonance in Medicine</i> (40), <i>IEEE Transactions on Medical Imaging</i> (17), <i>AJNR</i> (15), <i>Magnetic Resonance Imaging</i> (14), <i>JMRI</i> (12), <i>Scientific Reports</i> (12), <i>NeuroImage</i> (9), <i>Medical Image Analysis</i> (9).                                                                                                                                                                                                                                                                                                                                                                       |
| Predominant design                                                                      | Algorithmic / methodological reconstruction studies ( $n = 206$ ), alongside a verified observer/reader subset ( $n = 57$ ) (Radmanesh et al., 2022; Lang et al., 2024). Extraction recorded 61 reader designs; appraisal of their full text found 4 with no human reader, and those 4 were reclassified as algorithmic ( $202 + 4 = 206$ ; $61 - 4 = 57$ ). Of the 57: 34 are appraised and verified by all three reviewers; 18 were appraised by all three in a second round and await final adjudication of 6 split domain judgements; and 5 (S004, S020, S027, S034, S115) were re-appraised by the reviewers on full text recovered after the first round, with domain judgements verified against the source text. Every reader-derived statistic in this article uses 57. |
| Reference standard                                                                      | Expert reader consensus (70), an unspecified stated ground truth (62), a fully sampled acquisition (69), a conventional or standard-of-care acquisition (15); not stated in 47/263 (17.9%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Reproducibility                                                                         | Code availability stated in 97/263 (36.9%); a named public dataset or benchmark used in 109/263 (41.4%), most commonly fastMRI, HCP, ADNI and IXI. Availability is variable and remains an area for improvement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Sampling and setting                                                                    | Sampling frame not stated in 95/263 (36.1%); recruitment centre not stated in 174/263 (66.2%). Ethics approval or informed consent reported in 159/263 (60.5%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>QUADAS-2 Risk of Bias (<math>n = 57</math> Reader / Diagnostic Accuracy Studies)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Patient Selection                                                                       | Low: 15/57 (26.3%); Unclear: 19/57 (33.3%); High: 23/57 (40.4%). The weakest domain: sampling is predominantly purposive or pathology-enriched, and most of the appraised studies state no enrolment rule.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Index Test (AI Reconstruction)                                                          | Low: 26/57 (45.6%); Unclear: 9/57 (15.8%); High: 22/57 (38.6%). Low risk falls below half once the second round is included. High ratings arise where readers scored accelerated reconstructions with the reference visible, or where test parameters were set using reference-derived labels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Reference Standard                                                                      | Low: 26/57 (45.6%); Unclear: 15/57 (26.3%); High: 16/57 (28.1%). High ratings arise where no artifact-free reference exists, or where lesion truth is defined by the comparator images themselves (incorporation bias). Reading the recovered full text moved four of the re-appraised studies from Low to High here, each because the comparator images, or software applied to them, supplied the truth against which they were scored.                                                                                                                                                                                                                                                                                                                                        |

(Table 3 continued)

Flow and Timing Low: 45/57 (78.9%); Unclear: 7/57 (12.3%); High: 5/57 (8.8%). The strongest domain, since retrospective undersampling of the same raw  $k$ -space gives a zero interval between index test and reference.

---

**QUADAS-2 Applicability Concerns ( $n = 57$ )**

---

|                    |                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient Selection  | Low: 36/57 (63.2%); Unclear: 4/57 (7.0%); High: 17/57 (29.8%), chiefly healthy-volunteer, lesion-free, or non-clinical cohorts.                                                                                                               |
| Index Test         | Low: 30/57 (52.6%); Unclear: 4/57 (7.0%); High: 23/57 (40.4%), where the evaluated method is image-domain post-processing, cross-field synthesis, harmonization, or a quality index rather than reconstruction from under-sampled $k$ -space. |
| Reference Standard | Low: 39/57 (68.4%); Unclear: 3/57 (5.3%); High: 15/57 (26.3%).                                                                                                                                                                                |

---

Across all 228 risk-of-bias judgements: 112 (49.1%) Low, 50 (21.9%) Unclear, 66 (28.9%) High. Across all 171 applicability judgements: 105 (61.4%) Low, 11 (6.4%) Unclear, 55 (32.2%) High. Over the 364 judgements that rest on three independent ratings Fleiss'  $\kappa = 0.788$  (95% CI 0.739–0.833), but agreement is not uniform across the two appraisal rounds: the 238 retained first-round judgements were unanimous, the 126 of the second round reach  $\kappa = 0.390$  (95% CI 0.293–0.484); the 35 judgements of the 5 re-appraised studies are excluded, having been conducted subsequently on recovered full text. Section 3 gives the decomposition and Section 5.2 what it bounds. The high proportion of Unclear ratings reflects unreported blinding and sampling procedures, compounded by the one first-round study for which only abstract-level text is available. Six second-round judgements split three ways with no majority and are recorded as Unclear under the rule stated in Section 3. Per-study ratings, each with the verbatim supporting quotation, are released as `included_characteristics_supplement.csv`.

---

420     *Derived venue classification.* The analysis in Section 4.5 groups studies by  
421 publication community. That grouping is post-hoc, was not part of the fixed  
422 protocol, and is stated in full in Supplementary Table S13 so it can be re-  
423 produced or contested. Each study's journal title, and nothing else about the  
424 study, is matched case-insensitively against three ordered keyword lists — en-  
425 gineering and methods, then general neuroscience and neuroimaging, then clin-  
426 ical imaging — with anything unmatched left *unclassified* rather than forced  
427 into a group. The ordering is load-bearing, since titles such as *Medical Image*  
428 *Analysis* would otherwise match two lists, and it is applied identically to all  
429 263 studies. The rule as executed and the resulting per-study assignment are  
430 released as `5_Code/venue_classification.py` and `venue_community clas-`  
431 `sification.csv`.

432     *Synthesis.* Due to heterogeneity in acquisition protocols, reconstruction ar-  
433 chitectures, datasets, and outcome measures, evidence was synthesised narra-

tively in accordance with SWiM guidelines (Campbell et al., 2020). Studies were structured around the primary research questions (RQ1–RQ3) and grouped by reconstruction method family, field strength, pulse sequence, and failure mode. Quantitative values reflect named primary studies rather than pooled meta-analytic estimates; no effect measure is pooled, and no meta-analytic model is fitted. Percentages are reported to one decimal place and rounded half-up throughout, so 9/80 prints as 11.3% and not 11.2%. Sensitivity analysis was performed by recalculating headline prevalence counts on the 258 studies verified against complete full text (Section 5.2).

## 4. Results

All counts and prevalence statements in this section describe the 1,068 reports assessed in full and the 263 included studies; because 81 candidate reports remain unretrievable, they are lower bounds on the literature rather than estimates of it.

### 4.1. Selection and corpus

From 27,327 records identified across seven databases, multi-tier deduplication removed 403 duplicates, leaving 26,924 unique records screened at title and abstract. Of these, 1,149 proceeded to full-text eligibility, and full texts were obtained for 1,068 of them (93.0%; Section 3). Each of the 81 reports still out of reach was checked against its publisher platform and assigned a recorded reason: 53 sit behind title-level subscriptions our institution does not hold, across nineteen publishers; 13 are Springer conference chapters sold only per chapter; 4 are ISMRM proceedings abstracts issued without a PDF; 4 resolve to a whole proceedings volume or to front matter rather than a single report; and 7 carry no DOI and no locatable source. Supplementary Table S14 gives the per-publisher breakdown. Access difficulty is itself informative, though it bounds the missing evidence rather than characterising it. Of the 341 assessed reports obtainable only through title-level subscriptions and equivalent barriers,

294 (86.2%) proved ineligible on the same PICOS criteria, 2 were retained as background and 45 were eligible ( $294 + 2 + 45 = 341$ ; the 45 resolve to 44 studies, one being a preprint adjudicated a duplicate). Raw non-retrieval counts therefore overstate the missing evidence by roughly sevenfold. But those 45 were concentrated in subscription clinical journals, and Section 4.5 shows they differ systematically from the open literature in exactly the variable this review measures, so 86.2% is an upper bound on what non-retrieval costs us, not a demonstration that it costs us nothing. Of the 1,068 assessed reports, 787 were excluded (785 on PICOS grounds, 2 as duplicate reports of included studies) and 18 recorded as background, leaving **263 included primary studies** spanning 1995–2026 (Fig. 1). Each included study was verified against the fullest text obtainable for it: complete full text for 258, partial text for 1, and title, abstract and controlled vocabulary for 4 (Section 3). Their aggregate characteristics are given in Table 4, and their thematic distribution across the research questions is summarized in Supplementary Table S7; all 263 are listed, with full references, in Supplementary Tables S1–S3.

Corpus growth is recent: 34 studies appeared in 1995–2019, 55 in 2020–2022 and 174 in 2023–2026.

#### 4.2. RQ1: the metric–reader comparison is almost never made, and where readers appear they agree only partially

Across the 84 studies reporting objective fidelity metrics and the 57 verified reader studies, the comparisons that actually exist show agreement between fidelity metrics and radiologist assessment that is partial and condition-dependent (Table 5). The size of that evidence base should be stated in the text and not left to the table caption: only two of the four rows in Table 5 were collected on reconstructions of undersampled  $k$ -space. Tang et al. (2025) collected radiologist rankings on synthetically corrupted image pairs, and Sommer et al. (2020) on clinical scans corrupted by real patient motion rather than by undersampling. The table that carries RQ1 therefore rests on two on-target rows, which is the honest size of the direct evidence for the metric–reader comparison

Table 4: Characteristics of the 263 included studies.

| Demographic / Method Dimension                                                                                                                                                                                                                                                                                                                                                                                                                  | Category Breakdown                    | ( <i>n</i> )   | Percentage / Key Methodological Notes                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------|-------------------------------------------------------------------|
| <b>Publication Period</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 1995–2019                             | <i>n</i> = 34  | 12.9% (Classical CS/PI & early CNNs)                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020–2022                             | <i>n</i> = 55  | 20.9% (Unrolled physics & GAN priors)                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2023–2026                             | <i>n</i> = 174 | 66.2% (Diffusion priors & 3D/4D foundation models)                |
| <b>Reconstruction Paradigm</b><br>( <i>multi-label</i> ; 339 labels recorded)                                                                                                                                                                                                                                                                                                                                                                   | Compressed Sensing / Parallel Imaging | <i>n</i> = 93  | Iterative sparsity & coil sensitivity encoding                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Generative (GAN/Diffusion/VAE)        | <i>n</i> = 80  | Adversarial perception & posterior sampling                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Motion Correction                     | <i>n</i> = 57  | Intra-scan motion artifact suppression                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Physics-based Unrolled                | <i>n</i> = 37  | Explicit <i>k</i> -space data consistency layers                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Denoising                             | <i>n</i> = 30  | Image-domain noise suppression                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Super-resolution                      | <i>n</i> = 25  | Resolution enhancement post-acquisition                           |
| <b>MRI Pulse Sequences</b><br>( <i>multi-label</i> ; 490 labels)                                                                                                                                                                                                                                                                                                                                                                                | Self-supervised / Unsupervised        | <i>n</i> = 17  | Training without fully sampled reference scans                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | T1-weighted (T1w)                     | <i>n</i> = 163 | High-resolution anatomical cortical profiling                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | T2-weighted (T2w)                     | <i>n</i> = 122 | Ventricular & CSF pathology assessment                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | FLAIR                                 | <i>n</i> = 85  | Hyperintense peri-ventricular lesion detection                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Diffusion-Weighted (DWI)              | <i>n</i> = 72  | Acute ischemic lesion (DWI core) evaluation                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SWI / $T_2^*$ -weighted               | <i>n</i> = 48  | Punctate cerebral microbleed hypointensity voids                  |
| <b>Magnetic Field Strength</b><br>( <i>single-label</i> ; predominant field where several are used)                                                                                                                                                                                                                                                                                                                                             | 3.0 Tesla (3T)                        | <i>n</i> = 153 | 58.2% (Primary clinical scanner benchmark)                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7.0 Tesla (7T)                        | <i>n</i> = 35  | 13.3% (Ultra-high-field research acquisitions)                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1.5 Tesla (1.5T)                      | <i>n</i> = 21  | 8.0% (Routine clinical throughput imaging)                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Not stated</i>                     | <i>n</i> = 50  | 19.0% (Full text obtained; no field strength stated)              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Not ascertainable</i>              | <i>n</i> = 4   | 1.5% (Only abstract or partial text available)                    |
| <b>Evaluation Paradigm</b><br>( <i>multi-label</i> ; 177 labels recorded across 144 studies)                                                                                                                                                                                                                                                                                                                                                    | Pixel-wise (PSNR/SSIM/NRMSE)          | <i>n</i> = 84  | 47.5% of recorded (legacy fidelity reporting)                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Observer / Reader Studies             | <i>n</i> = 57  | 32.2% of recorded (task-based reader assessment; verified subset) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Uncertainty Estimation                | <i>n</i> = 18  | 10.2% of recorded (posterior variance maps)                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Learned / Perceptual                  | <i>n</i> = 18  | 10.2% of recorded (rank-trained metrics)                          |
| <b>Co-reporting.</b> Of the 144 studies with a recorded evaluation paradigm, 28 report more than one. Critically, only <b>18</b> studies in the entire corpus (6.8% of 263; 12.5% of the 144 recorded) report <i>both</i> a pixel-wise fidelity metric <i>and</i> a reader assessment on the same data, which is the minimum precondition for computing metric-versus-reader agreement on which safety claims implicitly depend. See Section 4. |                                       |                |                                                                   |

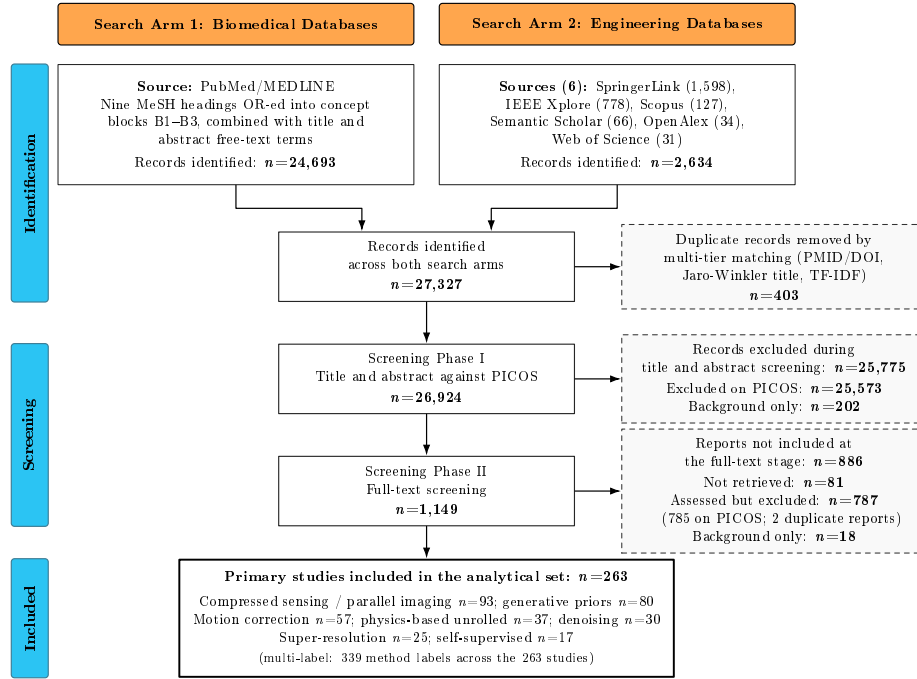


Figure 1: PRISMA 2020 flow diagram for study selection across the two search arms. Multi-tier deduplication reduced 27,327 identified records to 26,924 unique records; title and abstract screening advanced 1,149 candidates. Full texts were obtained for 1,068 of them (93.0%); 81 could not be retrieved. Following PRISMA 2020, exclusion and background retention are recorded as distinct dispositions: of the 1,068 reports assessed in full, 787 were excluded (785 on explicit PICOS grounds and 2 as duplicate reports of included studies) and a further 18 were retained as background material rather than excluded, leaving 263 primary studies in the analytical set ( $263 + 787 + 18 = 1,068$ ).

492 in this setting. In the fastMRI 2020 challenge’s reader phase, six radiologists  
493 ranked submissions for quality of pathology depiction; their concordance, by  
494 Kendall’s coefficient, was 0.457 in the  $4\times$  track and 0.386 in the  $8\times$  track, rising  
495 to 0.781 only in the transfer track, and the organisers report that concordance  
496 improved as SSIM scores diverged, which is to say the metric separated methods  
497 reliably only where their outputs already differed grossly (Muckley et al., 2021).  
498 When radiologist preference is instead used as the training signal, purpose-built  
499 quality networks predict expert rankings at  $75.2\% \pm 1.3\%$  and  $79.2\% \pm 1.7\%$ ,  
500 against radiologists’ own interobserver agreement of  $61.5\% \pm 5.5\%$ ; the same  
501 study finds rank-based reading far more reliable than Likert scoring (70.4% ver-  
502 sus 25% agreement between radiologists) (Tang et al., 2025). Reader tolerance

of acceleration is real but bounded: in a retrospective evaluation spanning accel-  
erations up to  $100\times$ , 94% of volumes (65 of 69) were rated sufficient for potential  
screening use at up to  $14\times$  (interreader intraclass correlation 0.875), with the  
threshold for full diagnostic quality lower still (Radmanesh et al., 2022). The  
one direct clinical observation of the failure this review is about comes from an  
emergency-department and inpatient comparison, and it is a single case. Across  
66 paired examinations in which a two-minute ultrafast protocol (T1w, T2/T2\*,  
FLAIR and DWI, machine-learning-assisted reconstruction) was read against a  
ten-minute reference protocol, two blinded neuroradiologists agreed on the main  
clinical diagnosis in 65 of 66 cases (98.5%), rated overall diagnostic quality no  
different between protocols ( $P > .05$ ), preferred the reference protocol for image  
noise and geometric distortion ( $P < .05$ ) and preferred the accelerated protocol  
for reduced motion artifact ( $P < .01$ ) (Lang et al., 2024). The single discrepant  
case is the one that matters here: a punctate focus of restricted diffusion, read  
as possible acute-to-subacute infarct on the reference DWI, was less conspicu-  
ous on the accelerated DWI, which the authors attribute to a combination of  
image-quality difference and section positioning, the lesion being smaller than  
the section thickness (4 mm reference against 5 mm accelerated) and so subject  
to partial-volume contamination. They draw the inference themselves, that the  
protocol’s sensitivity to small findings is an open question. One case in 66 is  
not a prevalence estimate and we do not offer it as one. It is a worked instance  
of the pattern this review predicts: an aggregate quality comparison that finds  
no difference, and inside it a focal finding that a reader saw and the aggregate  
did not. Where readers assess artifacts directly, blinded reading detects what  
reference-based metrics cannot express: a two-neuroradiologist blinded study of  
28 clinical scans with real patient motion found significant artifact-score reduc-  
tions after network correction ( $p < .03$ ), a setting in which the motion-corrupted  
reference invalidates PSNR and SSIM by construction (Sommer et al., 2020).  
Motion also interacts with the sampling scheme itself, appearing as ghosting  
under Cartesian trajectories and as streaking or blurring under radial ones,  
so the visual form of the same physical corruption differs by protocol (Schau-

man et al., 2026). The reference pipeline is not neutral either: conventional SWI homodyne phase filtering has been shown to create punctate artifactual microhaemorrhage-like foci, so processing can manufacture findings as well as remove them (Li et al., 2015).

No included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment for deep learning-based brain reconstruction, and none evaluated a model observer. The second statement needs a boundary that the first does not. Our search vocabulary (Table 1, block B3) contains reader-study and metric terms but no model-observer vocabulary, so a study that used a channelised Hotelling or non-prewhitening observer and described it only in those terms could have been missed at retrieval. What we can say is that none of the 263 studies retrieved, screened and extracted here reports one, and that the term does not appear in any extracted evaluation label; whether the true field-wide count is zero or merely very small, it is not a paradigm this corpus can be said to use. That absence is itself the RQ1 finding: only 18 of 263 studies (6.8%, a screening-level lower bound; Section 3) record a fidelity metric and a reader assessment on the same data, so the field’s central proxy relationship, that high fidelity scores imply diagnostic adequacy, is largely unmeasured rather than merely disputed.

Table 6 reports how evaluation practice varies with the acceleration factor actually stated in each study. It is built only from what the corpus records: the acceleration factor was extracted post hoc from the full texts of all 263 included studies with a verbatim supporting quotation for every value, and 63 studies (24.0%) state one. Studies are grouped by the highest factor at which they evaluate their method. The table deliberately carries no fidelity threshold, reader-score difference or metric–reader rank statistic per acceleration band, because no included study reports one: that absence is the finding of Section 4, not an omission from the table.



Table 5: What the reader studies in this corpus actually measured about metric–reader agreement (RQ1). Every value is quoted from the named study, with the reported coefficient and its definition stated. No included study reports a metric–reader rank correlation for deep learning-based brain reconstruction; the rows below are the closest measurements that exist, and the second column states for each row what the readers were actually shown, because two of the four rows were not collected on reconstructions of undersampled  $k$ -space.

| Study                                         | Readers and design                                                                                                                                                                            | Comparison performed                                                                 | Reported quantity (as defined in the study)                                                                                                                                                                                        |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fastMRI 2020 challenge (Muckley et al., 2021) | 6 radiologists ranking submissions for quality of pathology depiction                                                                                                                         | Radiologist concordance across accelerations; SSIM-based ranking vs reader ranking   | Kendall's $W = 0.457$ ( $4\times$ ), $0.386$ ( $8\times$ ), $0.781$ (transfer track); concordance improved as SSIM scores diverged                                                                                                 |
| Tang et al. (2025)                            | 7 radiologists, rank-based reading; 2 also Likert scoring. <b>Rankings were collected on synthetically corrupted image pairs, not on reconstructions of undersampled <math>k</math>-space</b> | Quality networks trained on radiologist rankings vs MSE/SSIM as optimisation targets | Ranking accuracy $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ with reference image; radiologist interobserver agreement $61.5\% \pm 5.5\%$ ; ranking vs Likert agreement $70.4\%$ vs $25\%$                                           |
| Radmanesh et al. (2022)                       | Retrospective; neuroradiologist scoring of diagnostic quality at accelerations up to $100\times$                                                                                              | Reader-assigned acceleration thresholds for diagnostic and screening quality         | $65/69$ volumes ( $94\%$ ) sufficient for potential screening at $\leq 14\times$ ; interreader ICC $0.875$ ; mean absolute interreader difference in threshold acceleration factor $0.7\times$ (level 1) and $5.4\times$ (level 2) |
| Sommer et al. (2020)                          | 2 neuroradiologists, blinded, 28 clinical cases with real motion                                                                                                                              | Artifact severity (5-point scale) before vs after network correction                 | Significant reduction in mean artifact scores ( $p < .03$ ); reference-based metrics inapplicable, as the reference itself is motion-corrupted                                                                                     |

Table 6: Evaluation practice by the highest acceleration factor ( $R$ ) each study reports, among the 63 of 263 included studies that state one. Bands are half-open, so a study reporting exactly  $R = 4$  falls in the second band and one reporting exactly  $R = 8$  in the third. Counts are studies; percentages are within band. *Reader*, any observer or reader assessment recorded; *M+R*, a pixel-wise fidelity metric and a reader assessment recorded on the same data; *Halluc.*, hallucination or fidelity instability among the recorded failure modes; *Artifacts*, aliasing, ghosting or Gibbs artifacts among them. Acceleration factors were extracted from full text after the registered extraction round, each with a verbatim supporting quotation and excluding matches inside reference lists and citation contexts; the 407 retained quotations come from the 63 studies that state a factor, a mean of 6.5 quotations per such study, and are released as `acceleration_factors_evidence.csv`. The extraction is screening-level, so counts are lower bounds. No study at any acceleration factor reports a rank correlation between a fidelity metric and reader judgement.

| Acc. ( $R$ )   | Studies   | Reader            | M+R             | Halluc.           | Artifacts         | Sequences most often evaluated |
|----------------|-----------|-------------------|-----------------|-------------------|-------------------|--------------------------------|
| $2 \leq R < 4$ | 32        | 7 (21.9%)         | 1 (3.1%)        | 7 (21.9%)         | 25 (78.1%)        | T1w 15, T2w 12, FLAIR 11       |
| $4 \leq R < 8$ | 17        | 5 (29.4%)         | 1 (5.9%)        | 2 (11.8%)         | 12 (70.6%)        | T1w 9, T2w 9, FLAIR 7          |
| $R \geq 8$     | 14        | 5 (35.7%)         | 4 (28.6%)       | 2 (14.3%)         | 8 (57.1%)         | T1w 9, T2w 8, FLAIR 8          |
| <b>All</b>     | <b>63</b> | <b>17 (27.0%)</b> | <b>6 (9.5%)</b> | <b>11 (17.5%)</b> | <b>45 (71.4%)</b> | T1w 33, T2w 29, FLAIR 26       |

The  $R$ -reporting subset is close to the whole corpus on reader assessment (27.0% against 21.7% for all 263), so it is not a systematically better-evaluated stratum. Reader assessment rises with acceleration (21.9%  $\rightarrow$  29.4%  $\rightarrow$  35.7%) and paired metric-reader reporting concentrates in the most accelerated band (4/14 against 1/32 in the  $2 \leq R < 4$  band), but the counts are small and most studies in every band (including 9 of the 14 with  $R \geq 8$ ) report no reader assessment at all. Artifact description is the most frequently recorded failure mode at every acceleration level, whereas hallucination or fidelity instability is recorded in 11 of 63.

#### 4.3. RQ2: four mechanisms explain the divergence

Four mechanisms account for why the metrics miss what the readers see. Table 7 states each one formally alongside the clinical failure it permits, and Supplementary Table S8 gives the fuller derivations and evidence sources.

*Spatial averaging.* Conventional  $L_2$  metrics average residual error uniformly over all  $H \times W \times D$  voxels, so the total decomposes over lesion and background regions:

$$\begin{aligned} \text{MSE}(\hat{X}, X) &= \frac{V_{\text{lesion}}}{V_{\text{total}}} \text{MSE}_{\text{lesion}} \\ &\quad + \left(1 - \frac{V_{\text{lesion}}}{V_{\text{total}}}\right) \text{MSE}_{\text{bg}}. \end{aligned} \quad (1)$$

With lesion volume fraction  $f = V_{\text{lesion}}/V_{\text{total}}$  and local error ratio  $r = \text{MSE}_{\text{lesion}}/\text{MSE}_{\text{bg}}$ , the global PSNR change caused by a localized edit follows

Table 7: The four mechanisms by which a whole-volume fidelity score fails to register a focal diagnostic error (RQ2). The third column names a failure the mechanism permits, not one this review has observed at a stated frequency; the fourth states why the score does not move. Numerical entries are computed from Equation (2) under the stated error model, not measured in any primary study.

| Mechanism                    | Formal statement                                                                                                                                            | Clinical failure it permits                                                                                                   | Why the score does not move                                                                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Spatial error averaging  | $L_2$ error is averaged uniformly over all $H \times W \times D$ voxels, so a lesion contributes with weight $f = V_{\text{lesion}}/V_{\text{total}}$       | Erasure of an acute lacunar infarct on DWI or FLAIR, or of a punctate microhaemorrhage on SWI or $T_2^*$                      | By (2), erasing a $100 \text{ mm}^3$ infarct at $r = 100$ moves global PSNR by $0.027\text{--}0.036 \text{ dB}$ ; a $5 \text{ mm}^3$ microhaemorrhage by $1.3\text{--}1.8 \times 10^{-3} \text{ dB}$ |
| (b) Reference dependence     | Every reference-based score presupposes a registered, artifact-free reference image                                                                         | A valid motion-corrected reconstruction is penalised against a corrupted reference, and genuine correction is scored as error | The comparison is no longer between reconstruction and truth; PSNR and SSIM become uninterpretable rather than merely noisy                                                                          |
| (c) Loss symmetry            | $(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$ : the objective is indifferent to the sign of the error                                                         | A suppressed hyperintense lesion is scored identically to added background noise of the same magnitude                        | Diagnosis is asymmetric where the metric is not: a missed finding and a spurious brightening carry the same penalty                                                                                  |
| (d) Realism–reward trade-off | Adversarial and perceptual objectives reward outputs typical of the training distribution, $\log D_\phi(\hat{X})$ , not outputs faithful to the measurement | Synthesis of plausible vessels, sulci or texture that was never acquired                                                      | Perceptual scores can rise as measurement fidelity falls, so the objective pays for the hallucination it should penalise                                                                             |

571 directly:

$$|\Delta\text{PSNR}| = 10 \log_{10} (1 + f(r - 1)) \quad (\text{dB}). \quad (2)$$

572 The consequence is quantitative rather than rhetorical, and it is worth anchoring  
573 on a lesion a neuroradiologist would expect to resolve rather than on the smallest  
574 arithmetically convenient one. An acute lacunar infarct of  $100 \text{ mm}^3$  is a sphere  
575  $5.8 \text{ mm}$  across: on a clinical DWI acquired at  $4\text{--}5 \text{ mm}$  section thickness (Lang  
576 et al., 2024) that is several voxels wide in-plane and spans one to two sections, so  
577 it is resolved, if only just, in the through-plane direction. It occupies a volume  
578 fraction  $f = 6.3\text{--}8.3 \times 10^{-5}$  of a  $1.2\text{--}1.6 \text{ L}$  brain. Even at a hundredfold local  
579 error ratio ( $r = 100$ ), Equation (2) gives  $|\Delta\text{PSNR}| = 0.027\text{--}0.036 \text{ dB}$ . Erasing

580 a resolvable acute infarct outright therefore moves global PSNR by about three  
 581 hundredths of a decibel, which is the same order as the third decimal place of a  
 582 score conventionally reported to two.

583 The limiting case is worth stating precisely, because it is easy to state loosely.  
 584 A punctate cerebral microhaemorrhage of  $5 \text{ mm}^3$  is 2.1 mm across. It is *not*  
 585 resolved on clinical DWI, whose voxels are roughly  $12\text{--}20 \text{ mm}^3$ , so no recon-  
 586 struction can be said to erase it there; it is not resolved in the first place. It  
 587 is several voxels across on the high-resolution SWI or  $T_2^*$ -weighted acquisition  
 588 on which microhaemorrhages are actually read, and on that sequence its era-  
 589 sure costs  $f = 3.1\text{--}4.2 \times 10^{-6}$  and  $|\Delta\text{PSNR}| = 1.3\text{--}1.8 \times 10^{-3} \text{ dB}$ , below the  
 590 precision at which the score is reported at all. Scaling in the other direction, a  
 591  $500 \text{ mm}^3$  cortical infarct costs  $\approx 0.15 \text{ dB}$  ( $0.13\text{--}0.18 \text{ dB}$  across the same brain-  
 592 volume range). Fig. 2 plots the identity across this range, with the sub-voxel  
 593 region marked. The argument does not need the extreme case: it holds by one  
 594 to two orders of magnitude at a lesion size that is unambiguously resolved and  
 595 unambiguously matters. These are computed bounds from the stated  $f$  and  $r$ ,  
 596 not measurements from any primary study, and the error model behind them, a  
 597 uniform background error and a free local error ratio, is an assumption we state  
 598 rather than a property of any reconstruction.

599 *Loss-function symmetry.* MSE and SSIM are indifferent to the sign of the  
 600 error, since  $(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$ , whereas diagnosis is not: suppressing a  
 601 true hyperintense lesion means the finding is missed, while adding background  
 602 noise costs nothing clinically. An evaluation respecting that asymmetry must  
 603 penalise the two directions differently, for example

$$\begin{aligned} \mathcal{L}_{\text{asym}}(\hat{X}, X) = & \alpha \sum_{i \in \Omega_{\text{lesion}}} \max(0, X_i - \hat{X}_i)^2 \\ & + \beta \sum_{i \in \Omega_{\text{bg}}} (\hat{X}_i - X_i)^2, \quad (\alpha \gg \beta). \end{aligned} \quad (3)$$

604 Several included studies already build objectives of this shape: lesion-mask-  
 605 weighted losses motivated explicitly by the dilution of lesion gradients among  
 606 background voxels (Lee et al., 2026a), overlap-weighted reconstruction losses

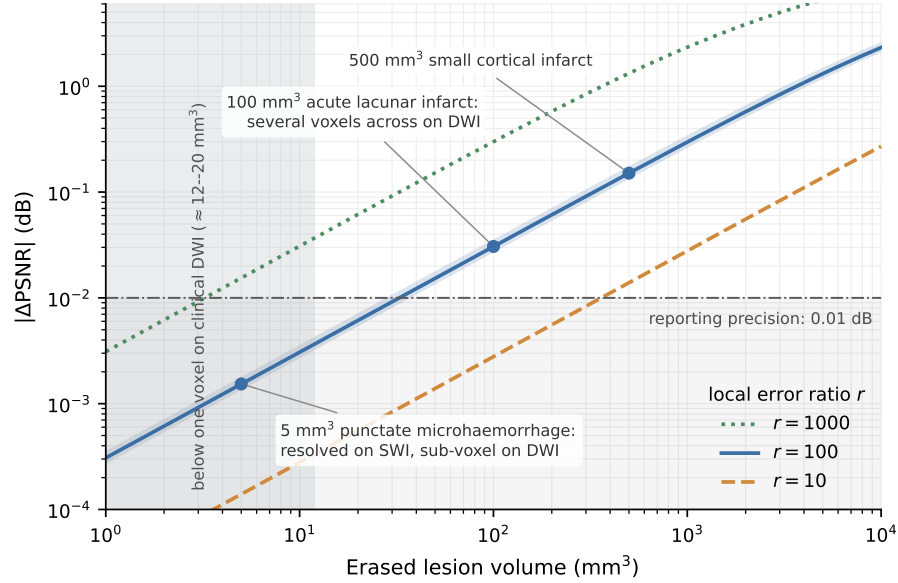


Figure 2: The PSNR cost of erasing a focal lesion, from Equation (2). Curves give  $|\Delta\text{PSNR}|$  against erased lesion volume for three local error ratios  $r$  at a 1.4 L brain; the shaded band around  $r = 100$  spans the 1.2–1.6 L range. Markers show the three clinical anchors used in the text. The horizontal line at 0.01 dB is the two-decimal precision at which PSNR is conventionally reported, and the shaded left-hand region is the volume below a single voxel on a clinical DWI acquisition, where erasure is not defined because the finding is not resolved. Every value is computed from the identity under the stated error model; nothing in this figure is measured.

607 that up-weight structure against background (Jayakumar et al., 2023), and  
 608 gradient-based structure-preservation terms (Zhang et al., 2026). Their exis-  
 609 tence is evidence that the symmetry problem is recognised in practice, though  
 610 none of these studies pairs the correction with a reader assessment.

611 *Realism–reward trade-off.* Adversarial and perceptual objectives reward out-  
 612 puts that look like the training distribution, which is not the same as outputs  
 613 faithful to the measurement, so a generative reconstruction can gain plausible  
 614 texture that was never acquired (Zhou et al., 2024; Bhadra et al., 2021).

615 *Reference dependence.* All four metrics presuppose a registered, artifact-free  
 616 reference. Intra-scan motion invalidates that assumption, and 69 of 263 studies  
 617 rest their reference standard solely on a fully sampled acquisition, which is valid  
 618 only where the subject did not move.

619 *4.4. RQ3: what current evaluation paradigms can and cannot establish*

620 Read against those mechanisms, the corpus supports a paradigm-by-  
 621 paradigm account of what each evaluation approach can demonstrate about  
 622 prior-induced diagnostic error. Table 8 states the account compactly; the  
 623 argument behind it is given here, because two of its six rows are analytical  
 624 claims we derive rather than findings we extracted, and an analysis presented  
 625 only in a table has not been argued.

626 *What the four empirical paradigms establish.* Pixel-wise fidelity metrics es-  
 627 tablish aggregate agreement with a reference and nothing about focal change:  
 628 by (2) the penalty for erasing a resolvable acute infarct is of the order of the  
 629 reporting precision, so a method can pass any threshold anyone would set while  
 630 failing on the finding that matters. Perceptual and rank-trained metrics track  
 631 reader preference better than MSE or SSIM as an optimisation target, but they  
 632 inherit the variability of the readers they were trained on and, in the one study  
 633 of this kind in the corpus, were calibrated on synthetically corrupted images  
 634 rather than on reconstructions of undersampled  $k$ -space (Tang et al., 2025), so  
 635 their transfer to the reconstruction setting is assumed rather than shown. Un-  
 636 certainty estimates flag where a model is unsure, which is not the same question:  
 637 a prior confident enough to remove a lesion is confident in the region where it  
 638 removed it, so posterior variance can be low exactly where the diagnostic error  
 639 is. Task-based observer assessment does settle the question directly, and it is  
 640 the only paradigm here that does, which is why its scarcity in this corpus (57 of  
 641 263 studies, and 18 paired with a metric) is the review’s central finding rather  
 642 than an incidental one.

643 *Two limits that follow from the physics, not from the corpus.* The first  
 644 concerns the check that reconstruction papers most often use to argue that an  
 645 output is faithful to the data: the measurement-space residual  $\|\mathbf{A}\hat{X} - Y\|_2 / \|Y\|_2$   
 646 against a tolerance. This check cannot detect prior-induced edits at all, and the  
 647 reason is structural rather than statistical. Write the reconstruction as the sum  
 648 of its measured and unmeasured components,  $\hat{X} = \mathbf{A}^+ \mathbf{A} \hat{X} + (\mathbf{I} - \mathbf{A}^+ \mathbf{A}) \hat{X}$ . The  
 649 second term lies in the null space  $\mathcal{N}(\mathbf{A})$ , and  $\mathbf{A}$  annihilates it by definition, so

any amount of structure written there, an erased lesion or an invented vessel  
 alike, leaves the residual exactly unchanged. Undersampling is what creates that  
 null space; the more aggressive the acceleration, the larger the subspace in which  
 a learned prior can write freely without moving the number the field uses to  
 certify data consistency. A residual check is therefore a two-sided measurement-  
 fidelity band and not a safety test, and its apparent reassurance grows least  
 trustworthy precisely where acceleration is highest.

The second limit is the constructive corollary. If prior-supplied content lives  
 in  $\mathcal{N}(\mathbf{A})$ , then isolating  $(\mathbf{I} - \mathbf{A}^+\mathbf{A})\hat{X}$  and inspecting it is the natural place to  
 look for erasure and hallucination, since that is where both must reside (Bhadra  
 et al., 2021; Gottschling et al., 2025). This is the one paradigm in Table 8 that  
 is not represented in the corpus at all, and its practical obstacle is the same one  
 that limits third-party auditing generally: computing the projection requires the  
 forward operator, hence the raw  $k$ -space and the coil sensitivities, which com-  
 mercial inline pipelines do not export (Section 5). Causal and counterfactual  
 auditing, which asks the most direct question of all, whether a specific finding  
 survives a controlled change of acquisition, appears in a small part of the cor-  
 pus and almost never for physics-constrained  $k$ -space reconstruction (Pawlowski  
 et al., 2020; Güngör et al., 2023; Peng et al., 2026).

The answer to RQ3 is therefore not that current evaluation is weak but  
 that it is mismatched: each paradigm in use answers a question adjacent to the  
 safety question, and no paradigm now in use establishes reconstruction safety  
 on its own. The two that could, task-based observer assessment and null-space  
 inspection, are respectively rare and absent, and the requirements in Section 4.7  
 follow from that pairing.

#### 4.5. Structure of the evaluation gap

The counts above establish that safety-relevant evaluation is scarce. Analysis  
 of the same per-study records establishes where the scarcity comes from, and  
 indicates a division of labour rather than uniform neglect.

Table 8: What each evaluation paradigm can and cannot establish about prior-induced diagnostic error (RQ3). Rows 1–4 summarize evaluation practice and performance as reported in the included studies; quoted values are representative results from the cited primary studies, not pooled estimates. Rows 5–6 state analytical limits derived by the authors from the acquisition physics (our own analysis; Section 4) and are labelled as such rather than attributed to any primary study.

| Evaluation Paradigm                   | Core Mechanism                                                                                                                                                                                                                | What It Can and Cannot Establish                                                                                                                                                                                                                                                                                                               | Key References                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Pixel-wise fidelity metrics        | $L_2$ / SSIM<br>whole-image<br>voxel averaging                                                                                                                                                                                | Aggregate agreement with a reference only. Cannot resolve focal change: by (2), erasing a resolvable $100 \text{ mm}^3$ lacunar infarct at $r = 100$ costs 0.03 dB. Requires a registered reference                                                                                                                                            | Recorded in 84 of the 144 evaluation-labelled studies                                                              |
| 2. Perceptual & learned metrics       | Networks trained on radiologist rankings                                                                                                                                                                                      | Tracks reader preference better than MSE/SSIM as an optimisation target ( $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ against readers' own $61.5\% \pm 5.5\%$ ). Inherits training-reader variability, and was calibrated on synthetically corrupted data rather than clinically undersampled acquisitions, so transfer is assumed (Section 4.4) | Tang et al. (2025)<br>( $n = 18$ in corpus)                                                                        |
| 3. Uncertainty estimation             | Posterior sampling / Bayesian variance $p(X   Y)$                                                                                                                                                                             | Flags out-of-distribution inputs and regions of low confidence. Does not attribute missing structure to prior vs sampling operator; high confidence in hallucinated regions is possible                                                                                                                                                        | Khawaled and Freiman (2024)<br>( $n = 18$ in corpus)                                                               |
| 4. Task-based observer assessment     | Human reader performance on a diagnostic task; model observers are its automated form, unrepresented here (Barrett et al., 1993; Barrett and Myers, 2004; International Commission on Radiation Units and Measurements, 1996) | Establishes diagnostic sufficiency directly; the gold standard in assessment theory. Costly and reader-variable ( $61.5\% \pm 5.5\%$ , falling to 25% under absolute Likert scoring), and rarely paired with a metric: only 18/263 report both                                                                                                 | Radmanesh et al. (2022),<br>Lang et al. (2024)<br>( $n = 57$ in corpus)                                            |
| 5. Measurement-space residual check   | $\ \mathbf{A}\tilde{X} - Y\ _2 / \ Y\ _2$ against tolerance                                                                                                                                                                   | <b>Cannot detect prior-induced edits:</b> the residual is invariant to anything in $\mathcal{N}(\mathbf{A})$ . A two-sided measurement-fidelity band, not a safety test                                                                                                                                                                        | Common diagnostic check in the reviewed studies                                                                    |
| 6. Null-space / hallucination mapping | Structure written into the null-space component $(\mathbf{I} - \mathbf{A}^+ \mathbf{A})\hat{X}$                                                                                                                               | Isolates prior-supplied content, where erasure and hallucination must reside. Requires $\mathbf{A}$ , hence raw $k$ -space and coil sensitivities                                                                                                                                                                                              | Bhadra et al. (2021),<br>Gottschling et al. (2025);<br>formalization for reconstruction auditing is author-derived |



679       *The benchmarks cannot show the failure the field is worried about.* One hun-  
 680       dred and nine of the 263 studies (41.4%) name a public dataset, and across  
 681       the whole corpus those references resolve to eleven datasets: fastMRI, the Hu-  
 682       man Connectome Project, ADNI, IXI, UK Biobank, BraTS, OASIS, Calgary–  
 683       Campinas, M4Raw, the developing Human Connectome Project and TCGA.  
 684       Dataset labels are multi-valued (136 cohort assignments across the 109 studies),  
 685       and of the 109, the cohorts named are healthy volunteers or population sam-  
 686       ples (55, 50.5%), chronic neurodegenerative cohorts (31, 28.4%), brain tumour  
 687       cohorts (16, 14.7%), or raw- $k$ -space benchmarks not curated for pathology (34,  
 688       31.2%). These four shares are multi-label and sum to 124.8%, a study naming  
 689       two cohorts being counted in both; every other percentage here is single-label.  
 690       *Within this corpus, no study benchmarks against a dataset curated for acute is-*  
 691       *chemic stroke or intracranial haemorrhage.* The sharper version of this finding  
 692       sits in the annotation layer the field has already built. fastMRI+ supplies 7,570  
 693       expert bounding-box annotations across thirty brain pathology categories (Zhao  
 694       et al., 2022), and not one of the thirty is a haemorrhage category of any kind,  
 695       microhaemorrhage included. The apparent counterexample should be stated  
 696       plainly rather than left for a reader to find, because it is the category closest  
 697       to this article’s own worked example: fastMRI+ *does* carry an image-level *La-*  
 698       *cunar Infarct* label (113 annotations across 32 subjects), alongside study-level  
 699       *Small Vessel Chronic White Matter Ischemic Change* and a *Global Ischemia*  
 700       label recorded for a single subject. It does not rescue the benchmark, and the  
 701       reasons are the ones that matter for reconstruction safety. No fastMRI+ cat-  
 702       egory distinguishes acute from chronic infarction, so a lacunar label carries no  
 703       acuity and cannot identify the acute presentation whose erasure is at issue. The  
 704       limit is in the source images as much as in the label set: the annotated se-  
 705       ries are axial T2-FLAIR, T1w or contrast-enhanced T1w only, so neither DWI,  
 706       on which infarct acuity is established, nor SWI or  $T_2^*$ -weighted imaging, on  
 707       which microhaemorrhage is read, was annotated at all. Acute vascular pathol-  
 708       ogy is therefore absent even where annotation infrastructure exists, and it is  
 709       absent for a reason that adding labels alone would not repair. Supplementary

710 Table S10 sets out the full derivation, including the sub-selection of 1,001 of  
711 the 5,847 fastMRI brain studies, the category listing, and the single-annotator  
712 limitation that travels with these annotations. Stroke-lesion datasets do exist,  
713 ISLES (Hernandez Petzsche et al., 2022) and ATLAS (Liew et al., 2022) among  
714 them, but they are segmentation benchmarks distributed as processed images  
715 without raw  $k$ -space, so they cannot serve reconstruction evaluation as released.  
716 This matters mechanically: lesion erasure cannot be measured on cohorts that  
717 contain no acute lesions, so a model can lead every public leaderboard in the  
718 field while remaining untested against the failure mode that would matter most  
719 in an emergency department. Where a pathology-labelled dataset is used at all  
720 it is BraTS, whose tumours are large, high-contrast mass lesions, close to the  
721 easiest possible case for lesion preservation and the least informative about the  
722 lacunar and punctate findings at issue here. The benchmark infrastructure and  
723 the metric therefore fail in the same direction.

724 *Rigour is divided between two literatures.* Of the 263 studies, 84 (31.9%)  
725 release code but report no reader assessment, 44 (16.7%) report a reader assess-  
726 ment but release no code, 122 (46.4%) do neither, and only 13 (4.9%) do both.  
727 The division tracks publication venue. Applying the post-hoc classification rule  
728 stated in Section 3 to all 263 venues, clinical imaging journals ( $n = 131$ ) carry  
729 reader assessment at nearly three times the rate of engineering and methods  
730 venues ( $n = 78$ ), which in turn release code at nearly twice the clinical rate;  
731 general neuroscience and neuroimaging venues ( $n = 37$ ) sit between. Supple-  
732 mentary Table S12 gives the full cross-tabulation. The rule leaves 17 of the 263  
733 journal titles unmatched, and those are reported as a fourth *unclassified* group  
734 rather than forced into one of the three; the four groups between them account  
735 for all 57 reader studies and all 97 code releases. The tendency for the two  
736 practices to fall in different studies rather than the same ones gives a Fisher  
737 exact odds ratio of 0.43 ( $p = 0.013$ ). The estimate is insensitive to the one  
738 labelling choice that could move it: taking the reader subset as the 61 studies  
739 originally labelled at extraction rather than the 57 verified at appraisal gives an  
740 odds ratio of 0.43 and  $p = 0.010$ . We nonetheless report the association descrip-

tively rather than as a hypothesis test, because the venue grouping it rests on is a post-hoc keyword rule that was not part of the fixed protocol, and a  $p$ -value computed on a partition chosen after seeing the data does not carry its nominal meaning. The direction and magnitude are what we rely on. The magnitude is the part that does not depend on the choice: computational verifiability and diagnostic verifiability are largely held by different communities, only one study in twenty holds both, and a safety argument requires both.

*Awareness of a failure mode does not produce the evaluation that would detect it.* Generative studies record hallucination or fidelity instability in 31/80 (38.8%) of cases, the highest share among the four largest method families, against 15.1% for compressed-sensing and parallel-imaging work and 12.3% for motion correction. Yet at 9/80 (11.3%) they are also among the two least reader-evaluated of those families, while the small self-supervised group records the corpus’s highest hallucination share (8/17, 47.1%) and no reader assessment at all (0/17); per-family shares are in Table 9 and Fig. 3. The sub-fields that most clearly recognise that their models can synthesise structure are the least likely to run the test capable of detecting it.

*Evaluation is not broadening as the field grows.* Reader-assessment share fell from 32.4% (1995–2019) to 18.2% (2020–2022), recovering only partially to 20.7% (2023–2026) while the corpus grew fivefold (Table 9 and Fig. 4). What recovery there is comes entirely from the subscription-walled clinical literature: among 2023–2026 studies obtainable only through title-level library subscriptions, reader assessment runs at 52.0% (13 of 25), against 15.4% (23 of 149) in the openly accessible remainder of that era. Reader evidence, where it exists, sits disproportionately behind paywalls, and a review that cannot retrieve those texts will understate it; Section 5.2 takes the direction of that bias seriously rather than dismissing it. Among the 144 studies recording any evaluation paradigm, four in five record exactly one, and the share recording two or more fell across the same periods (Supplementary Table S12). Of the 84 studies recording a pixel-wise metric, 60 (71.4%) record no reader, uncertainty or perceptual measure beside it.

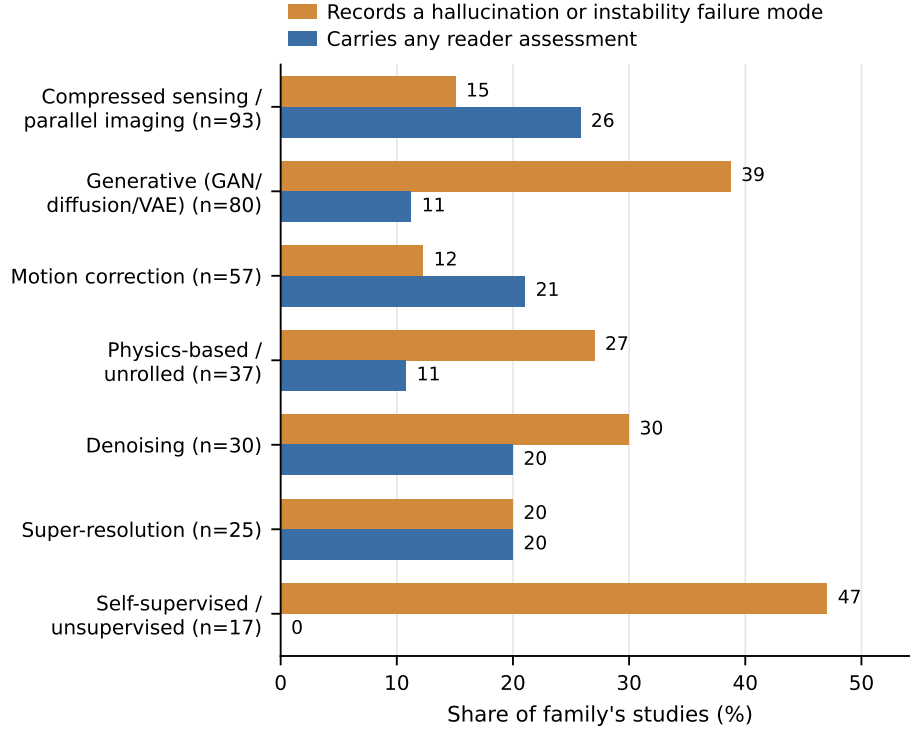


Figure 3: Reader assessment against hallucination awareness, by method family (recorded extraction labels; lower bounds). Bars are the share of each family’s studies that record a hallucination or instability failure mode (orange) and the share that carry any reader assessment (blue), using the verified reader subset ( $n = 57$ ). Bar labels are rounded to the nearest integer; the one-decimal values quoted in the text are the same quantities (for example 15 against 15.1%, 39 against 38.8%). Generated from the released per-study file (`included_characteristics.csv`) by `5_Code/make_figures.py`; every value is recomputable from the deposit.

772 *Readers are not displaced away from the sequences where the stakes are*  
773 *highest.* One expectation is not borne out. The comparison must be made  
774 on disjoint strata to mean anything, because pulse sequence is a multi-label  
775 field and the naive contrast double-counts the 107 studies that acquire both  
776 an acute-pathology and a structural sequence (Supplementary Table S11 gives  
777 the correction). That the two disjoint strata below sum to  $23 + 84 = 107$ , the  
778 same number, is a third coincidence of unrelated tallies rather than a tran-  
779 scription error, verified against the released per-study file as the two flagged in  
780 Section `efsec:requirements` are. Restricted to studies acquiring one group and

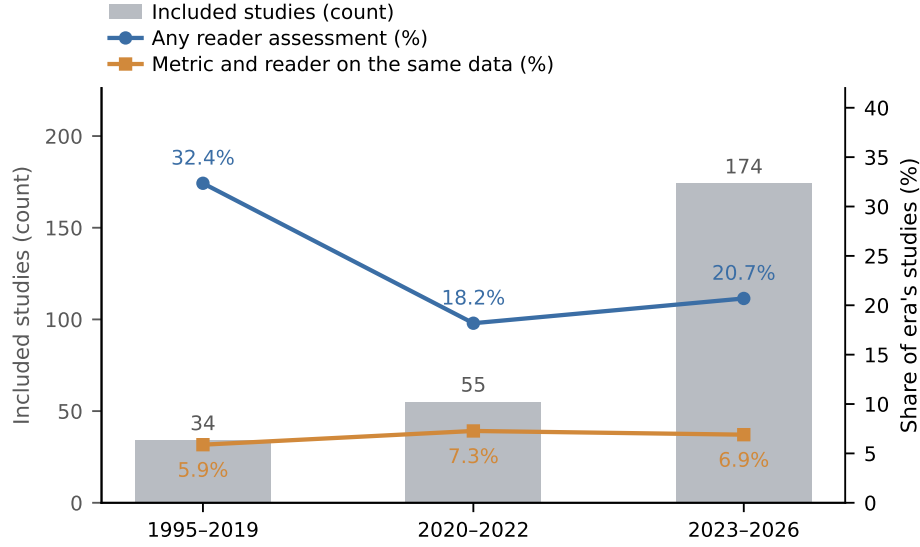


Figure 4: Evaluation practice across publication eras. Grey bars give the number of included studies per era (34, 55, 174); lines give the share with any reader assessment and the share co-reporting a fidelity metric and a reader assessment on the same data, using the verified reader subset ( $n = 57$ ). What recovery there is in 2023–2026 comes from subscription-walled clinical journals (52.0% reader assessment in that stratum against 15.4% in the open remainder; Section 5.2). Generated from the released per-study file by `5_Code/make_figures.py`; counts are lower bounds from recorded extraction labels.

not the other, reader assessment runs at 5/23 (21.7%) for DWI or FLAIR without T1w or T2w against 10/84 (11.9%) for T1w or T2w without DWI or FLAIR — a difference in the direction one would hope for, resting on 23 studies in the smaller stratum and not approaching significance (Fisher exact odds ratio 2.06,  $p = 0.31$ ). We report it as descriptive and draw only the negative conclusion it supports: there is no evidence that reader assessment is *displaced away* from the sequences on which acute pathology is read. The deficit is one of scarcity and of distribution across research communities, not of misdirection within the clinical literature.

#### 4.6. Methodological quality and risk of bias

Across all 57 QUADAS-2 appraised diagnostic and reader studies, methodological quality was highest for Flow and Timing (45/57 low risk of bias, 78.9%), reflecting the standard retrospective undersampling paradigm where reconstruc-

Table 9: Evaluation practice across publication eras and method families (recorded labels; lower bounds). Reader-assessment share fell as the corpus grew and recovers only partially in the latest era, and that recovery is carried by subscription-walled clinical journals. Among the four largest method families the generative group, the one most associated with hallucination, is one of the two least reader-evaluated and is indistinguishable from physics-based unrolled work; the small self-supervised family ( $n = 17$ ) records no reader assessment at all. Reader shares use the verified reader subset ( $n = 57$ ; Section 3), so counts sum to 57 across the era rows and are multi-label across the family rows.

| <b>Era</b>                            | <b>n</b> | <b>Reader</b>       | <b>Uncert.</b> | <b>Co-report</b> |
|---------------------------------------|----------|---------------------|----------------|------------------|
| 1995–2019                             | 34       | 32.4% (11/34)       | 5.9%           | 5.9%             |
| 2020–2022                             | 55       | 18.2% (10/55)       | 5.5%           | 7.3%             |
| 2023–2026                             | 174      | 20.7% (36/174)      | 7.5%           | 6.9%             |
| <b>Method family (multi-label)</b>    |          |                     |                |                  |
| <b>Family</b>                         | <b>n</b> | <b>Reader share</b> |                |                  |
| Compressed sensing / parallel imaging | 93       | 25.8% (24/93)       |                |                  |
| Motion correction                     | 57       | 21.1% (12/57)       |                |                  |
| Super-resolution                      | 25       | 20.0% (5/25)        |                |                  |
| Denoising                             | 30       | 20.0% (6/30)        |                |                  |
| Generative (GAN/diffusion/VAE)        | 80       | 11.3% (9/80)        |                |                  |
| Physics-based / unrolled              | 37       | 10.8% (4/37)        |                |                  |
| Self-supervised / unsupervised        | 17       | 0.0% (0/17)         |                |                  |

tion and reference share the same acquisition timepoint without temporal degradation. Reference Standard and Index Test both sit at 26/57 low risk (45.6%), each now below half, with high ratings arising where radiologists scored accelerated images with reference ground truth visible or where lesion truth was defined by the comparator images themselves (incorporation bias). Patient Selection remains the weakest domain (15/57 low risk, 26.3%; 19/57 unclear, 33.3%; 23/57 high risk, 40.4%), as cohorts were predominantly retrospective, single-centre, and pathology-enriched without explicit prospective enrolment rules. Applicability concerns follow a similar distribution, with low concerns in Patient Selection for 36/57 (63.2%), Index Test for 30/57 (52.6%), and Reference Standard for 39/57 (68.4%).

The two rounds do not place the weaknesses in quite the same order, and the pooled shares above differ from the first round’s in every one of the seven domains. Low-risk shares fell for Index Test (53.8% to 45.6%), applicability of the Index Test (59.0% to 52.6%), Flow and Timing (82.1% to 78.9%), Reference Standard (46.2% to 45.6%) and applicability of Patient Selection (64.1% to 63.2%); they rose for Patient Selection risk of bias (23.1% to 26.3%) and applica-

bility of the Reference Standard (59.0% to 68.4%). The two largest movements concern the index test and the applicability of the reference standard, and the first of these runs against the appraised studies. The second-round studies are therefore not a favourable addition to the appraisal, which is why the pooled figures are the ones reported here. The first-round shares quoted here already incorporate the five studies re-appraised on recovered full text.

Corpus-wide reporting transparency evaluated against STARD principles revealed substantial omissions across the broader 263 studies: 109/263 (41.4%) provided no identifiable reference standard (47 stating none at all, Table 3, and a further 62 naming a ground truth they left unspecified), scanner field strength was unreported in 54/263 (20.5%), and 166/263 (63.1%) did not make code openly available. Overall, 220/263 studies (83.7%) omitted at least one key methodological indicator (field strength, reference standard, or open code). Two directions must be kept apart here, and conflating them has caused confusion elsewhere in this literature. With respect to the *reconstruction methods*, this reporting and selection pattern is anti-conservative: retrospective single-centre evaluation on curated data flatters reported performance relative to what prospective clinical deployment would show. With respect to *this review’s own conclusion*, the same pattern is conservative: if published performance is flattered and failure is under-reported, then the scarcity of safety-relevant evaluation we document is an understatement, and the true gap is at least as large as the one measured. The safety argument turns on the second direction and is not weakened by the first. Detailed domain-level judgments and supporting quotations are available in the open project repository.

#### 4.7. Requirements for safety-oriented evaluation

Read together, the synthesis implies five requirements that any evaluation intended to establish reconstruction safety must meet, each stated with the corpus evidence that motivates it and none met by more than a small minority of included studies. **R1, focal task-based measurement:** every global fidelity score must be accompanied by a region-of-interest or task-based observer

measure, because spatial averaging is the first mechanism by which lesion-scale error disappears (60/84, or 71.4%, of pixel-metric studies record nothing beside the metric; this equals the artifact share of the *R*-reporting subset in Table 6 (45/63) to the decimal, which is a coincidence of two unrelated tallies rather than a transcription error, verified against the released per-study file). **R2, paired metric–reader reporting:** metric validity claims require a reader comparison on the same data (6.8% of studies provide one). **R3, reference-independent validity:** evaluation must remain interpretable when no registered artifact-free reference exists (26.2% rest solely on a fully sampled acquisition). **R4, robustness under acquisition shift as a primary outcome,** reported as a result rather than a caveat; robustness or distribution shift is recorded as a failure-mode concern in 76/263 studies (28.9%), almost always in the discussion rather than as a pre-specified endpoint. **R5, hallucination-specific detection,** operating on the image side, since measurement-space residual checks are provably blind to null-space edits. To these the analysis in Section 4.5 adds a precondition: benchmarks containing the pathology whose erasure is at issue, without which R1, R2 and R5 cannot be exercised at all.

## 5. Discussion

Three conclusions follow. First, the comparison on which the field’s central proxy rests is almost never made: no included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment, and only 18 of 263 record the two on the same data at all. Where readers do appear, what is measured is mostly agreement between readers, and it is weak and condition-dependent. The finding therefore rests more on the absence of measurement than on the measurements that exist. Second, the reason is structural, through the four mechanisms of Section 4: background voxels average focal error away, the loss functions are sign-symmetric where diagnosis is not, adversarial terms pay for invented texture, and every one of these metrics assumes a registered reference that motion destroys. The mechanisms are math-



870 ematical properties of the metrics themselves; how often each matters clinically  
871 is the part the thin reader evidence leaves less certain. Third, no evaluation  
872 paradigm now in use settles the safety question by itself, a conclusion resting  
873 on the paradigm-by-paradigm synthesis of Section 4.

874 The analysis of where the gap comes from, which was not part of the fixed  
875 protocol and is reported as exploratory throughout, changes what would follow  
876 from these findings if it holds. The shortfall does not look like a uniform over-  
877 sight; it co-occurs with two structural features. Evaluation practice is divided  
878 between a community that releases code and a community that runs readers,  
879 with only 4.9% of studies doing both (an association we interpret descriptively  
880 in Section 4.5, its venue partition being post-hoc), and the benchmark datasets  
881 named in this corpus contain no acute pathology, so the failure mode of great-  
882 est clinical consequence is not merely under-measured but unmeasurable on the  
883 datasets in common use. If that association is causal, which this design cannot  
884 establish, recommendations addressed to individual authors will not close the  
885 gap; what would be required is benchmark construction and reporting conven-  
886 tions that force the two halves of the evidence together.

887 This matters because accelerated reconstruction has left the research setting.  
888 Deep-learning acceleration, denoising and image enhancement ship inside com-  
889 mercial MRI products, several of them cleared as software as a medical device:  
890 one such tool, an FDA-cleared vendor-agnostic DICOM-domain enhancement  
891 product, is evaluated in this corpus by a prospective multicentre multireader  
892 trial (Bash et al., 2021), which is also a reminder that a cleared product can  
893 be validated on one sequence family and deployed across others. Published  
894 guidance in both jurisdictions increasingly asks manufacturers to demonstrate  
895 performance for the device’s stated intended use and under conditions repre-  
896 sentative of deployment rather than on curated development data: the FDA,  
897 Health Canada and MHRA guiding principles on good machine learning practice  
898 ask that datasets be representative of the intended patient population and that  
899 testing be independent of training (U.S. Food and Drug Administration et al.,  
900 2021), the EU AI Act imposes accuracy and robustness obligations on high-risk

901 AI systems (European Parliament and Council of the European Union, 2024),  
 902 and the Medical Device Regulation requires clinical evidence appropriate to the  
 903 device’s intended purpose (European Parliament and Council of the European  
 904 Union, 2017). Our reading of that guidance is that aggregate pixel-wise fidelity  
 905 cannot support a claim about small-lesion detection, because (2) shows the ag-  
 906 gregate does not move when the small lesion is removed. That reading is ours:  
 907 no regulator has reviewed or endorsed it, and none of the three instruments  
 908 names an image-quality metric.

909 A practical obstacle compounds it, and it is the reason the requirements in  
 910 Section 4.7 are addressed to infrastructure rather than to authors. Commer-  
 911 cial inline pipelines export DICOM images after undisclosed normalisation, coil  
 912 combination and vendor post-processing, and the raw complex  $k$ -space typi-  
 913 cally never leaves the scanner. Every method in Table 8 that could detect a  
 914 prior-induced edit needs something the exported DICOM does not carry: the  
 915 null-space projection needs the forward operator and the coil sensitivities, a  
 916 counterfactual audit needs the ability to re-run the acquisition under a con-  
 917 trolled change, and even a measurement-space residual needs  $Y$ . Independent  
 918 third-party auditing of a deployed reconstruction is therefore not merely difficult  
 919 but structurally impossible without an open raw-data interface, and no amount  
 920 of reporting discipline by authors of research papers can supply one.

921 Three things follow that are worth stating as claims rather than as recom-  
 922 mendations.

923 *The cheapest useful change costs no new data.* A reconstruction paper that  
 924 claims its method preserves small lesions can state, from (2) and its own brain  
 925 volume and assumed local error ratio, the change in global PSNR that erasing  
 926 the smallest lesion it claims to preserve would produce. For a lacunar infarct  
 927 that is about 0.03 dB, which is smaller than the difference the paper is likely to  
 928 be reporting between its method and its baseline. This is arithmetic, not a new  
 929 experiment, and it converts an implicit claim into an explicit one at the point  
 930 where the claim is made. Nothing in this review suggests such a line would be  
 931 difficult; the finding is that nobody currently reports it.

932       *The two-community split is self-reinforcing, which is why exhortation has*  
933 *not worked.* Each community’s evaluation is close to the best it can do with  
934 the data it can obtain. Engineering venues evaluate on the datasets that carry  
935 raw  $k$ -space, and those contain healthy volunteers, chronic cohorts and tumours  
936 rather than acute pathology, so a reader study run on them would not test lesion  
937 preservation even if it were run. Clinical venues have the pathology, the readers  
938 and the acute presentations, and no access to raw  $k$ -space or to the vendor’s  
939 reconstruction, so a reproducibility claim is not available to them. The 4.9% of  
940 studies doing both are not more diligent than the rest; they are the studies that  
941 happened to sit where both kinds of access coincide. An instruction to individual  
942 authors to run more reader studies asks most of them to do something the data  
943 available to them makes uninformative.

944       *The missing object is a dataset, and its requirements are now specifiable.*  
945 The public resources discussed in this review supply each necessary compo-  
946 nent separately, and the distinction between what the corpus uses and what  
947 merely exists is part of the point: raw multi-coil  $k$ -space is available in three  
948 of the eleven datasets the included studies actually name (fastMRI, Calgary–  
949 Campinas, M4Raw); acute cerebrovascular pathology with expert segmentation  
950 exists in ISLES and ATLAS, which no included study uses because they are dis-  
951 tributed as processed images; and reader-level pathology annotation on brain  
952 MRI exists in fastMRI+, which no included study uses either. None combines  
953 them, and the combination is what every requirement in Section 4.7 presup-  
954 poses: raw  $k$ -space so the forward operator is known, acute lesions confirmed  
955 against a standard independent of the images being evaluated, lesion-level an-  
956 notation so a focal figure of merit can be computed, and a licence permitting  
957 redistribution so the benchmark can be used by people who did not collect it.  
958 That object does not exist, and until it does the safety question is not merely  
959 unanswered but unaskable on public data.

### 960 5.1. *Priorities for future research*

961 Three of the five requirements in Section 4.7 translate directly into research  
962 priorities, and we do not restate them here: paired metric–reader reporting  
963 (R2), robustness under acquisition shift reported as a primary endpoint rather  
964 than a caveat (R4), and benchmarks carrying the pathology whose erasure is at  
965 issue, which is the precondition on which R1, R2 and R5 all depend. Calibrated  
966 voxel-level uncertainty that tracks local reconstruction error (Khawaled and  
967 Freiman, 2024) belongs with R4 and is the form in which R4 is most likely to  
968 be actionable for generative and self-supervised architectures.

969 A fourth priority does not follow from anything this review extracted, and we  
970 separate it for that reason. Many current pipelines operate slice by slice and in-  
971 troduce inter-slice discontinuities on isotropic 3D acquisitions, so reconstruction  
972 and auditing frameworks should model 3D anatomical coherence directly (Peng  
973 et al., 2026) while meeting scanner-side latency constraints. Slice-wise versus  
974 volumetric processing was not an extraction field in this review. We state the  
975 priority on mechanistic grounds and on the cited work, not on a corpus count.

### 976 5.2. *Threats to validity*

977 The strongest conclusions rest on a small subset of reader-based studies, and  
978 we state that prominently rather than in passing: 57 of 263 studies carry any  
979 reader assessment and 18 pair it with a metric. Four threat classes bound what  
980 follows from that base.

981 *How to read the limitations below.* They are not all bounded in the same  
982 way, and treating them as one undifferentiated list of caveats would misrepresent  
983 what this review knows about itself. Three kinds appear. Some are *bounded*  
984 *by released evidence*: the non-retrieval analysis, the paywall stratification and  
985 the appraisal-agreement decomposition are each quantified from files a reader  
986 can recompute. Some are *bounded by argument*: the direction of publication  
987 bias and of the reporting asymmetry is reasoned rather than measured, and we  
988 say which way each points. One is simply *unbounded*, and it is the sharpest:  
989 extraction has no reliability estimate.

990 Three properties of the extraction layer follow from that and should be  
 991 weighed together. First, its depth is deliberately shallow. Method, failure-  
 992 mode and evaluation categories were assigned from title, abstract and controlled  
 993 vocabulary, so the prevalence counts are floors on what studies *report at that*  
 994 *level* rather than measurements of what they did. Second, the shallowness is  
 995 visible in the counts themselves: only 84 of 263 studies are recorded as reporting  
 996 a pixel-wise fidelity metric, and 39 of the 57 verified reader studies are recorded  
 997 as reporting none, which for this literature is implausible as a description of  
 998 practice and entirely plausible as a description of abstracts. A further 119 of  
 999 263 studies (45.2%) carry no evaluation-paradigm label at all, which is why  
 1000 we report both the /263 and the /144 denominators wherever the distinction  
 1001 matters. Third, the headline 18/263 mixes two depths: the reader label was  
 1002 verified against full text and corrected at appraisal, while the metric label was  
 1003 not. That asymmetry biases the intersection downward by an unknown factor,  
 1004 and it is the single respect in which our headline figure is least defensible as a  
 1005 point estimate — though the direction is conservative, since correcting it could  
 1006 only widen the measured gap.

1007 What we offer in place of a reliability coefficient is auditability, and the  
 1008 substitution is deliberate rather than a shortfall we failed to notice. Every  
 1009 extracted value is released with the verbatim quotation that supports it, so a  
 1010 reader can check whether an assignment is *correct*, which is a stronger and more  
 1011 useful guarantee than a coefficient showing that two people were *consistent*. It  
 1012 is not a substitute for reliability in the statistical sense, and we do not claim  
 1013 it is: consistency and correctness bound different failure modes, and only the  
 1014 first is estimable from our records. A duplicate-extraction exercise on a random  
 1015 subsample, reported with a  $\kappa$ , is the obvious next step and we name it as an  
 1016 outstanding task rather than a completed one.

1017 *Construct validity.* We searched with broad, high-sensitivity terms for  
 1018 architectures, artifacts and evaluation paradigms, and the corpus spans the  
 1019 reconstruction-safety construct, from pixel-wise metrics through perceptual  
 1020 losses to reader studies. What the corpus cannot guarantee is that a metric

1021 means the same thing wherever it appears: studies differ in what they treat as  
1022 the quantity of interest, so a rank correlation with reader scores in one paper  
1023 is not straightforwardly the same measurement as in another, and grouping  
1024 such values under one heading is an interpretation we make, not a property of  
1025 the data. The venue classification underlying Section 4.5 is likewise a post-hoc  
1026 keyword assignment, released with the data rather than extracted from the  
1027 studies.

1028     *Internal validity.* A protocol fixed before screening began, a documented  
1029 search across seven databases, and independent screening and appraisal by three  
1030 reviewers provide robust protection, and every extracted value is released with  
1031 the verbatim snippet supporting it. Five specific threats bound these findings.  
1032 First, 81 of the 1,149 candidate reports remain without obtainable text through  
1033 structural licensing barriers. Screening the nearest obtainable stratum — the  
1034 341 reports recovered only through title-level subscriptions — found 86.2% ineli-  
1035 gible on the same criteria, so at that rate the 81 would contribute on the order of  
1036 11 studies rather than 81. That bounds the missing evidence without establish-  
1037 ing that the bias is nil. Non-retrieval is demonstrably *not* random with respect  
1038 to the variable this review measures: within the same stratum, 2023–2026 stud-  
1039 ies carry reader assessment at 52.0% against 15.4% in the openly accessible  
1040 literature of that era. Both findings point the same way. Reader evidence is  
1041 over-represented behind paywalls, so a review that cannot retrieve those texts  
1042 understates how much reader assessment the field performs, and our headline  
1043 scarcity figures are conservative bounds rather than point estimates. Second,  
1044 four included studies were assessed on abstract and controlled vocabulary alone  
1045 and one on partial text; a sensitivity analysis recomputing all headline metrics  
1046 on the 258 studies verified against complete full text leaves every finding direc-  
1047 tionally stable (e.g., metric–reader co-reporting remains 7.0% vs 6.8%). Third,  
1048 every QUADAS-2 tally in this article rests on the whole verified reader subset  
1049 ( $n = 57, 399$  domain judgements), so none is computed on a partial base. The  
1050 second round moved all seven domain distributions, and the movements that  
1051 most concern the index test run against the appraised studies: low risk of bias

1052 for the Index Test falls from 53.8% to 45.6%, and high applicability concern  
 1053 for the Index Test rises from 33.3% to 40.4% (Section 4). Four studies remain  
 1054 abstract-only and are appraised on that basis. Fourth, reliability is measured  
 1055 for screening and for appraisal but not for extraction, and extraction is the  
 1056 sharpest methodological limitation of the review. The screening and appraisal  
 1057 coefficients, and the decomposition that must be read with the appraisal figure,  
 1058 are given in Section 3 and not restated here; the consequence for this section  
 1059 is that 126 of the 399 QUADAS-2 judgements were reached at a level of agree-  
 1060 ment a single rater could not be relied on to reproduce, so the domain shares  
 1061 in Table 3 carry that uncertainty. It does not propagate to the prevalence find-  
 1062 ings, which rest on extraction labels rather than QUADAS-2 ratings, but it  
 1063 bounds how firmly the risk-of-bias picture can be stated. Neither screening nor  
 1064 appraisal agreement transfers to extraction. The duplicate-screening sample is  
 1065 also smaller than the 300–400 records a referee might reasonably ask for, and  
 1066 the interval reflects that. Fifth, four studies recorded at extraction as observer  
 1067 or reader designs were found at appraisal to contain no human reader and were  
 1068 reclassified as algorithmic, moving the reader subset from 61 to 57 and the al-  
 1069 gorithmic set from 202 to 206. The effect on the headline numbers is small and  
 1070 in the direction of a larger gap: metric–reader co-reporting moves from 7.6%  
 1071 to 6.8%, the 2023–2026 reader share from 23.0% to 20.7%, the code-and-reader  
 1072 intersection from 5.3% to 4.9%, and the Fisher exact  $p$  from 0.010 to 0.013 at  
 1073 an odds ratio that moves only from 0.43 to 0.43. Those four corrections are  
 1074 also the source of the 4/61 (6.6%) error rate quoted above, and the design field  
 1075 is the only one that received three-reviewer full-text verification: a comparable  
 1076 rate elsewhere would not overturn any finding here, since the headline shares are  
 1077 separated from their comparators by more than that margin, but it is a source  
 1078 of variability measured in one place and assumed in the rest.

1079 *External validity.* Among the 209 studies stating a field strength the corpus  
 1080 covers 1.5, 3 and 7 Tesla (21, 153 and 35 respectively), together with several  
 1081 vendors’ coil configurations and the major pulse sequences; the remaining 54  
 1082 (20.5%) state none, and nothing about field-strength coverage should be read

1083 into them. Tasks, protocols, architectures, datasets and metrics otherwise vary  
 1084 substantially, which is why we report the direction and consistency of find-  
 1085 ings rather than their magnitudes. Many algorithmic studies evaluate on retro-  
 1086 spectively undersampled single-centre data, which flatters performance relative  
 1087 to prospective multi-centre use; where the public data were themselves pre-  
 1088 processed before undersampling, that inflation is larger again (Shimron et al.,  
 1089 2022). Prospective multicentre reader evidence is thin, and the examples this re-  
 1090 view leans on each carry a boundary that must travel with them: the emergency-  
 1091 department comparison is single-centre and its one discrepant finding is a single  
 1092 case (Lang et al., 2024); the widest-ranging acceleration reader study is retro-  
 1093 spective and single-centre (Radmanesh et al., 2022); and the largest prospective  
 1094 multicentre multireader trial evaluated 3D T1-weighted acquisitions only, with  
 1095 an image-domain DICOM enhancement tool rather than reconstruction from un-  
 1096 dersampled  $k$ -space, reported improved rather than degraded lesion conspicuity,  
 1097 and has several authors affiliated with the vendor of the evaluated tool (Bash  
 1098 et al., 2021) — we cite it for what it does establish, that such tools are cleared  
 1099 and deployed. Finally, the benchmark concentration of Section 4.5 — eleven  
 1100 public datasets named across the corpus, not one curated for acute ischaemic  
 1101 stroke or intracranial haemorrhage — caps how far any leaderboard result can  
 1102 be carried into emergency imaging.

1103 *Conclusion validity.* Because pixel-wise metrics agree only loosely with ra-  
 1104 diologist judgement (RQ1), a study resting on PSNR and SSIM alone supports  
 1105 a safety claim less well than one with readers or uncertainty-aware observers,  
 1106 and we weight the evidence accordingly. Publication bias favouring successful  
 1107 reconstructions is a material limitation and works in the same direction, mak-  
 1108 ing the reported prevalence of failure evidence an understatement: a literature  
 1109 with little incentive to report failure is exactly the corpus a review about failure  
 1110 must draw on, and the absence of a common effect measure rules out funnel-plot  
 1111 and small-study asymmetry tests. We applied no certainty framework such as  
 1112 GRADE, since the review produces no pooled effect estimates for such a frame-  
 1113 work to rate. Prevalence counts derive from recorded extraction labels and



understate true practice, and no range we quote should be read as an interval estimate rather than as values from named studies.

## 6. Conclusion

We reviewed 263 primary studies on the safety, artifacts and fidelity of deep learning-based brain MRI reconstruction. Wherever metrics and readers have been compared, agreement between conventional fidelity scores and radiologist judgement is weak, and at most 18 of the 263 studies record the paired metric–reader evaluation capable of exposing that divergence. Every prevalence figure here is a floor derived from screening-level labels rather than a measurement of practice, and although screening agreement was high, the extraction labels those floors rest on carry no reliability estimate of their own. The weakness is structural rather than incidental: the four mechanisms identified under RQ2 explain why, and the safety question is not one that any current evaluation paradigm answers alone.

The contribution we would emphasise is the account of why the gap persists, and we offer it as a hypothesis these data are consistent with rather than as an established structural fact: the venue grouping it rests on is post-hoc, so although the association between code release and reader assessment is strong and would be conventionally significant on a pre-specified partition, we do not read it as a test (Section 4.5). What does not depend on that classification is the magnitude. Only 4.9% of studies both release code and run readers, so computational verifiability and diagnostic verifiability are, as recorded, held by different studies roughly nineteen times in twenty; and no dataset named in this corpus is curated for acute stroke or intracranial haemorrhage. A reconstruction model can therefore satisfy every metric and every public benchmark in current use while the question of whether it preserves a small acute lesion has never been asked. Closing that gap requires benchmarks that contain the pathology at issue and reporting conventions under which no pixel-wise metric appears alone, and until both exist, claims that accelerated reconstruction is diagnostically safe will

1143 continue to rest on measurements that cannot see the failures that matter.  
1144       One change is available immediately and needs no new data. Equation (2)  
1145 converts any lesion volume into the PSNR movement its erasure would cause, so  
1146 an author reporting a reconstruction at a stated acceleration can state, alongside  
1147 the global score, the smallest focal change that score is capable of registering  
1148 at their image size. For a 100 mm<sup>3</sup> infarct that figure is around 0.03 dB, which  
1149 is below the precision at which PSNR is conventionally reported. Publishing  
1150 that bound next to the metric costs one line, requires no reader and no new  
1151 acquisition, and makes explicit what the metric is silent about. It is the cheap-  
1152 est useful step towards evaluation that could detect the failure this review is  
1153 concerned with.

## 1154 **Funding Statement**

1155 This research was supported by institutional research computing infrastruc-  
1156 ture and scholarship funding from RMIT University. The funder had no role in  
1157 the design of the review, the search strategy, the eligibility criteria, the screening  
1158 or appraisal of studies, the synthesis or interpretation of findings, the writing  
1159 of this report, or the decision to submit it for publication. No external, com-  
1160 mercial, or industry funding was received, and no vendor of MRI hardware or  
1161 reconstruction software contributed financial or non-financial support to this  
1162 work.

## 1163 **Ethics statement**

1164 This review synthesises secondary data from published, de-identified study  
1165 reports. It involved no human participants, no animal experimentation and  
1166 no collection of private health data, so institutional review board approval and  
1167 patient informed consent were not required.

## 1168 **Data and code availability**

1169 The review protocol, executed search strategy, screening criteria, every  
1170 title/abstract and full-text decision with its recorded reason, the com-  
1171 pleted PRISMA 2020 checklist, the per-study extraction and QUADAS-  
1172 2 worksheets for all 263 included studies, the analysis code, and the  
1173 duplicate-screening sample and per-reviewer appraisal worksheets with their  
1174 agreement computations (`kappa_worksheet_214_HUMAN_*.csv`, `kappa_hu-`  
1175 `man_results.csv`, `reviewer_worksheet_R*.csv` and `quadas2_three_re-`  
1176 `viewer_worksheet.csv`) are openly available on the Open Science Frame-  
1177 work at <https://doi.org/10.17605/OSF.IO/832Z6> and mirrored at <https://github.com/dmai287/brain-mri-recon-safety-slr>, archived at Zenodo  
1178 under DOI 10.5281/zenodo.22100024 (release v1.0.3, commit 8c856ba;  
1179 concept DOI 10.5281/zenodo.21836966). Every count reported here is regen-  
1180 erable from `included_characteristics.csv` in that release, and the deposit  
1181

1182 encodes the same reader/algorithmic partition (57/206) that this article re-  
1183 ports, with the four reclassified studies flagged individually. The two figures  
1184 generated from that file and the  $\Delta$ PSNR identity figure are produced by  
1185 `5_Code/make_figures.py` and `5_Code/make_dpsnr_figure.py`. The deposit  
1186 is retrospective and is documented as such. Full-text PDFs of the included  
1187 studies are not redistributed, as publisher licence terms do not permit it; the  
1188 DOI list in the released bibliography (`included_studies_263.bib`) identifies  
1189 the corpus in full.

#### 1190 **CRediT authorship contribution statement**

1191 **Dat Tat Mai:** Conceptualization, Methodology, Software, Investigation,  
1192 Data curation, Formal analysis, Visualization, Project administration, Writing  
1193 – original draft, Writing – review & editing. **Thai Viet Pham:** Investiga-  
1194 tion, Data curation, Validation, Resources, Writing – review & editing. **Thu**  
1195 **Nguyen Thi Dang:** Investigation, Data curation, Validation, Writing – re-  
1196 view & editing. **James Jin Kang:** Supervision, Validation, Writing – review  
1197 & editing.

#### 1198 **Declaration of competing interest**

1199 The authors declare that they have no known competing financial interests or  
1200 personal relationships that could have appeared to influence the work reported  
1201 in this paper.

#### 1202 **Declaration of generative AI and AI-assisted technologies in the** 1203 **manuscript preparation process**

1204 During the preparation of this work the authors used generative AI technolo-  
1205 gies solely to assist with language editing and proofreading of the manuscript  
1206 text. After using these tools, the authors reviewed and edited the content as  
1207 needed and take full responsibility for the content of the published article. All

study screening, data extraction, and quality appraisals reported in this review were conducted manually by the authors.

## Appendix A. The included studies and how to trace a claim to them

All 263 included primary studies appear in the reference list, so the evidence base is part of the article of record rather than of the supplement alone. Because the synthesis is narrative and most claims are grouped rather than attached to a single study, the reference list alone does not show which studies produced a given number. That mapping is carried by `included_characteristics.csv`, which holds one row per included study with its extracted method family, evaluation paradigm, failure modes, sequences, field strength, reference standard and code availability, each with the verbatim quotation supporting it, so any count, share or crosstab in Section 4 can be reproduced and the contributing studies listed by filtering that file. Supplementary Tables S1–S3 give the same 263 studies as a numbered index by publication year, and `included_characteristics_supplement.csv` adds the per-study QUADAS-2 and reproducibility ratings.

The reference list therefore contains two kinds of entry, and a reader needs a rule to tell them apart. It prints 299: the 263 included studies, carried into the list under PRISMA 2020 item 17, and 36 narrative-only works cited in argument. **An entry appearing in the numbered index of Supplementary Tables S1–S3 is a member of the reviewed corpus; an entry that does not is a narrative citation.** Equivalently, and checkable without the supplement, every narrative citation is cited at a specific point in the running text of Sections 1–6, whereas corpus entries not discussed by name appear without an in-text citation. The two lists were generated by independent mechanisms and agree exactly as sets. We would be glad to move the corpus to a separately labelled supplementary reference list if the editor prefers that production route.

## 1235 **Appendix B. Supplementary material**

1236       Supplementary material associated with this article (Supplementary Ta-  
1237 bles S1–S12 and Supplementary Figure S1) is provided as a separate document.  
1238 Tables S1–S3 list all 263 included primary studies with full bibliographic details;  
1239 Tables S4–S9 give the comparison against related secondary studies, the search  
1240 vocabulary and its derivation, the extraction form, the thematic evidence ar-  
1241 chitecture, the four metric-failure mechanisms with their evidence sources, and  
1242 the provenance of every released data file. Tables S10–S12 carry the support-  
1243 ing detail for Section 4.5: the annotation-layer derivation for fastMRI+ (S10),  
1244 the disjoint-strata correction for the sequence contrast (S11), and the venue  
1245 cross-tabulation with evaluation-paradigm multiplicity by era (S12).

## 1246 **References**

- 1247 Aali, A., Arvinte, M., Kumar, S., Arefeen, Y.I., Tamir, J.I., 2025. Robust multi-coil  
1248 MRI reconstruction via self-supervised denoising. *Magnetic resonance in medicine*  
1249 doi:10.1002/mrm.30591. PMID: 40457510.
- 1250 Abed, M.A., Demir, S., Groteklaes, A., Germani, E., Faghihroohi, S., Sabir, H., Albar-  
1251 qouni, S., 2026. Mriqt: Physics-aware diffusion model for image quality transfer in  
1252 neonatal ultra-low-field MRI. 2026 IEEE 23rd International Symposium on Biomed-  
1253 ical Imaging (ISBI) doi:10.1109/isbi61048.2026.11515815.
- 1254 Aggarwal, H.K., Mani, M.P., Jacob, M., 2018. MoDL: Model-based deep learning  
1255 architecture for inverse problems. *IEEE Transactions on Medical Imaging* 38, 394–  
1256 405.
- 1257 Al-Masni, M.A., Lee, S., Yi, J., Kim, S., Gho, S.M., Choi, Y.H., Kim, D.H., 2022.  
1258 Stacked u-nets with self-assisted priors towards robust correction of rigid motion  
1259 artifact in brain MRI. *NeuroImage* doi:10.1016/j.neuroimage.2022.119411.
- 1260 Alçalar, Y.U., Akçakaya, M., 2025. Sparsity-driven parallel imaging consistency for  
1261 improved self-supervised MRI reconstruction. 2025 IEEE International Conference  
1262 on Image Processing (ICIP) doi:10.1109/icip55913.2025.11084355.
- 1263 Altmann, S., Grauhan, N.F., Brockstedt, L., Kondova, M., Schmidtman, I., Paul,  
1264 R., Clifford, B., Feiweier, T., Hosseini, Z., Uphaus, T., Groppa, S., Brockmann,  
1265 M.A., Othman, A.E., 2024. Ultrafast brain MRI with deep learning reconstruction  
1266 for suspected acute ischemic stroke. *Radiology* doi:10.1148/radiol.231938. PMID:  
1267 38376403.
- 1268 Altmann, S., Grauhan, N.F., Mercado, M.A.A., Almansour, H., Paul, R.H., Schöffing,  
1269 V.I., Ottenhausen, M., Brockmann, M.A., Othman, A.E., 2026. Ultra-fast sub-  
1270 minute brain MRI with deep-learning-reconstruction for anesthesia-free emergency  
1271 imaging in children. *Clinical Neuroradiology* doi:10.1007/s00062-026-01627-7.

1272 Angella, P., Balbi, L., Ferrando, F., Traverso, P., Varriale, R., Pastore, V.P., Santace-  
1273 saria, M., 2025a. Dima: Diffusing motion artifacts for unsupervised correction in  
1274 brain MRI images. *IEEE Access* doi:10.1109/access.2025.3634749.

1275 Angella, P., Pastore, V.P., Santacesaria, M., 2025b. Assessing the use of diffusion  
1276 models for motion artifact correction in brain MRI. 2025 IEEE 22nd International  
1277 Symposium on Biomedical Imaging (ISBI) doi:10.1109/isbi60581.2025.10981000.

1278 Antun, V., Renna, F., Poon, C., Adcock, B., Hansen, A.C., 2020. On instabilities  
1279 of deep learning in image reconstruction and the potential costs of AI. *Proceed-*  
1280 *ings of the National Academy of Sciences* 117, 30088–30095. doi:10.1073/pnas  
1281 .1907377117.

1282 Barrett, H.H., Myers, K.J., 2004. *Foundations of Image Science*. Wiley, Hoboken, NJ.

1283 Barrett, H.H., Yao, J., Rolland, J.P., Myers, K.J., 1993. Model observers for as-  
1284 sessment of image quality. *Proceedings of the National Academy of Sciences* 90,  
1285 9758–9765.

1286 Bash, S., Wang, L., Airriess, C., Zaharchuk, G., Gong, E., Shankaranarayanan, A.,  
1287 Tanenbaum, L.N., 2021. Deep learning enables 60% accelerated volumetric brain  
1288 MRI while preserving quantitative performance: A prospective, multicenter, multi-  
1289 reader trial. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7358.  
1290 pMID: 34824098.

1291 Bathla, G., Rai, P., Chan, V.E.Y., Nickel, M.D., Kollasch, P., Dueker, K., Fagan, A.J.,  
1292 Benson, J.C., Huston, J., Messina, S.A., 2025. Deep-learning accelerated vessel wall  
1293 imaging using T1-SPACE at ultra-high-field strength MRI. *American Journal of*  
1294 *Neuroradiology* 47, 1424–1431. doi:10.3174/ajnr.a9097.

1295 Bauman, G., Afshari, R., Bieri, O., 2025. Ultra-high-resolution brain MRI at 0.55t:  
1296 bstar and its application to magnetization transfer ratio imaging. *Zeitschrift fur*  
1297 *medizinische Physik* doi:10.1016/j.zemedi.2024.12.001. pMID: 39824686.

1298 Baz, A.M., Bendella, Z., Katemann, C., Sprinkart, A.M., Weiss, K., Weber, O.M.,  
1299 Peeters, J.M., Lehn, N.C., Clauberg, R., Luetkens, J.A., Radbruch, A., Wicht-  
1300 mann, B.D., 2026. Deep learning improves image quality in motion-robust and  
1301 sedation-free pediatric brain MRI. *European radiology* doi:10.1007/s00330-026-  
1302 12482-y. pMID: 41925834.

1303 Begnoche, J.P., Schilling, K.G., Boyd, B.D., Cai, L.Y., Taylor, W.D., Landman, B.A.,  
1304 2022. Epi susceptibility correction introduces significant differences far from local  
1305 areas of high distortion. *Magnetic resonance imaging* doi:10.1016/j.mri.2022.05.  
1306 016. pMID: 35644448.

1307 Behrendt, F., Bhattacharya, D., Mielsing, R., Maack, L., Krüger, J., Opfer, R., Schlae-  
1308 fer, A., 2025. Guided reconstruction with conditioned diffusion models for un-  
1309 supervised anomaly detection in brain mris. *Computers in biology and medicine*  
1310 doi:10.1016/j.combiomed.2025.109660.

1311 Beljaards, L., Nagtegaal, M., Rao, C., Dong, Y., Osch, M.J.P.v., Pezzotti, N., Doneva,  
1312 M., Staring, M., 2026. Deep-disorder: Motion correction in 3D MRI via segment re-  
1313 construction and registration. *NMR in biomedicine* doi:10.1002/nbm.70286. pMID:  
1314 41958180.

1315 Beljaards, L., Pezzotti, N., Rao, C., Doneva, M., Osch, M.J.P.v., Staring, M., 2024.  
1316 AI-based motion artifact severity estimation in undersampled MRI allowing for  
1317 selection of appropriate reconstruction models. *Medical physics* doi:10.1002/mp  
1318 .16918. PMID: 38167996.

1319 Benjamin, A.J.V., Bano, W., Mair, G., Thompson, G., Casado, A., Perri, C.D., Davies,  
1320 M., Marshall, I., 2020. Diagnostic quality assessment of ir-prepared 3D magnetic  
1321 resonance neuroimaging accelerated using compressed sensing and k-space sampling  
1322 order optimization. *Magnetic resonance imaging* doi:10.1016/j.mri.2020.08.025.  
1323 PMID: 32890675.

1324 Bhadra, S., Kelkar, V.A., Brooks, F.J., Anastasio, M.A., 2021. On hallucinations  
1325 in tomographic image reconstruction. *IEEE Transactions on Medical Imaging* 40,  
1326 3249–3260. doi:10.1109/TMI.2021.3077857.

1327 Bian, W., Jang, A., Liu, F., 2024. Improving quantitative MRI using self-supervised  
1328 deep learning with model reinforcement: Demonstration for rapid t1 mapping. *Mag-*  
1329 *netic resonance in medicine* doi:10.1002/mrm.30045. PMID: 38342980.

1330 Bjørkeli, E.B., Geitung, J.T., Esmaeili, M., 2025. Artificial intelligence-powered four-  
1331 fold upscaling of human brain synthetic metabolite maps. *The Journal of interna-*  
1332 *tional medical research* doi:10.1177/03000605251330578. PMID: 40257058.

1333 Bollmann, S., Kristensen, M.H., Larsen, M.S., Olsen, M.V., Pedersen, M.J., Øster-  
1334 gaard, L.R., O'Brien, K., Langkammer, C., Fazlollahi, A., Barth, M., 2019. Shar-  
1335 qnet - sophisticated harmonic artifact reduction in quantitative susceptibility map-  
1336 ping using a deep convolutional neural network. *Zeitschrift fur medizinische Physik*  
1337 doi:10.1016/j.zemedi.2019.01.001. PMID: 30773331.

1338 Bossuyt, P.M., Reitsma, J.B., Bruns, D.E., Gatsonis, C.A., Glasziou, P.P., Irwig, L.,  
1339 Lijmer, J.G., Moher, D., Rennie, D., de Vet, H.C.W., 2015. STARD 2015: An  
1340 updated list of essential items for reporting diagnostic accuracy studies. *BMJ* 351,  
1341 h5527.

1342 Brunátová, J., Löcke, M., Uribe, S., Bertoglio, C., 2025. Denoising of dual-venc pc-  
1343 MRI with large high/low venc ratios. *Magnetic resonance in medicine* doi:10.1002/  
1344 mrm.30278.

1345 Byanju, R., Klein, S., Cristobal-Huerta, A., Hernandez-Tamames, J.A., Poot, D.H.J.,  
1346 2025. Myelin water imaging from accelerated 3D-grase acquisitions using subspace  
1347 constrained reconstruction. *Magma (New York, N.Y.)* doi:10.1007/s10334-025-  
1348 01276-w. PMID: 40679773.

1349 Campbell, M., McKenzie, J.E., Sowden, A., Katikireddi, S.V., Brennan, S.E., Ellis,  
1350 S., Hartmann-Boyce, J., Ryan, R., Shepperd, S., Thomas, J., Welch, V., Thomson,  
1351 H., 2020. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting  
1352 guideline. *BMJ* 368, l6890.

1353 Cao, Y., Cao, J., Li, B., 2026. Loddpm: High-fidelity pseudo-healthy brain MRI  
1354 synthesis via longitudinal differential guidance. *Biomedical Signal Processing and*  
1355 *Control* 123, 110600. doi:10.1016/j.bspc.2026.110600.



1356 Cao, Y., Levin, D., 1995. Using an image database to constrain the acquisition and  
1357 reconstruction of MR images of the human head. *IEEE Transactions on Medical*  
1358 *Imaging* doi:10.1109/42.387716.

1359 Castro, D.C., Walker, I., Glocker, B., 2020. Causality matters in medical imaging.  
1360 *Nature Communications* 11, 3673.

1361 Cerjanic, A.M., Diano, A.M., Johnson, C.L., 2025. Elastographic magnetization pre-  
1362 pared imaging with rapid encoding. *Magnetic resonance in medicine* doi:10.1002/  
1363 *mrm*.30482. PMID: 40065646.

1364 Chan, K.L., Ziegls, T., Henning, A., 2022. Improved signal-to-noise performance of  
1365 multinet GRAPPA 1h FID MRSI reconstruction with semi-synthetic calibration  
1366 data. *Magnetic Resonance in Medicine* 88, 1500–1515. doi:10.1002/*mrm*.29314.

1367 Chen, G., Xie, H., Rao, X., Liu, X., Otikovs, M., Frydman, L., Sun, P., Zhang, Z., Pan,  
1368 F., Yang, L., Zhou, X., Liu, M., Bao, Q., Liu, C., 2025a. MRI motion correction  
1369 through disentangled CycleGAN based on multi-mask k-space subsampling. *IEEE*  
1370 *Transactions on Medical Imaging* 44, 1907–1921. doi:10.1109/*tmi*.2024.3523949.

1371 Chen, L., Lin, J., Fu, G., Yang, P., Liu, M., 2025b. Pw-GAN: Pseudo-warping field  
1372 guided GAN for unsupervised denoising of fetal brain MRI images. *bioRxiv* doi:10.  
1373 1101/2025.04.30.651598.

1374 Chen, X., Peng, Z., Valadez, G.H., 2023. Dd-cisenet: Dual-domain cross-iteration  
1375 squeeze and excitation network for accelerated MRI reconstruction. *arXiv.org*  
1376 doi:10.48550/*arxiv*.2305.00088.

1377 Choi, J.W., Cho, Y.J., Lee, S.B., Lee, S., Hwang, J.Y., Choi, Y.H., Cheon, J.E.,  
1378 Lee, J., 2025a. Accelerated brain magnetic resonance imaging with deep learning  
1379 reconstruction: a comparative study on image quality in pediatric neuroimaging.  
1380 *Pediatric Radiology* 55, 1903–1914. doi:10.1007/*s00247-025-06314-2*.

1381 Choi, K.S., Park, C., Lee, J.Y., Lee, K.H., Jeon, Y.H., Hwang, I., Yoo, R.E., Yun, T.J.,  
1382 Lee, M.J., Jung, K.H., Kang, K.M., 2025b. Prospective evaluation of accelerated  
1383 brain MRI using deep learning-based reconstruction: Simultaneous application to  
1384 2D spin-echo and 3D gradient-echo sequences. *Korean journal of radiology* doi:10.  
1385 3348/*kjr*.2024.0653. PMID: 39780631.

1386 Chow, L.S., Rajagopal, H., 2017. Modified-BRISQUE as no reference image quality  
1387 assessment for structural MR images. *Magnetic Resonance Imaging* 43, 74–87.  
1388 doi:10.1016/*j.mri*.2017.07.016.

1389 Chung, K.J., Souza, R., Frayne, R., 2020. Restoration of lossy JPEG-compressed brain  
1390 MR images using cross-domain neural networks. *IEEE Signal Processing Letters* 27,  
1391 141–145. doi:10.1109/*lsp*.2019.2961072.

1392 Ciceri, T., Squarcina, L., Pigoni, A., Ferro, A., Montano, F., Bertoldo, A., Persico,  
1393 N., Boito, S., Triulzi, F.M., Conte, G., Brambilla, P., Peruzzo, D., 2023. Geomet-  
1394 ric reliability of super-resolution reconstructed images from clinical fetal MRI in  
1395 the second trimester. *Neuroinformatics* doi:10.1007/*s12021-023-09635-5*. PMID:  
1396 37284977.

1397 Conklin, J., Longo, M.G.F., Tabari, A., Filho, A.L.G., Liu, W., Splitthoff, D.N.,  
 1398 Lo, W.C., Cauley, S.F., Setsompop, K., Schaefer, P.W., Kirsch, J.E., Rapalino,  
 1399 O., Huang, S.Y., 2022. Clinical validation of wave-caipi susceptibility-weighted  
 1400 imaging for routine brain MRI at 1.5 t. *European radiology* doi:10.1007/s00330-  
 1401 022-08871-8. pMID: 35925387.

1402 Cordero-Grande, L., Ortuño-Fisac, J.E., del Hoyo, A.A., Uus, A., Deprez, M., Santos,  
 1403 A., Hajnal, J.V., Ledesma-Carbayo, M.J., 2023. Fetal MRI by robust deep generative  
 1404 prior reconstruction and diffeomorphic registration. *IEEE Transactions on*  
 1405 *Medical Imaging* doi:10.1109/tmi.2022.3217725.

1406 Dabrowski, O., Falcone, J.L., Klauser, A., Songeon, J., Kocher, M., Chopard, B.,  
 1407 Lazeyras, F., Courvoisier, S., 2025. Sismik for brain MRI: Deep-learning-based  
 1408 motion estimation and model-based motion correction in k-space. *IEEE transactions*  
 1409 *on medical imaging* doi:10.1109/tmi.2024.3446450. pMID: 39159019.

1410 Delattre, B.M.A., Boudabbous, S., Hansen, C., Neroladaki, A., Hachulla, A.L., Vargas,  
 1411 M.I., 2019. Compressed sensing MRI of different organs: ready for clinical daily  
 1412 practice? *European Radiology* 30, 308–319. doi:10.1007/s00330-019-06319-0.

1413 Desalegn, L., Jifara, W., 2024. Hara-GAN: Hybrid attention and relative average  
 1414 discriminator based generative adversarial network for MR image reconstruction.  
 1415 *IEEE Access* doi:10.1109/access.2024.3364699.

1416 Dewey, B.E., Remedios, S.W., Sanjayan, M., Rjeily, N.B., Lee, A.Z., Wyche, C.,  
 1417 Duncan, S., Prince, J.L., Calabresi, P.A., Fitzgerald, K.C., Mowry, E.M., 2025.  
 1418 Super-resolution in clinically available spinal cord mris enables automated atrophy  
 1419 analysis. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a8526.  
 1420 pMID: 39366765.

1421 Dieckmeyer, M., Roy, A.G., Senapati, J., Wachinger, C., Grundl, L., Döpfert, J.,  
 1422 Bertran, P.F., Lemke, A., Zimmer, C., Kirschke, J.S., Hedderich, D.M., 2021. Effect  
 1423 of MRI acquisition acceleration via compressed sensing and parallel imaging on brain  
 1424 volumetry. *Magma (New York, N.Y.)* doi:10.1007/s10334-020-00906-9. pMID:  
 1425 33502667.

1426 Ding, J., Duan, Y., Zhuo, Z., Yuan, Y., Zhang, G., Song, Q., Gao, B., Zhang, B.,  
 1427 Wang, M., Yang, L., Hou, Y., Yuan, J., Feng, C., Wang, J., Lin, L., Liu, Y., 2021.  
 1428 Acceleration of brain tof-mra with compressed sensitivity encoding: A multicenter  
 1429 clinical study. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7091.  
 1430 pMID: 33858820.

1431 Dolui, S., Wang, Z., Wolf, R.L., Nabavizadeh, A., Xie, L., Tosun, D., Nasrallah, I.M.,  
 1432 Wolk, D.A., Detre, J.A., Initiative, A.D.N., 2024. Automated quality evaluation  
 1433 index for arterial spin labeling derived cerebral blood flow maps. *Journal of magnetic*  
 1434 *resonance imaging : JMRI* doi:10.1002/jmri.29308. pMID: 38400805.

1435 Dong, Z., Reese, T.G., Lee, H.H., Huang, S.Y., Polimeni, J.R., Wald, L.L., Wang,  
 1436 F., 2025. Romer-epti: Rotating-view motion-robust super-resolution epti for snr-  
 1437 efficient distortion-free in-vivo mesoscale diffusion MRI and microstructure imaging.  
 1438 *Magnetic resonance in medicine* doi:10.1002/mrm.30365. pMID: 39552568.

1439 Du, J., Wang, L., Liu, Y., Zhou, Z., He, Z., Jia, Y., 2020. Brain MRI super-resolution  
1440 using 3D dilated convolutional encoder-decoder network. *IEEE Access* doi:10.1109/  
1441 access.2020.2968395.

1442 Duffy, B.A., Zhao, L., Sepehrband, F., Min, J., Wang, D.J., Shi, Y., Toga, A.W., Kim,  
1443 H., Initiative, A.D.N., 2021. Retrospective motion artifact correction of structural  
1444 MRI images using deep learning improves the quality of cortical surface reconstruc-  
1445 tions. *NeuroImage* doi:10.1016/j.neuroimage.2021.117756. pMID: 33460797.

1446 Duong, S.T.M., Phung, S.L., Bouzerdoun, A., Ang, S.P., Schira, M.M., 2021. Correct-  
1447 ing susceptibility artifacts of MRI sensors in brain scanning: A 3D anatomy-guided  
1448 deep learning approach. *Sensors (Basel, Switzerland)* doi:10.3390/s21072314.  
1449 pMID: 33810289.

1450 Eeden Risager, K., Gholamalizadeh, T., Mehdipour Ghazi, M., 2025. Non-reference  
1451 quality assessment for medical imaging: Application to synthetic brain mris. *Deep*  
1452 *Generative Models* doi:10.1007/978-3-031-72744-3\_19.

1453 Eggl, M.F., D. Santis, S., 2026. Simulation-based inference at the theoretical limit  
1454 for fast, robust microstructural MRI with minimal diffusion data. *Communications*  
1455 *Medicine* doi:10.1038/s43856-026-01614-6.

1456 Elhabian, S., Vachet, C., Piven, J., Styner, M., Gerig, G., 2016. Compressive sensing  
1457 based q-space resampling for handling fast bulk motion in hardi acquisitions. 2016  
1458 IEEE 13th International Symposium on Biomedical Imaging (ISBI) doi:10.1109/is  
1459 bi.2016.7493412.

1460 Elsaid, N.M.H., Wu, Y.C., 2019. Super-resolution diffusion tensor imaging using sr-  
1461 cnn: A feasibility study. *Annual International Conference of the IEEE Engineer-  
1462 ing in Medicine and Biology Society. IEEE Engineering in Medicine and Biology*  
1463 *Society. Annual International Conference* doi:10.1109/embc.2019.8857125. pMID:  
1464 31946482.

1465 Ernst, P., Chatterjee, S., Rose, G., Speck, O., Nürnberger, A., 2023. Sinogram upsam-  
1466 pling using primal-dual unet for undersampled CT and radial MRI reconstruction.  
1467 *Neural networks : the official journal of the International Neural Network Society*  
1468 doi:10.1016/j.neunet.2023.08.004.

1469 Esteban, O., Birman, D., Schaer, M., Kiar, G., Gorgolewski, K.J., Poldrack, R.A.,  
1470 2017. MRIQC: Advancing the automatic prediction of image quality in MRI from  
1471 unseen sites. *PLoS ONE* 12, e0184661. doi:10.1371/journal.pone.0184661.

1472 European Parliament and Council of the European Union, 2017. Regulation (EU)  
1473 2017/745 on medical devices (Medical Device Regulation). *Official Journal of the*  
1474 *European Union*, L 117, 5 May 2017.

1475 European Parliament and Council of the European Union, 2024. Regulation (EU)  
1476 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelli-  
1477 gence Act). *Official Journal of the European Union*, 12 July 2024.

1478 Fadeev, A., Eltonsy, N., Tourassi, G., Elmaghraby, A., 2005. Quality evaluation of  
1479 fast morphing interpolation model for 3D volume reconstruction. *Proceedings of*  
1480 *the Fifth IEEE International Symposium on Signal Processing and Information*  
1481 *Technology*, 2005. doi:10.1109/isspit.2005.1577201.

1482 Fantini, I., Rittner, L., Yasuda, C., Lotufo, R., 2018. Automatic detection of mo-  
1483 tion artifacts on MRI using deep CNN. 2018 International Workshop on Pattern  
1484 Recognition in Neuroimaging (PRNI) doi:10.1109/prni.2018.8423948.

1485 Feizollah, S., Tardif, C.L., 2025. 3D mermaid: 3D multi-shot enhanced recovery motion  
1486 artifact insensitive diffusion for submillimeter, multi-shell, and snr-efficient diffusion  
1487 imaging. Magnetic resonance in medicine doi:10.1002/mrm.30436. pMID: 40035173.

1488 Fujita, S., Polak, D., Nickel, D., Splitthoff, D.N., Huang, Y., Gil, N., Buathong,  
1489 S., Chiang, C.H., Lo, W.C., Clifford, B., Cauley, S.F., Conklin, J., Huang, S.Y.,  
1490 2025. Motion-informed 3d deep learning reconstruction in patients with cognitive  
1491 impairment. American Journal of Neuroradiology 47, 446–453. doi:10.3174/ajnr  
1492 .a8977.

1493 Gao, Y., Zhu, X., Moffat, B.A., Glarin, R., Wilman, A.H., Pike, G.B., Crozier, S.,  
1494 Liu, F., Sun, H., 2021. xqsm: quantitative susceptibility mapping with octave  
1495 convolutional and noise-regularized neural networks. NMR in biomedicine doi:10.  
1496 1002/nbm.4461. pMID: 33368705.

1497 Gardner, E.A., Ellis, J.H., Hyde, R.J., Aisen, A.M., Quint, D.J., Carson, P.L., 1995.  
1498 Detection of degradation of magnetic resonance (MR) images: comparison of an  
1499 automated MR image-quality analysis system with trained human observers. Aca-  
1500ademic radiology doi:10.1016/s1076-6332(05)80184-9. pMID: 9419562.

1501 Garma-Oehmichen, A., Acuna-Luna, K.P., Santos-Diaz, A., 2021. Evaluation of echo  
1502 planar imaging (EPI) distortion correction using synb0-disco and reversed phase  
1503 encoding acquisition, in: 2021 43rd Annual International Conference of the IEEE  
1504 Engineering in Medicine & Biology Society (EMBC), pp. 4023–4026. doi:10.1109/  
1505 embc46164.2021.9630596.

1506 Giannakopoulos, I.I., Muckley, M.J., Kim, J., Breen, M., Johnson, P.M., Lui, Y.W.,  
1507 Lattanzi, R., 2024. Accelerated MRI reconstructions via variational network and  
1508 feature domain learning. Scientific Reports doi:10.1038/s41598-024-59705-0.

1509 Gil, N., Tabari, A., Lo, W.C., Clifford, B., Lang, M., Awan, K., Gaudet, K., Split-  
1510 thoff, D.N., Polak, D., Cauley, S., Huang, S.Y., 2024. Quantitative evaluation of  
1511 scout accelerated motion estimation and reduction (samer) mprage for morpho-  
1512 metric analysis of brain tissue in patients undergoing evaluation for memory loss.  
1513 NeuroImage doi:10.1016/j.neuroimage.2024.120865. pMID: 39349147.

1514 Glessgen, C., Gallichan, D., Moor, M., Hainc, N., Federau, C., 2019. Evaluation  
1515 of 3D fat-navigator based retrospective motion correction in the clinical setting  
1516 of patients with brain tumors. Neuroradiology doi:10.1007/s00234-019-02160-w.  
1517 pMID: 30671618.

1518 Goffi, F., Bianchi, A.M., Schiena, G., Brambilla, P., Maggioni, E., 2025. Multi-metric  
1519 approach for the comparison of denoising techniques for resting-state fMRI. Human  
1520 brain mapping doi:10.1002/hbm.70080. pMID: 40309965.

1521 Gong, E., Pauly, J.M., Wintermark, M., Zaharchuk, G., 2018. Deep learning enables  
1522 reduced gadolinium dose for contrast-enhanced brain MRI. Journal of Magnetic  
1523 Resonance Imaging 48, 330–340. doi:10.1002/jmri.25970.

1524 Gottschling, N.M., Antun, V., Hansen, A.C., Adcock, B., 2025. The troublesome  
1525 kernel: On hallucinations, no free lunches, and the accuracy–stability trade-off in  
1526 inverse problems. *SIAM Review* 67, 73–104. doi:10.1137/23M1568739.

1527 Güngör, A., Dar, S.U., Öztürk, c., Korkmaz, Y., Bedel, H.A., Elmas, G., Ozbey,  
1528 M., Çukur, T., 2023. Adaptive diffusion priors for accelerated MRI reconstruction.  
1529 *Medical image analysis* doi:10.1016/j.media.2023.102872. pMID: 37384951.

1530 Gurbani, S.S., Schreibmann, E., Maudsley, A.A., Cordova, J.S., Soher, B.J., Poptani,  
1531 H., Verma, G., Barker, P.B., Shim, H., Cooper, L.A.D., 2018. A convolutional neural  
1532 network to filter artifacts in spectroscopic MRI. *Magnetic resonance in medicine*  
1533 doi:10.1002/mrm.27166. pMID: 29520831.

1534 Hagiwara, A., Otsuka, Y., Hori, M., Tachibana, Y., Yokoyama, K., Fujita, S., Andica,  
1535 C., Kamagata, K., Irie, R., Koshino, S., Maekawa, T., Chougar, L., Wada, A.,  
1536 Takemura, M.Y., Hattori, N., Aoki, S., 2019. Improving the quality of synthetic  
1537 FLAIR images with deep learning using a conditional generative adversarial network  
1538 for pixel-by-pixel image translation. *AJNR. American journal of neuroradiology*  
1539 doi:10.3174/ajnr.a5927. pMID: 30630834.

1540 Hales, P.W., Pfeuffer, J., Clark, C.A., 2020. Combined denoising and suppression  
1541 of transient artifacts in arterial spin labeling MRI using deep learning. *Journal of*  
1542 *magnetic resonance imaging : JMRI* doi:10.1002/jmri.27255.

1543 Hammernik, K., Klatzer, T., Kobler, E., Recht, M.P., Sodickson, D.K., Pock, T.,  
1544 Knoll, F., 2018. Learning a variational network for reconstruction of accelerated  
1545 MRI data. *Magnetic Resonance in Medicine* 79, 3055–3071.

1546 Hampe, N., Katscher, U., Berg, C.A.T.v.d., Tha, K.K., Mandija, S., 2020. Investigat-  
1547 ing the challenges and generalizability of deep learning brain conductivity mapping.  
1548 *Physics in medicine and biology* doi:10.1088/1361-6560/ab9356. pMID: 32408291.

1549 Harada, S., Takatsu, Y., Murayama, K., Sano, Y., Ikedo, M., 2025. Verification of reso-  
1550 lution and imaging time for high-resolution deep learning reconstruction techniques.  
1551 *Magnetic Resonance Imaging* 123, 110463. doi:10.1016/j.mri.2025.110463.

1552 Harper, J.R., Cherukuri, V., O'Reilly, T., Yu, M., Mbabazi-Kabachelor, E., Mu-  
1553 lando, R., Sheth, K.N., Webb, A.G., Warf, B.C., Kulkarni, A.V., Monga, V.,  
1554 Schiff, S.J., 2021. Assessing the utility of low resolution brain imaging: treatment  
1555 of infant hydrocephalus. *NeuroImage. Clinical* doi:10.1016/j.nicl.2021.102896.  
1556 pMID: 34911199.

1557 Haskell, M.W., Cauley, S.F., Bilgic, B., Hossbach, J., Splittthoff, D.N., Pfeuffer, J., Set-  
1558 sompop, K., Wald, L.L., 2019. Network accelerated motion estimation and reduction  
1559 (namer): Convolutional neural network guided retrospective motion correction using  
1560 a separable motion model. *Magnetic resonance in medicine* doi:10.1002/mrm.27771.  
1561 pMID: 31045278.

1562 Heo, H.Y., Singh, M., Yedavalli, V., Jiang, S., Zhou, J., 2024. Cest and nuclear over-  
1563 hauser enhancement imaging with deep learning-extrapolated semisolid magnetiza-  
1564 tion transfer reference: Scan-rescan reproducibility and reliability studies. *Magnetic*  
1565 *resonance in medicine* doi:10.1002/mrm.29937. pMID: 38009996.

1566 Hernandez Petzsche, M.R., de la Rosa, E., Hanning, U., Wiest, R., Valenzuela, W.,  
1567 Reyes, M., Meyer, M., Liew, S.L., Kofler, F., Ezhov, I., et al., 2022. ISLES 2022: A  
1568 multi-center magnetic resonance imaging stroke lesion segmentation dataset. *Scientific Data* 9, 762. doi:10.1038/s41597-022-01875-5.

1570 Hewlett, M., Petrov, I., Johnson, P.M., Drangova, M., 2024. Deep-learning-based  
1571 motion correction using multichannel MRI data: a study using simulated artifacts  
1572 in the fastmri dataset. *NMR in biomedicine* doi:10.1002/nbm.5179.

1573 Ho, B.C., Kim, D., Kumar, A., Weiss, S., Vossler, H., Mormino, E., Zaharchuk, G.,  
1574 Initiative, A.D.N., 2026. Evaluation of image-level harmonization methods for multi-  
1575 center MR neuroimaging. *Journal of magnetic resonance imaging : JMRI* doi:10.  
1576 1002/jmri.70221. PMID: 41489091.

1577 Hossbach, J., Splitthoff, D.N., Cauley, S., Clifford, B., Polak, D., Lo, W.C., Meyer,  
1578 H., Maier, A., 2023. Deep learning-based motion quantification from k-space for  
1579 fast model-based magnetic resonance imaging motion correction. *Medical physics*  
1580 doi:10.1002/mp.16119. PMID: 36433748.

1581 Hosseini, M.S., Aghamiri, S.M.R., Ardekani, A.F., BagheriMofidi, S.M., Safari, M.,  
1582 2025. Autocornn: An unsupervised physics-aware deep learning model for geo-  
1583 metric distortion correction of brain MRI images towards MR-only stereotactic ra-  
1584 diosurgery. *Journal of imaging informatics in medicine* doi:10.1007/s10278-024-  
1585 01171-1. PMID: 39080159.

1586 Hu, Y., Xu, Y., Tian, Q., Chen, F., Shi, X., Moran, C.J., Daniel, B.L., Hargreaves,  
1587 B.A., 2021. Run-up: Accelerated multishot diffusion-weighted MRI reconstruction  
1588 using an unrolled network with u-net as priors. *Magnetic resonance in medicine*  
1589 doi:10.1002/mrm.28446. PMID: 32783339.

1590 Hu, Y., Zhang, X., Chen, D., Yan, Z., Shen, X., Yan, G., Ou-Yang, L., Lin, J.,  
1591 Dong, J., Qu, X., 2022. Spatiotemporal flexible sparse reconstruction for rapid  
1592 dynamic contrast-enhanced MRI. *IEEE transactions on bio-medical engineering*  
1593 doi:10.1109/tbme.2021.3091881. PMID: 34166181.

1594 Huang, J., Zhou, G., Yu, Z., Hu, W., 2022. [deep parallel MRI reconstruction based  
1595 on a complex-valued loss function]. *Nan fang yi ke da xue xue bao = Journal of*  
1596 *Southern Medical University* doi:10.12122/j.issn.1673-4254.2022.12.02. PMID:  
1597 36651242.

1598 Huang, Y., Wang, Z., Zhang, X., Cao, J., Tu, Z., Lin, M., Li, L., Jiang, X., Guo, D.,  
1599 Qu, X., 2025. Improve robustness to mismatched sampling rate: An alternating  
1600 deep low-rank approach for exponential function reconstruction and its biomedical  
1601 magnetic resonance applications. *Journal of Magnetic Resonance* 376, 107898.  
1602 doi:10.1016/j.jmr.2025.107898.

1603 Huber, L.R., Stirnberg, R., Morgan, A.T., Feinberg, D.A., Ehse, P., Knudsen, L.,  
1604 Gulban, O.F., Koiso, K., Gephart, I., Swegle, S., Wardle, S.G., Persichetti, A.S.,  
1605 Beckett, A.J.S., Stöcker, T., Boulant, N., Poser, B.A., Bandettini, P.A., 2025. Short-  
1606 term gradient imperfections in high-resolution epi lead to fuzzy ripple artifacts.  
1607 *Magnetic resonance in medicine* doi:10.1002/mrm.30489. PMID: 40173320.

Huijben, E.M.C., Amirrajab, S., Pluim, J.P.W., 2025. Enhancing reconstruction-based out-of-distribution detection in brain MRI with model and metric ensembles. *Computer methods and programs in biomedicine* doi:10.1016/j.cmpb.2025.109045.

International Commission on Radiation Units and Measurements, 1996. *Medical Imaging — The Assessment of Image Quality*. Technical Report ICRU Report 54. ICRU. Bethesda, MD.

Islam, K.T., Dayarathna, S., Zhong, S., Zakavi, P., Kavnoudias, H., Farquharson, S., Durbridge, G., Sun, H., Bacchi, S., Egan, G.F., Barth, M., Dwyer, A., McMahon, K.L., Parizel, P.M., Law, M., Chen, Z., 2026. Synpoc: a high-quality generative diffusion model for transforming ultra-low-field point-of-care MRI using high-field MRI representations. *Scientific reports* doi:10.1038/s41598-025-33162-9. pMID: 41580445.

Ismail, A.R., Azhary, M.Z.R., Azwan, N.A.Z.N., Ismail, A., Alsaiani, N.A., 2024. Performance evaluation of medical image denoising using convolutional autoencoders, in: *2024 3rd International Conference on Creative Communication and Innovative Technology (ICCIT)*, pp. 1–6. doi:10.1109/iccit62134.2024.10701250.

Jayadeepthi, K., Yakshitha, K., Reddy, K.P.K., Gowtham, K., Mahesh, K., 2026. Deep learning based denoising framework for high-fidelity brain MRI reconstruction, in: *2026 4th International Conference on Inventive Computing and Informatics (ICICI)*, pp. 696–702. doi:10.1109/icici68773.2026.11580971.

Jayakumar, N., Hossain, T., Zhang, M., 2023. SADIR: Shape-aware diffusion models for 3D image reconstruction. *Shape in Medical Imaging* doi:10.1007/978-3-031-46914-5\_23.

Jeong, S., 2026. Evaluating reliability of automated quantitative brain morphometry from fetal t2-weighted MRI. *Frontiers in Neuroscience* doi:10.3389/fnins.2026.1817732.

Ji, Y., Wu, W., Buck, M.H.S.d., Okell, T., Jezzard, P., 2023. Highly accelerated intracranial time-of-flight magnetic resonance angiography using wave-encoding. *Magnetic resonance in medicine* doi:10.1002/mrm.29647. pMID: 37010811.

Jiang, Y., Zhang, S., Chi, J., 2023. Multi-modal brain tumor data completion based on reconstruction consistency loss. *Journal of digital imaging* doi:10.1007/s10278-022-00697-6. pMID: 36856903.

Jochmann, T., Salman, F., Dwyer, M.G., Bergsland, N., Zivadinov, R., Hauelsen, J., Schweser, F., 2025. Quantitative susceptibility mapping in magnetically inhomogeneous tissues. *Magnetic resonance in medicine* doi:10.1002/mrm.30537. pMID: 40312865.

Joshi, D., Gautam, A., Borotikar, B., 2025. Arterial spin labeling-based CBF map generation using a vector quantized variational autoencoder, in: *2025 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, pp. 1–5. doi:10.1109/conecct65861.2025.11306510.

1648 Jun, Y., Arefeen, Y., Cho, J., Fujita, S., Wang, X., Grant, P.E., Gagoski, B., Jaimes,  
1649 C., Gee, M.S., Bilgic, B., 2024. Zero-deepsub: Zero-shot deep subspace reconstruction  
1650 for rapid multiparametric quantitative MRI using 3D-qalas. *Magnetic resonance*  
1651 *in medicine* doi:10.1002/mrm.30018. pMID: 38282270.

1652 Jung, W., Jeong, G., Kim, S., Hwang, I., Choi, S.H., Jeon, Y.H., Choi, K.S., Lee, J.Y.,  
1653 Yoo, R.E., Yun, T.J., Kang, K.M., 2025. Reliability of brain volume measures of  
1654 accelerated 3D t1-weighted images with deep learning-based reconstruction. *Neu-*  
1655 *roradiology* doi:10.1007/s00234-024-03461-5. pMID: 39316090.

1656 Kamesh Iyer, S., Moon, B.F., Josselyn, N., Kurtz, R.M., Song, J.W., Ware, J.B.,  
1657 Nabavizadeh, S.A., Witschey, W.R., 2022. Quantitative susceptibility mapping  
1658 using plug-and-play alternating direction method of multipliers. *Scientific Reports*  
1659 doi:10.1038/s41598-022-22778-w.

1660 Kanemaru, N., Takao, H., Amemiya, S., Abe, O., 2022. The effect of a post-scan  
1661 processing denoising system on image quality and morphometric analysis. *Journal of*  
1662 *neuroradiology* = *Journal de neuroradiologie* doi:10.1016/j.neurad.2021.11.007.

1663 Kang, S.H., Chung, J.Y., Lee, Y., Initiative, A.D.N., 2026. Simulation data-based dual  
1664 domain network (sim-ddnet) for motion artifact reduction in MR images. *MAGNE-*  
1665 *TOCHEMISTRY* doi:10.3390/magnetochemistry12010014.

1666 Kang, S.K., Shin, S.A., Seo, S., Byun, M.S., Lee, D.Y., Kim, Y.K., Lee, D.S., Lee, J.S.,  
1667 2021. Deep learning-based 3D inpainting of brain MR images. *Scientific Reports*  
1668 doi:10.1038/s41598-020-80930-w.

1669 Karimi, D., Jaimes, C., Machado-Rivas, F., Vasung, L., Khan, S., Warfield,  
1670 S.K., Gholipour, A., 2021a. Deep learning-based parameter estimation in fetal  
1671 diffusion-weighted MRI. *NeuroImage* doi:10.1016/j.neuroimage.2021.118482.  
1672 pMID: 34455242.

1673 Karimi, D., Vasung, L., Machado-Rivas, F., Jaimes, C., Khan, S., Gholipour, A.,  
1674 2021b. Accurate parameter estimation in fetal diffusion-weighted MRI, in: *MICCAI*  
1675 *2021, LNCS 12907*. doi:10.1007/978-3-030-87234-2/46.

1676 Karthik, A., Aggarwal, K., Kapoor, A., Singh, D., Hu, L., Gandhamal, A., Kumar, D.,  
1677 2024. Comprehensive assessment of imaging quality of artificial intelligence-assisted  
1678 compressed sensing-based MR images in routine clinical settings. *BMC medical*  
1679 *imaging* doi:10.1186/s12880-024-01463-6. pMID: 39434010.

1680 Karuna, Y., Syamala, N., Ravikumar, C.V., Thakur, P., Saladi, S., 2025. Modified  
1681 energy-based GAN for intensity in homogeneity correction in brain MR images.  
1682 *Scientific reports* doi:10.1038/s41598-025-08552-8. pMID: 40691452.

1683 Kayvanrad, M., Lin, A., Joshi, R., Chiu, J., Peters, T., 2016. Diagnostic quality  
1684 assessment of compressed sensing accelerated magnetic resonance neuroimaging.  
1685 *Journal of Magnetic Resonance Imaging* 44, 433–444. doi:10.1002/jmri.25149.

1686 Kazerouni, A., Aghdam, E.K., Heidari, M., Azad, R., Fayyaz, M., Hacihaliloglu, I.,  
1687 Merhof, D., 2023. Diffusion models in medical imaging: A comprehensive survey.  
1688 *Medical Image Analysis* 88, 102846.



1689 Kharaji, M., Canton, G., Guo, Y., Mosi, M.H., Zhou, Z., Balu, N., Mossa-Basha, M.,  
1690 2025. Dante-caipi accelerated contrast-enhanced 3D t1: Deep learning-based image  
1691 quality improvement for vessel wall MRI. *AJNR. American journal of neuroradiol-*  
1692 *ogy* doi:10.3174/ajnr.a8424. pMID: 39038956.

1693 Khawaled, S., Freiman, M., 2024. NPB-rec: A non-parametric bayesian deep-learning  
1694 approach for undersampled MRI reconstruction with uncertainty estimation. *Artifi-*  
1695 *cial intelligence in medicine* doi:10.1016/j.artmed.2024.102798. pMID: 38462289.

1696 Kiani Kalejahi, B., 2026. GAN-based cross-modality brain MRI synthesis: Paired  
1697 versus unpaired training and comparison with diffusion and transformer models.  
1698 *Biomimetics* doi:10.3390/biomimetics11030175.

1699 Kim, K.H., Park, S.H., 2017. Artificial neural network for suppression of banding  
1700 artifacts in balanced steady-state free precession MRI. *Magnetic Resonance Imaging*  
1701 37, 139–146. doi:10.1016/j.mri.2016.11.020.

1702 Kim, S., Park, S., 2021. Simultaneous multislice brain MRI t1 mapping with im-  
1703 proved low-rank modeling. *Tomography (Ann Arbor, Mich.)* doi:10.3390/tomogr  
1704 aphy7040047. pMID: 34698294.

1705 Kim, S.H., Choi, Y.H., Lee, J.S., Lee, S.B., Cho, Y.J., Lee, S.H., Shin, S.M., Cheon,  
1706 J.E., 2022. Deep learning reconstruction in pediatric brain MRI: comparison of  
1707 image quality with conventional T2-weighted MRI. *Neuroradiology* 65, 207–214.  
1708 doi:10.1007/s00234-022-03053-1.

1709 Kim, T., Ji, S., Min, K., Kim, M., Youn, J., Oh, C., Kim, J., Lee, J., 2026. Vessel  
1710 segmentation for  $\chi$ -separation in quantitative susceptibility mapping. *Magnetic*  
1711 *resonance in medicine* doi:10.1002/mrm.70054. pMID: 40891385.

1712 Knoll, F., Hammernik, K., Zhang, C., Moeller, S., Pock, T., Sodickson, D.K.,  
1713 Akçakaya, M., 2020a. Deep-learning methods for parallel magnetic resonance imag-  
1714 ing reconstruction: A survey of the current approaches, trends, and issues. *IEEE*  
1715 *Signal Processing Magazine* 37, 128–140. doi:10.1109/MSP.2019.2950640.

1716 Knoll, F., Zbontar, J., Sriram, A., Muckley, M.J., Bruno, M., Defazio, A., Parente,  
1717 M., Geras, K.J., Katsnelson, J., Chandarana, H., Zhang, Z., Drozdal, M., Romero,  
1718 A., Rabbat, M., Vincent, P., Pinkerton, J., Wang, D., Yakubova, N., Owens, E.,  
1719 Zitnick, C.L., Recht, M.P., Sodickson, D.K., Lui, Y.W., 2020b. fastmri: A publicly  
1720 available raw  $k$ -space and DICOM dataset of knee images for accelerated MR image  
1721 reconstruction using machine learning. *Radiology: Artificial Intelligence* 2, e190007.

1722 Kofler, A., Kerkering, K.M., Göschel, L., Fillmer, A., Kolbitsch, C., 2024. Quanti-  
1723 tative MR image reconstruction using parameter-specific dictionary learning with  
1724 adaptive dictionary-size and sparsity-level choice. *IEEE Transactions on Biomedical*  
1725 *Engineering* doi:10.1109/tbme.2023.3300090.

1726 Kulkarni, P.H., Merchant, S.N., Awate, S.P., 2021. Dictionary+wavelet model with  
1727 nested-minorized vb-em for sms-caipi r-fMRI reconstruction. *IEEE Open Journal*  
1728 *of Signal Processing* doi:10.1109/oj-sp.2021.3100751.

1729 Kumar, V., Sharma, M.K., Jehadeesan, R., Venkatraman, B., Suman, G., Patra, A.,  
1730 Goenka, A.H., Sheet, D., 2021. Learning to generate missing pulse sequence in MRI  
1731 using deep convolution neural network trained with visual turing test, in: 2021 43rd  
1732 Annual International Conference of the IEEE Engineering in Medicine & Biology  
1733 Society (EMBC), pp. 3419–3422. doi:10.1109/embc46164.2021.9630435.

1734 Lacerda, L.M., Sperl, J.I., Menzel, M.I., Sprenger, T., Barker, G.J., Dell'Acqua, F.,  
1735 2016. Diffusion in realistic biophysical systems can lead to aliasing effects in diffusion  
1736 spectrum imaging. *Magnetic resonance in medicine* doi:10.1002/mrm.26080. PMID:  
1737 26714794.

1738 Lan, H., Varghese, B.A., Sheikh-Bahaei, N., Sepehrband, F., Toga, A.W., Choupan, J.,  
1739 Initiative, A.D.N., 2025. Diffusion based multi-domain neuroimaging harmonization  
1740 method with preservation of anatomical details. *NeuroImage* doi:10.1016/j.neur  
1741 oimage.2025.121297.

1742 Lang, M., Clifford, B., Lo, W.C., Applewhite, B.P., Tabari, A., Goncalves Filho,  
1743 A.L.M., Hosseini, Z., Figueiro Longo, M.G., Cauley, S.F., Setsompop, K., Bilgic,  
1744 B., Feiweier, T., Lev, M.H., Schaefer, P.W., Rapalino, O., Huang, S.Y., Conklin, J.,  
1745 2024. Clinical evaluation of a 2-minute ultrafast brain MR protocol for evaluation of  
1746 acute pathology in the emergency and inpatient settings. *AJNR. American Journal*  
1747 *of Neuroradiology* 45, 379–385. doi:10.3174/ajnr.A8143.

1748 Lang, M., Tabari, A., Polak, D., Ford, J., Clifford, B., Lo, W., Manzoor, K., Split-  
1749 thoff, D.N., Wald, L.L., Rapalino, O., Schaefer, P., Conklin, J., Cauley, S., Huang,  
1750 S.Y., 2023. Clinical evaluation of scout accelerated motion estimation and reduction  
1751 technique for 3D MR imaging in the inpatient and emergency department set-  
1752 tings. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7777. PMID:  
1753 36702502.

1754 Lau, V., Xiao, L., Zhao, Y., Su, S., Ding, Y., Man, C., Wang, X., Tsang, A., Cao, P.,  
1755 Lau, G.K.K., Leung, G.K.K., Leong, A.T.L., Wu, E.X., 2023. Pushing the limits of  
1756 low-cost ultra-low-field MRI by dual-acquisition deep learning 3D superresolution.  
1757 *Magnetic resonance in medicine* doi:10.1002/mrm.29642.

1758 Lee, J., Jung, W., Yang, S., Park, J.H., Hwang, I., Chung, J.W., Choi, S.H., Choi,  
1759 K.S., 2024. Deep learning-based super-resolution and denoising algorithm improves  
1760 reliability of dynamic contrast-enhanced MRI in diffuse glioma. *Scientific reports*  
1761 doi:10.1038/s41598-024-76592-7. PMID: 39455814.

1762 Lee, J., Kim, H., Chung, H., Eom, H., Jang, J., Sohn, C.H., Choi, K.S., 2026a. Lesion-  
1763 aware post-training of latent diffusion models for synthesizing diffusion MRI from  
1764 CT perfusion, in: *Medical Image Computing and Computer Assisted Intervention*  
1765 – MICCAI 2025, Springer. doi:10.1007/978-3-032-04937-7\_27.

1766 Lee, N.G., Cui, S.X., Nayak, K.S., 2026b. Cartesian maxgirt: Model-based epi recon-  
1767 struction incorporating gradient nonlinearity and concomitant field effects. *Magnetic*  
1768 *resonance in medicine* doi:10.1002/mrm.70113. PMID: 41039908.

1769 Li, H., Chiew, M., Dragonu, I., Jezzard, P., Okell, T.W., 2026. Few-shot learning  
1770 for highly accelerated 3D time-of-flight mra reconstruction. *Magnetic resonance in*  
1771 *medicine* doi:10.1002/mrm.70072. PMID: 40931540.

Li, M., Chen, C., Xiong, Z., Liu, Y., Rong, P., Shan, S., Liu, F., Sun, H., Gao, Y., 2025a. Quantitative susceptibility mapping via deep neural networks with iterative reverse concatenations and recurrent modules. *Medical physics* doi:10.1002/mp.17747.

Li, N., Wang, W.T., Pham, D.L., Butman, J.A., 2015. Artifactual microhemorrhage generated by susceptibility weighted image processing. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri.24728. PMID: 25143262.

Li, Y., Liao, Y.P., Wang, J., Lu, W., Zhang, Y., 2025b. Patient-specific MRI super-resolution via implicit neural representations and knowledge transfer. *Physics in medicine and biology* doi:10.1088/1361-6560/adbed4. PMID: 40064110.

Li, Z., Fan, Q., Bilgic, B., Wang, G., Wu, W., Polimeni, J.R., Miller, K.L., Huang, S.Y., Tian, Q., 2023. Diffusion MRI data analysis assisted by deep learning synthesized anatomical images (deepanat). *Medical image analysis* doi:10.1016/j.media.2023.102744. PMID: 36867912.

Liang, D., Liu, Y., Wang, H., Zhu, Y., 2020. Deep learning for magnetic resonance image reconstruction: An overview. *IEEE Transactions on Radiation and Plasma Medical Sciences* 4, 641–655.

Liang, Z., Guo, X., Xu, W., Ibrahim, Y., Voets, N., Pretorius, P.M., Initiative, A.D.N., Noble, J.A., Kamnitsas, K., 2026. Itermask3d: Unsupervised anomaly detection and segmentation with test-time iterative mask refinement in 3D brain MRI. *Medical image analysis* doi:10.1016/j.media.2025.103763.

Liebrand, L.C., Karkalousos, D., Poirion, É., Emmer, B.J., Roosendaal, S.D., Marquering, H.A., Majoie, C.B.L.M., Savatovsky, J., Caan, M.W.A., 2025. Deep learning for efficient reconstruction of highly accelerated 3D FLAIR MRI in neurological deficits. *Magma (New York, N.Y.)* doi:10.1007/s10334-024-01200-8. PMID: 39212832.

Liew, S.L., Tavenner, B.P., Donnelly, M.R., Zavaliangos-Petropulu, A., Jeong, J.N., Barisano, G., Hutton, A., Simon, J.P., Juliano, J.M., Suri, A., et al., 2022. A large, curated, open-source stroke neuroimaging dataset to improve lesion segmentation algorithms. *Scientific Data* 9, 320. doi:10.1038/s41597-022-01401-7.

Liu, G., Ueguchi, T., Ogawa, S., 2026a. Block-interleaved segmented echo-planar imaging for improved activity detection in submillimeter high-resolution functional MRI at 7 t. *Magnetic resonance in medicine* doi:10.1002/mrm.30569. PMID: 40432253.

Liu, H., Chen, Z., Law, L.H., Liu, Y., Wang, Z., Wang, J., Zhang, Y., Shen, D., Huang, J., Chan, K.W.Y., 2026b. Accelerating cest MRI through complementary under-sampling and multi-offset transformer reconstruction. *Communications Engineering* doi:10.1038/s44172-025-00580-6.

Liu, H., Versteeg, E., Fuderer, M., Heide, O.v.d., Schilder, M.B., Berg, C.A.T.v.d., Sbrizzi, A., 2025a. Time-efficient, high-resolution 3t whole-brain relaxometry using cartesian 3D MR spin tomography in time-domain (MR-stat) with cerebrospinal fluid suppression. *Magnetic resonance in medicine* doi:10.1002/mrm.30384. PMID: 39607873.

1814 Liu, J., Kocak, M., Supanich, M., Deng, J., 2020. Motion artifacts reduction in brain  
1815 MRI by means of a deep residual network with densely connected multi-resolution  
1816 blocks (DRN-DCMB). *Magnetic Resonance Imaging* 71, 69–79. doi:10.1016/j.mri.  
1817 2020.05.002.

1818 Liu, J., Lin, Q., Xiong, Z., Shan, S., Liu, C., Li, M., Liu, F., Pike, G.B., Sun, H., Gao,  
1819 Y., 2026c. Highly undersampled MRI reconstruction via a single posterior sampling  
1820 of diffusion models. *IEEE Transactions on Medical Imaging* doi:10.1109/tmi.2026.  
1821 3654585.

1822 Liu, Z., Patel, V., Zhou, X., Tao, S., Yu, T., Ma, J., Nickel, D., Liebig, P., West-  
1823 erhold, E.M., Mojahed, H., Gupta, V., Middlebrooks, E.H., 2025b. Deep learn-  
1824 ing reconstruction for 7t MP2RAGE and SPACE MRI: Improving image quality  
1825 at high acceleration factors. *American Journal of Neuroradiology* 46, 2446–2454.  
1826 doi:10.3174/ajnr.a8841.

1827 Liu, Z., Zhou, X., Tao, S., Ma, J., Nickel, D., Liebig, P., Mostapha, M., Patel, V.,  
1828 Westerhold, E.M., Mojahed, H., Gupta, V., Middlebrooks, E.H., 2025c. Application  
1829 of deep learning accelerated image reconstruction in t2-weighted turbo spin-echo  
1830 imaging of the brain at 7t. *AJNR. American journal of neuroradiology* doi:10.  
1831 3174/ajnr.a8662. PMID: 39832954.

1832 Lustig, M., Donoho, D., Pauly, J.M., 2007. Sparse MRI: The application of compressed  
1833 sensing for rapid MR imaging. *Magnetic Resonance in Medicine* 58, 1182–1195.

1834 Ma, Q., Lai, Z., Wang, Z., Qiu, Y., Zhang, H., Qu, X., 2024. MRI reconstruction with  
1835 enhanced self-similarity using graph convolutional network. *BMC medical imaging*  
1836 doi:10.1186/s12880-024-01297-2. PMID: 38760778.

1837 Ma, X., Ugurbil, K., Wu, X., 2022. Mitigating transmit-b1 artifacts by predicting  
1838 parallel transmission images with deep learning: A feasibility study using high-  
1839 resolution whole-brain diffusion at 7 tesla. *Magnetic resonance in medicine* doi:10.  
1840 1002/mrm.29238. PMID: 35403237.

1841 Mallardi, G., Calefato, F., Lanubile, F., Logroscino, G., Tafuri, B., 2025. Diffusion  
1842 models for neuroimaging data augmentation: Assessing realism and clinical rele-  
1843 vance. *Journal of Medical Systems* doi:10.1007/s10916-025-02300-1.

1844 Marchetto, E., Eichhorn, H., Gallichan, D., Schnabel, J.A., Ganz, M., 2025. Agree-  
1845 ment of image quality metrics with radiological evaluation in the presence of mo-  
1846 tion artifacts. *Magma (New York, N.Y.)* doi:10.1007/s10334-025-01266-y. PMID:  
1847 40493331.

1848 Marchetto, E., Flassbeck, S., Mao, A., Assländer, J., 2026. Contrast-optimized ba-  
1849 sis functions for self-navigated motion correction in quantitative MRI. *Magnetic*  
1850 *resonance in medicine* doi:10.1002/mrm.70090. PMID: 40961378.

1851 Marie, H.S., Draz, M.M., Said, W., Elbaz, M., 2026. Bald uakari algorithm: a novel  
1852 nature-inspired metaheuristic for missing pixel imputation in medical images. *Clus-*  
1853 *ter Computing* doi:10.1007/s10586-026-06180-z.

1854 Mason, A., Rioux, J., Clarke, S.E., Costa, A., Schmidt, M., Keough, V., Huynh, T.,  
1855 Beyea, S., 2020. Comparison of objective image quality metrics to expert radiolo-  
1856 gists' scoring of diagnostic quality of MR images. *IEEE Transactions on Medical*  
1857 *Imaging* 39, 1064–1072.

1858 Matsuo, K., Nakaura, T., Morita, K., Uetani, H., Nagayama, Y., Kidoh, M., Hoka-  
1859 mura, M., Yamashita, Y., Shinoda, K., Ueda, M., Mukasa, A., Hirai, T., 2023.  
1860 Feasibility study of super-resolution deep learning-based reconstruction using k-  
1861 space data in brain diffusion-weighted images. *Neuroradiology* 65, 1619–1629.  
1862 doi:10.1007/s00234-023-03212-y.

1863 McIlvain, G., 2022. Quantitative effects of off-resonance related distortion on brain  
1864 mechanical property estimation with magnetic resonance elastography. *NMR in*  
1865 *Biomedicine* doi:10.1002/nbm.4616.

1866 Mello, V.B.B., McKinley, R., Wiest, R., Rummel, C., 2026. Fast surface reconstruction  
1867 of human brain MRI: benchmarking deep-learning based morphometry tools.  
1868 *Scientific reports* doi:10.1038/s41598-026-55397-w. PMID: 42286078.

1869 Meng, Y., Allen, J.W., Sharghi, V.K., Qiu, D., 2025. Motion and temporal b0-shift cor-  
1870 rections for qsm and r 2 \* mapping using dual-echo spiral navigators and conjugate-  
1871 phase reconstruction. *Magnetic resonance in medicine* doi:10.1002/mrm.30266.

1872 Meng, Z., Guo, R., Li, Y., Guan, Y., Wang, T., Zhao, Y., Sutton, B., Li, Y., Liang, Z.,  
1873 2020. Accelerating t2mapping of the brain by integrating deep learning priors with  
1874 low?rank and sparse modeling. *Magnetic Resonance in Medicine* 85, 1455–1467.  
1875 doi:10.1002/mrm.28526.

1876 Mercier, C., Faisan, S., Pron, A., Girard, N., Auzias, G., Chonavel, T., Rousseau, F.,  
1877 2025. Intersection-based slice motion estimation for fetal brain imaging. *Computers*  
1878 *in biology and medicine* doi:10.1016/j.combiomed.2025.110005. PMID: 40112563.

1879 Miao, J., Huang, F., Narayan, S., Wilson, D.L., 2013. A new perceptual difference  
1880 model for diagnostically relevant quantitative image quality evaluation: a prelimi-  
1881 nary study. *Magnetic resonance imaging* doi:10.1016/j.mri.2012.09.009. PMID:  
1882 23218792.

1883 Miao, J., Huang, F., Wilson, D.L., 2010. Comprehensive quantitative image quality  
1884 evaluation of compressed sensing MRI reconstructions using a weighted perceptual  
1885 difference model (case-PDM): selective evaluation, disturbance calibration, and ag-  
1886 gregative evaluation of noise, blur, aliasing, and oil-painting artifacts, in: *SPIE*  
1887 *Proceedings*, p. 762709. doi:10.1117/12.845507.

1888 Milovic, C., Tejos, C., Silva, J., Shmueli, K., Irarrazaval, P., 2024. XSIM : A structural  
1889 similarity index measure optimized for MRI QSM. *Magnetic Resonance in Medicine*  
1890 93, 411–421. doi:10.1002/mrm.30271.

1891 Mirza, M.U., Dalmaz, O., Bedel, H.A., Elmas, G., Korkmaz, Y., Gungor, A., Dar,  
1892 S.U.H., Oguz, K.K., Çukur, T., 2026. Learning fourier-constrained diffusion bridges  
1893 for MRI reconstruction. *IEEE Transactions on Medical Imaging* doi:10.1109/tmi.  
1894 2026.3673956.

1895 Mohsen, M., Tamer, R., Hossam, H., Ahmed, O., Sherif, S., 2025. Deep super-  
1896 resolution and age prediction for brain MRI using SRCNN: A lightweight frame-  
1897 work for high-fidelity reconstruction and regression, in: 2025 Twelfth International  
1898 Conference on Intelligent Computing and Information Systems (ICICIS), pp. 38–45.  
1899 doi:10.1109/icicis66182.2025.11313176.

1900 Montalt-Tordera, J., Muthurangu, V., Hauptmann, A., Steeden, J.A., 2021. Machine  
1901 learning in magnetic resonance imaging: Image reconstruction. *Physica Medica* 83,  
1902 79–87.

1903 Moshe, Y.H., Buchsweiler, Y., Teicher, M., Artzi, M., 2024. Handling missing MRI  
1904 data in brain tumors classification tasks: Usage of synthetic images vs. duplicate  
1905 images and empty images. *Journal of magnetic resonance imaging : JMRI* doi:10.  
1906 1002/jmri.29072.

1907 Muckley, M.J., Riemenschneider, B., Radmanesh, A., Kim, S., Jeong, G., Ko, J., Jun,  
1908 Y., Shin, H., Hwang, D., Mostapha, M., Arberet, S., Nickel, D., Ramzi, Z., Ciuciu,  
1909 P., Starck, J.L., Teuwen, J., Karkalousos, D., Zhang, C., Sriram, A., Huang, Z.,  
1910 Yakubova, N., Lui, Y.W., Knoll, F., 2021. Results of the 2020 fastMRI challenge  
1911 for machine learning MR image reconstruction. *IEEE Transactions on Medical*  
1912 *Imaging* 40, 2306–2317. doi:10.1109/tmi.2021.3075856.

1913 Müller, S.J., Khadhraoui, E., Voit, D., Riedel, C.H., Frahm, J., Romero, J.M., Ernst,  
1914 M., 2023. Comparison of epi DWI and steam DWI in early postoperative MRI con-  
1915 trols after resection of tumors of the central nervous system. *Clinical Neuroradiology*  
1916 doi:10.1007/s00062-023-01261-7.

1917 Muro, I., Isoiwa, T., Shibukawa, S., Usui, K., Otsuka, Y., 2025. Using deep learning  
1918 to simultaneously reduce noise and motion artifacts in brain MR imaging. *Magnetic*  
1919 *resonance in medical sciences : MRMS : an official journal of Japan Society of Mag-*  
1920 *netic Resonance in Medicine* doi:10.2463/mrms.mp.2024-0098. pMID: 39938896.

1921 Muro, I., Shimizu, S., Tsukamoto, H., 2022. [improvement of motion artifacts in brain  
1922 MRI using deep learning by simulation training data]. *Nihon Hoshasen Gijutsu*  
1923 *Gakkai zasshi* doi:10.6009/jjrt.780108. pMID: 35046218.

1924 Nagaraj, U.D., Greer, J.S., Debnath, P., Tkach, J.A., Kline-Fath, B.M., 2025a. Slice-  
1925 to-volume reconstruction of fetal brain MR imaging in clinical practice. *AJNR.*  
1926 *American journal of neuroradiology* doi:10.3174/ajnr.a8728.

1927 Nagaraj, U.D., Meineke, J., Sriastwa, A., Tkach, J.A., Leach, J.L., Doneva, M.,  
1928 2025b. Evaluation of synthetic images derived from a neural network in pedi-  
1929 atric brain magnetic resonance imaging. *Magnetic Resonance Imaging* 121, 110427.  
1930 doi:10.1016/j.mri.2025.110427.

1931 Nimje, S., Artières, T., Guye, M., de Rochefort, L., 2025. Insights on scan-specific  
1932 deep-learning strategies for brain MRI parallel imaging reconstruction. *NMR in*  
1933 *Biomedicine* 38, e70079. doi:10.1002/nbm.70079. pMID: 40545734.

1934 Ning, L., Laun, F., Gur, Y., DiBella, E.V.R., Deslauriers-Gauthier, S., Megherbi,  
1935 T., Ghosh, A., Zucchelli, M., Menegaz, G., Fick, R., St-Jean, S., Paquette, M.,  
1936 Aranda, R., Descoteaux, M., Deriche, R., O'Donnell, L., Rathi, Y., 2015. Sparse  
1937 reconstruction challenge for diffusion MRI: Validation on a physical phantom to

determine which acquisition scheme and analysis method to use? Medical image analysis doi:10.1016/j.media.2015.10.012. PMID: 26606457.

Nishioka, N., Shimizu, Y., Kaneko, Y., Shirai, T., Suzuki, A., Amemiya, T., Ochi, H., Bito, Y., Takizawa, M., Ikebe, Y., Kameda, H., Harada, T., Fujima, N., Kudo, K., 2025. Accelerating FLAIR imaging via deep learning reconstruction: potential for evaluating white matter hyperintensities. Japanese journal of radiology doi:10.1007/s11604-024-01666-5. PMID: 39316286.

Nonninger, J.N., Kienast, P., Pogledic, I., Mallouhi, A., Barkhof, F., Trattnig, S., Robinson, S.D., Kasprian, G., Haider, L., 2025. Assessment of AI-accelerated t2-weighted brain MRI, based on clinical ratings and image quality evaluation. European journal of radiology doi:10.1016/j.ejrad.2025.112123. PMID: 40315626.

Okell, T.W., Woods, J.G., Chiew, M., 2026. Combined angiography and perfusion using radial imaging and arterial spin labeling with structural contrast. Magnetic resonance in medicine doi:10.1002/mrm.70073. PMID: 40955094.

Olsson, H., Millward, J.M., Starke, L., Gladysz, T., Klein, T., Fehr, J., Lai, W.C., Lippert, C., Niendorf, T., Waiczies, S., 2024. Simulating rigid head motion artifacts on brain magnitude MRI data-outcome on image quality and segmentation of the cerebral cortex. PloS one doi:10.1371/journal.pone.0301132. PMID: 38626138.

Ortiz-Gonzalez, A., Kobler, E., Schletter, L., Effland, A., 2026. Motiondps: Motion-compensated 3D brain MRI reconstruction. IEEE Transactions on Medical Imaging doi:10.1109/tmi.2026.3705847.

Osadebey, M.E., Pedersen, M., Arnold, D.L., Wendel-Mitoraj, K.E., 2018. Blind blur assessment of MRI images using parallel multiscale difference of gaussian filters. Biomedical engineering online doi:10.1186/s12938-018-0514-4. PMID: 29898715.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., et al., 2021. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ 372, n71.

Park, S., Beckett, A., Häkkinen, S., Walker, E., Ma, S.J., Kim, S., Kim, H., Feinberg, D.A., 2025a. Higher spatial resolution and sensitivity in whole brain functional MRI at 7t using 3D epi accelerated with variable density caipi sampling and temporal random walk. Magnetic resonance in medicine doi:10.1002/mrm.30512.

Park, Y.W., Yoo, R.E., Shin, I., Jeon, Y.H., Singh, K.P., Lee, M.D., Kim, S., Yang, K., Jeong, G., Ryu, L., Han, K., Ahn, S.S., Lee, S.K., Jain, R., Choi, S.H., 2025b. A multinational study of deep learning-based image enhancement for multiparametric glioma MRI. Scientific Reports doi:10.1038/s41598-025-17993-0.

Pawar, K., Chen, Z., Shah, N.J., Egan, G.F., 2022. Suppressing motion artefacts in MRI using an inception-resnet network with motion simulation augmentation. NMR in biomedicine doi:10.1002/nbm.4225. PMID: 31865624.

Pawlowski, N., Castro, D.C., Glocker, B., 2020. Deep structural causal models for medical imaging. Advances in Neural Information Processing Systems (NeurIPS) 33, 15061–15073.

1979 Peng, W., Xia, T., Ribeiro, F.D.S., Bosschieter, T., Adeli, E., Zhao, Q., Glocker, B.,  
1980 Pohl, K.M., 2026. Latent causal modeling for 3D brain MRI counterfactuals. Deep  
1981 Generative Models doi:10.1007/978-3-032-05472-2\_19.

1982 Pirkel, C.M., Cencini, M., Kurzawski, J.W., Waldmannstetter, D., Li, H., Sekuboyina,  
1983 A., Endt, S., Peretti, L., Donatelli, G., Pasquariello, R., Costagli, M., Buonincontri,  
1984 G., Tosetti, M., Menzel, M.I., Menze, B.H., 2022. Learning residual motion  
1985 correction for fast and robust 3D multiparametric MRI. Medical image analysis  
1986 doi:10.1016/j.media.2022.102387. PMID: 35180675.

1987 Pizarro, R., Assemblal, H.E., Jegathambal, S.K.B., Jubault, T., Antel, S., Arnold, D.,  
1988 Shmuel, A., 2023. Deep learning, data ramping, and uncertainty estimation for  
1989 detecting artifacts in large, imbalanced databases of MRI images. Medical image  
1990 analysis doi:10.1016/j.media.2023.102942. PMID: 37797482.

1991 Pruessmann, K.P., Weiger, M., Scheidegger, M.B., Boesiger, P., 1999. SENSE: Sensi-  
1992 tivity encoding for fast MRI. Magnetic Resonance in Medicine 42, 952–962.

1993 Qiao, Y., Shi, Y., 2022. Unsupervised deep learning for fod-based susceptibility  
1994 distortion correction in diffusion MRI. IEEE transactions on medical imaging  
1995 doi:10.1109/tmi.2021.3134496. PMID: 34882551.

1996 Qiu, Z., Hu, S., Zhao, W., Sakaie, K., Sun, J.E.P., Griswold, M.A., Jones, D.K.,  
1997 Ma, D., 2024. Self-calibrated subspace reconstruction for multidimensional MR  
1998 fingerprinting for simultaneous relaxation and diffusion quantification. Magnetic  
1999 resonance in medicine doi:10.1002/mrm.29969. PMID: 38102776.

2000 Radmanesh, A., Muckley, M.J., Murrell, T., Lindsey, E., Sriram, A., Knoll, F., Sod-  
2001 ickson, D.K., Lui, Y.W., 2022. Exploring the acceleration limits of deep learning  
2002 variational network-based two-dimensional brain MRI. Radiology. Artificial intelli-  
2003 gence doi:10.1148/ryai.210313. PMID: 36523647.

2004 Ramos-Llordén, G., Lobos, R.A., Kim, T.H., Tian, Q., Witzel, T., Lee, H.H., Scholz,  
2005 A., Keil, B., Yendiki, A., Bilgiç, B., Halder, J.P., Huang, S.Y., 2023. High-fidelity,  
2006 high-spatial-resolution diffusion magnetic resonance imaging of ex vivo whole human  
2007 brain at ultra-high gradient strength with structured low-rank echo-planar imaging  
2008 ghost correction. NMR in biomedicine doi:10.1002/nbm.4831. PMID: 36106429.

2009 Rassmann, S., Kugler, D., Ewert, C., Reuter, M., 2026. Regression is all you need for  
2010 medical image translation. IEEE transactions on medical imaging doi:10.1109/tm  
2011 i.2025.3650412. PMID: 41477795.

2012 Rauland, A., Meisler, S.L., Alexander-Bloch, A.F., Bagautdinova, J., Baller, E.B.,  
2013 Gur, R.E., Gur, R.C., Luo, A.C., Moore, T.M., Popovych, O.V., Reetz, K., Roalf,  
2014 D.R., Shinohara, R.T., Sotardi, S., Sydnor, V.J., Vossough, A., Eickhoff, S.B.,  
2015 Cieslak, M., Satterthwaite, T.D., 2025. White matter bundle reconstruction from  
2016 single-shell diffusion magnetic resonance imaging: Test-retest reliability and pre-  
2017 dictive capability across orientation distribution function reconstruction methods.  
2018 Human brain mapping doi:10.1002/hbm.70429. PMID: 41376618.

2019 Raymond, C., Yao, J., Clifford, B., Feiweier, T., Oshima, S., Telesca, D., Zhong, X.,  
2020 Meyer, H., Everson, R.G., Salamon, N., Cloughesy, T.F., Ellingson, B.M., 2025.



2021 Leveraging physics-based synthetic MR images and deep transfer learning for arti-  
2022 fact reduction in echo-planar imaging. *AJNR. American journal of neuroradiology*  
2023 doi:10.3174/ajnr.a8566. PMID: 39947682.

2024 Rezvani, S., Moazami, S., Azevedo, C.J., Oberai, A.A., 2025. Probabilistic brain  
2025 MR image transformation using generative models. *Scientific Reports* doi:10.1038/  
2026 s41598-025-31958-3.

2027 Risk, B.B., Kociuba, M.C., Rowe, D.B., 2018. Impacts of simultaneous multislice  
2028 acquisition on sensitivity and specificity in fMRI. *NeuroImage* doi:10.1016/j.neur  
2029 oimage.2018.01.078. PMID: 29408461.

2030 Rizzuti, G., Schakel, T., Huttinga, N.R.F., Dankbaar, J.W., Leeuwen, T.v., Sbrizzi,  
2031 A., 2024. Towards retrospective motion correction and reconstruction for clinical 3D  
2032 brain MRI protocols with a reference contrast. *Magma (New York, N.Y.)* doi:10.  
2033 1007/s10334-024-01161-y. PMID: 38758490.

2034 Ryu, K., Nam, Y., Gho, S.M., Jang, J., Lee, H.J., Cha, J., Baek, H.J., Park, J., Kim,  
2035 D.H., 2019. Data-driven synthetic MRI FLAIR artifact correction via deep neural  
2036 network. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri.26712.

2037 Saeed, S., Nickel, D., Weiland, E., Frohwein, L.J., Niehoff, J.H., Reiman, G., Schreck,  
2038 J., Haag, N., Boriesosdick, J., Schoenbeck, D., Katz, M., Shahzadi, I., Woeltjen,  
2039 M.M., Kroeger, J.R., Borggreffe, J., Moenninghoff, C., 2026. Improved image quality  
2040 and reduced acquisition time in brain MRI using deep learning-based reconstruction:  
2041 A quantitative and subjective assessment compared to standard mprage in 0.55 t  
2042 MRI. *Magnetic resonance imaging* doi:10.1016/j.mri.2026.110656.

2043 Safari, M., Wang, S., Eidex, Z., Qiu, R.L.J., Chang, C.W., Yu, D.S., Yang, X., 2025a.  
2044 A physics-informed deep learning model for MRI brain motion correction. *Medical*  
2045 *physics* doi:10.1002/mp.70197. PMID: 41395855.

2046 Safari, M., Wang, S., Li, Q., Eidex, Z., Qiu, R.L.J., Chang, C.W., Mao, H., Yang, X.,  
2047 2025b. Res-mocodiff: residual-guided diffusion models for motion artifact correction  
2048 in brain MRI. *Physics in medicine and biology* doi:10.1088/1361-6560/ae1110.  
2049 PMID: 41061726.

2050 Sagawa, H., Fushimi, Y., Nakajima, S., Fujimoto, K., Miyake, K.K., Numamoto,  
2051 H., Koizumi, K., Nambu, M., Kataoka, H., Nakamoto, Y., Saga, T., 2021. Deep  
2052 learning-based noise reduction for fast volume diffusion tensor imaging: Assessing  
2053 the noise reduction effect and reliability of diffusion metrics. *Magnetic resonance*  
2054 *in medical sciences : MRMS : an official journal of Japan Society of Magnetic*  
2055 *Resonance in Medicine* doi:10.2463/mrms.tn.2020-0061. PMID: 32963184.

2056 Sakitis, C.J., Rowe, D.B., 2025. Bayesian merged utilization of GRAPPA and SENSE  
2057 (bmugs) for in-plane accelerated reconstruction increases fMRI detection power.  
2058 *Magnetic resonance imaging* doi:10.1016/j.mri.2024.110252.

2059 Sanchez, P., Kascenas, A., Liu, X., O'Neil, A.Q., Tsiftaris, S.A., 2022. What is  
2060 healthy? generative counterfactual diffusion for lesion localization, in: *Deep Gener-*  
2061 *ative Models (DGM4MICCAI), MICCAI Workshop, Springer.* pp. 34-44.

2062 Sanchez, T., Zalevskyi, V., Mihailov, A., Martí Juan, G., Eixarch, E., Jakab, A.,  
2063 Dunet, V., Koob, M., Auzias, G., Bach Cuadra, M., 2026. Automatic quality  
2064 control in multi-centric fetal brain MRI super-resolution reconstruction. *Perinatal,*  
2065 *Preterm and Paediatric Image Analysis* doi:10.1007/978-3-032-05997-0\_1.

2066 Sannananja, B., Zhu, C., Colip, C.G., Somasundaram, A., Ibrahim, M., Khrisat, T.,  
2067 Mossa-Basha, M., 2022. Image-quality assessment of 3D intracranial vessel wall MRI  
2068 using dante or dante-caipi for blood suppression and imaging acceleration. *AJNR.*  
2069 *American journal of neuroradiology* doi:10.3174/ajnr.a7531. PMID: 35618420.

2070 Schauman, S., Niekerk, A.v., Norbeck, O., Rydén, H., Avventi, E., Skare, S., 2026. An  
2071 exploration of motion-sampling interactions in 3D MRI for neuroimaging. *Magnetic*  
2072 *resonance in medicine* doi:10.1002/mrm.70132. PMID: 41204061.

2073 Schlemper, J., Caballero, J., Hajnal, J.V., Price, A.N., Rueckert, D., 2017. A deep  
2074 cascade of convolutional neural networks for dynamic MR image reconstruction.  
2075 *IEEE Transactions on Medical Imaging* 37, 491–503.

2076 Shan, S., Gao, Y., Liu, P.Z.Y., Whelan, B., Sun, H., Dong, B., Liu, F., Wadding-  
2077 ton, D.E.J., 2023. Distortion-corrected image reconstruction with deep learning  
2078 on an MRI-linac. *Magnetic resonance in medicine* doi:10.1002/mrm.29684. PMID:  
2079 37125656.

2080 Shan, S., Gao, Y., Waddington, D., Chen, H., Whelan, B., Liu, P., Wang, Y., Liu,  
2081 C., Gan, H., Gao, M., Liu, F., 2024. Image reconstruction with b<sup>2</sup> inhomogeneity  
2082 using a deep unrolled network on an open-bore MRI-linac. *IEEE Transactions on*  
2083 *Instrumentation and Measurement* 73, 1–9. doi:10.1109/tim.2024.3481545.

2084 Sheng, H., Wang, X., Jiang, M., Zhang, Z., 2022. Deep learning-based diffusion-  
2085 weighted magnetic resonance imaging in the diagnosis of ischemic penumbra in  
2086 early cerebral infarction. *Contrast media & molecular imaging* doi:10.1155/2022/  
2087 6270700. PMID: 35291425.

2088 Shimron, E., Tamir, J.I., Wang, K., Lustig, M., 2022. Implicit data crimes: Ma-  
2089 chine learning bias arising from misuse of public data. *Proceedings of the National*  
2090 *Academy of Sciences* 119, e2117203119. doi:10.1073/pnas.2117203119.

2091 Simkó, A., Ruiter, S., Löfstedt, T., Garpebring, A., Nyholm, T., Bylund, M., Jonsson,  
2092 J., 2023. Improving MR image quality with a multi-task model, using convolutional  
2093 losses. *BMC medical imaging* doi:10.1186/s12880-023-01109-z. PMID: 37784039.

2094 Singh, M., Mahmud, S.Z., Yedavalli, V., Zhou, J., Kamson, D.O., Zijl, P.v., Heo,  
2095 H.Y., 2025. Learning-based motion artifact correction in the z-spectral domain  
2096 for chemical exchange saturation transfer MRI. *Magnetic resonance in medicine*  
2097 doi:10.1002/mrm.30440. PMID: 39834104.

2098 Solak, M., Tören, M., Asan, B., Kaba, E., Beyazal, M., Çeliker, F.B., 2026. Braingan  
2099 framework for generating synthetic contrast-enhanced multisequence MRI in brain  
2100 metastases. *Journal of Imaging Informatics in Medicine* doi:10.1007/s10278-026-  
2101 02059-y.

2102 Sommer, K., Saalbach, A., Brosch, T., Hall, C., Cross, N.M., Andre, J.B., 2020.  
2103 Correction of motion artifacts using a multiscale fully convolutional neural net-  
2104 work. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a6436. PMID:  
2105 32054615.

2106 Song, W., Zeng, C., Zhang, X., Wang, Z., Huang, Y., Lin, J., Wei, W., Qu, X.,  
2107 2022. Jointly estimating bias field and reconstructing uniform MRI image by deep  
2108 learning. *Journal of Magnetic Resonance* 343, 107301. doi:10.1016/j.jmr.2022.  
2109 107301.

2110 Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B., 2021.  
2111 Score-based generative modeling through stochastic differential equations, in: In-  
2112 ternational Conference on Learning Representations (ICLR).

2113 Starck, S., Sideri-Lampretsa, V., Kainz, B., Menten, M.J., Mueller, T.T., Rueckert,  
2114 D., 2026. Diff-def: Diffusion-generated deformation fields for conditional atlases.  
2115 *IEEE transactions on medical imaging* doi:10.1109/tmi.2025.3595421.

2116 Sterne, J.A.C., Hernán, M.A., Reeves, B.C., Savović, J., Berkman, N.D., Viswanathan,  
2117 M., Henry, D., White, I.R., Higgins, J.P.T., 2016. Robins-i: A tool for assessing  
2118 risk of bias in non-randomised studies of interventions. *BMJ* 355, i4919.

2119 Sui, Y., Afacan, O., Jaimes, C., Gholipour, A., Warfield, S.K., 2022. Scan-specific gen-  
2120 erative neural network for MRI super-resolution reconstruction. *IEEE Transactions*  
2121 *on Medical Imaging* doi:10.1109/tmi.2022.3142610.

2122 Sujit, S.J., Coronado, I., Kamali, A., Narayana, P.A., Gabr, R.E., 2019. Automated  
2123 image quality evaluation of structural brain MRI using an ensemble of deep learning  
2124 networks. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri.26693.  
2125 PMID: 30811739.

2126 Tabari, A., Lang, M., Awan, K., Liu, W., Clifford, B., Lo, W.C., Splitthoff, D.N.,  
2127 Cauley, S., Rapalino, O., Schaefer, P., Huang, S.Y., Conklin, J., 2023. Optimized  
2128 flow compensation for contrast-enhanced t1-weighted wave-caipi 3D mprage imaging  
2129 of the brain. *European radiology experimental* doi:10.1186/s41747-023-00351-y.  
2130 PMID: 37394534.

2131 Tabari, A., Vejdani-Jahromi, M., Lang, M., Nickel, D., Lo, W.C., Clifford, B., Con-  
2132 klin, J., Huang, S.Y., 2026. Clinical validation of deep learning-accelerated versus  
2133 wave-CAIPI postcontrast 3d T1-MPRAGE for evaluation of intracranial enhancing  
2134 lesions. *American Journal of Neuroradiology* 47, 394–400. doi:10.3174/ajnr.a8992.

2135 Takada, S., Nakaura, T., Yoshida, N., Uetani, H., Shiraishi, K., Kobayashi, N., Mat-  
2136 suo, K., Morita, K., Nagayama, Y., Kidoh, M., Yamashita, Y., Takayanagi, R.,  
2137 Hirai, T., 2025. Impact of super-resolution deep learning-based reconstruction for  
2138 hippocampal MRI: A volunteer and phantom study. *European Journal of Radiology*  
2139 191, 112289. doi:10.1016/j.ejrad.2025.112289.

2140 Tang, C., Eisenmenger, L.B., Rivera-Rivera, L., Huo, E., Junn, J.C., Kuner, A.D.,  
2141 Oechtering, T.H., Peret, A., Starekova, J., Johnson, K.M., 2025. Incorporating  
2142 radiologist knowledge into MRI quality metrics for machine learning using rank-  
2143 based ratings. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri  
2144 .29672. PMID: 39690114.

2145 Tang, H., Hu, N., Yuan, Y., Xia, C., Liu, X., Zuo, P., Stalder, A.F., Schmidt, M., Zhou,  
2146 X., Song, B., Sun, J., 2019. Accelerated time-of-flight magnetic resonance angiog-  
2147 raphy with sparse undersampling and iterative reconstruction for the evaluation  
2148 of intracranial arteries. Korean journal of radiology doi:10.3348/kjr.2017.0634.  
2149 pMID: 30672166.

2150 Tanno, R., Worrall, D.E., Kaden, E., Ghosh, A., Grussu, F., Bizzi, A., Sotiropoulos,  
2151 S.N., Criminisi, A., Alexander, D.C., 2021. Uncertainty modelling in deep learning  
2152 for safer neuroimage enhancement: Demonstration in diffusion MRI. NeuroImage  
2153 doi:10.1016/j.neuroimage.2020.117366.

2154 Taymourtash, A., Schwartz, E., Nenning, K.H., Licandro, R., Kienast, P., Hiele,  
2155 V., Prayer, D., Kasprian, G., Langs, G., 2025. Measuring the effects of motion  
2156 corruption in fetal fMRI. Human brain mapping doi:10.1002/hbm.26806. pMID:  
2157 39846325.

2158 Tessema, A.W., Jin, S., Cho, H., Han, S., Cho, H., 2026. Technical feasibility of  
2159 an efficient deep learning-based super-resolution method for 3D MR angiography.  
2160 IEEE Access doi:10.1109/access.2026.3712275.

2161 Thaler, C., Sedlacik, J., Forkert, N.D., Stellmann, J.P., Schön, G., Fiehler, J., Gel-  
2162 lißen, S., 2023. Effect of geometric distortion correction on thickness and volume  
2163 measurements of cortical parcellations in 3D t1w gradient echo sequences. PloS one  
2164 doi:10.1371/journal.pone.0284440. pMID: 37058493.

2165 Tisdall, M.D., Atkins, M.S., 2006. Using human and model performance to com-  
2166 pare MRI reconstructions. IEEE transactions on medical imaging doi:10.1109/tm  
2167 i.2006.881374. pMID: 17117780.

2168 Trivedi, V.K., Sharma, B., Balamurugan, P., 2024. Mcddpm: Multichannel conditional  
2169 denoising diffusion model for unsupervised anomaly detection in brain MRI. 2024  
2170 17th International Congress on Image and Signal Processing, BioMedical Engineer-  
2171 ing and Informatics (CISP-BMEI) doi:10.1109/cisp-bmei64163.2024.10906217.

2172 Uher, D., Drenthen, G.S., Poser, B.A., Hofman, P.A.M., Wagner, L.G., Lanen,  
2173 R.H.G.J.v., Hoeberigs, C.M., Colon, A.J., Schijns, O.E.M.G., Jansen, J.F.A.,  
2174 Backes, W.H., 2024. Deepflair: A neural network approach to mitigate signal and  
2175 contrast loss in temporal lobes at 7 tesla FLAIR images. Magnetic resonance imag-  
2176 ing doi:10.1016/j.mri.2024.04.013. pMID: 38621552.

2177 U.S. Food and Drug Administration, Health Canada, Medicines and Healthcare  
2178 products Regulatory Agency, 2021. Good machine learning practice for medical  
2179 device development: Guiding principles. Joint guiding principles, October 2021.  
2180 URL: [https://www.fda.gov/medical-devices/software-medical-device-samd](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles)  
2181 [/good-machine-learning-practice-medical-device-development-guiding-pri](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles)  
2182 [nciples](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles).

2183 Usui, K., Muro, I., Shibukawa, S., Goto, M., Ogawa, K., Sakano, Y., Kyogoku, S.,  
2184 Daida, H., 2023. Evaluation of motion artefact reduction depending on the artefacts'  
2185 directions in head MRI using conditional generative adversarial networks. Scientific  
2186 reports doi:10.1038/s41598-023-35794-1. pMID: 37237139.

2187 Vakli, P., Weiss, B., Szalma, J., Barsi, P., Gyuricza, I., Kemenczky, P., Somogyi, E.,  
2188 Ádám Nárai, Gál, V., Hermann, P., Vidnyánszky, Z., 2023. Automatic brain MRI  
2189 motion artifact detection based on end-to-end deep learning is similarly effective  
2190 as traditional machine learning trained on image quality metrics. Medical image  
2191 analysis doi:10.1016/j.media.2023.102850.

2192 Vanderhasselt, T., Naeyaert, M., Watté, N., Allemeersch, G., Raeymaeckers, S.,  
2193 Dudink, J., Mey, J.d., Raeymaekers, H., 2020. Synthetic MRI of preterm in-  
2194 fants at term-equivalent age: Evaluation of diagnostic image quality and auto-  
2195 mated brain volume segmentation. AJNR. American journal of neuroradiology  
2196 doi:10.3174/ajnr.a6533. PMID: 32299803.

2197 Verclytte, S., Beaugrard, G., Nickel, D., Muñoz-Ramírez, V., Norberciak, L., Boissel, M., Chaton, V., Kwiatkowski, A., 2026. Deep learning-accelerated 3d FLAIR enables reliable MS lesion detection. American Journal of Neuroradiology 47, 1560–1568. doi:10.3174/ajnr.a9140.

2201 Verdicchio, M., Grassi, F., Esposito, G., Cavaliere, C., Alfano, V., Cerrone, M., Pappatà, S., Salvatore, E., Cecca, A.D., Salvatore, M., Aiello, M., 2025. Reliability assessment of accelerated MR sequences for the estimation of structural and vascular markers in dementia. Scientific reports doi:10.1038/s41598-025-87224-z. PMID: 40240759.

2206 Vranic, J.E., Cross, N.M., Wang, Y., Hippe, D.S., Weerd, E.d., Mossa-Basha, M., 2019. Compressed sensing-sensitivity encoding (cs-SENSE) accelerated brain imaging: Reduced scan time without reduced image quality. AJNR. American journal of neuroradiology doi:10.3174/ajnr.a5905. PMID: 30523142.

2210 Wada, A., Akashi, T., Hagiwara, A., Nishizawa, M., Shimoji, K., Kikuta, J., Maekawa, T., Sano, K., Kamagata, K., Nakanishi, A., Aoki, S., 2024. Deep learning-driven transformation: A novel approach for mitigating batch effects in diffusion MRI beyond traditional harmonization. Journal of magnetic resonance imaging : JMIR doi:10.1002/jmri.29088.

2215 Wang, C.J., Rost, N.S., Golland, P., 2023. Spatial-intensity transforms for medical image-to-image translation. IEEE transactions on medical imaging doi:10.1109/tmi.2023.3283948. PMID: 37285247.

2218 Wang, G., Gong, E., Banerjee, S., Martin, D., Tong, E., Choi, J., Chen, H., Wintermark, M., Pauly, J.M., Zaharchuk, G., 2020. Synthesize high-quality multi-contrast magnetic resonance imaging from multi-echo acquisition using multi-task deep generative model. IEEE Transactions on Medical Imaging 39, 3089–3099. doi:10.1109/tmi.2020.2987026.

2223 Wang, N., Liao, C., Cao, X., Nishimura, M., Brackner, Y.W.E., Yurt, M., Gao, M., Abraham, D., Alkan, C., Iyer, S.S., Zhou, Z., Jeong, H., Kerr, A., Haldar, J.P., Setsompop, K., 2025a. Spherical echo-planar time-resolved imaging (septi) for rapid 3D quantitative  $t_2^*$  and susceptibility imaging. Magnetic resonance in medicine doi:10.1002/mrm.30255. PMID: 39250435.

2228 Wang, X., Cabezas, M., Tang, Z., D'Souza, A., Liu, D., Wang, C., Cai, W., 2025b. Estimating uncertainty on deep learning-driven fibre orientation distribution enhancement, in: ICONIP 2024, CCIS 2285. doi:10.1007/978-981-96-6957-8/25.

2231 Wang, X., Fan, H., Tan, Z., Vasylechko, S., Yang, E., Didier, R., Afacan, O., Uecker,  
2232 M., Warfield, S.K., Gholipour, A., 2025c. Rapid, high-resolution and distortion-free  
2233  $R_2^*$  mapping of fetal brain using multi-echo radial flash and model-based reconstruction.  
2234 Magnetic resonance in medicine doi:10.1002/mrm.30604. pMID: 40534142.

2235 Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P., 2004. Image quality assessment:  
2236 from error visibility to structural similarity. IEEE Transactions on Image Processing  
2237 13, 600–612. doi:10.1109/TIP.2003.819861.

2238 Wang, Z., Ramasawmy, R., Javed, A., Mugler, J.P., Meyer, C.H., Campbell-Washburn,  
2239 A.E., 2025d. Variable-flip-angle 3D spiral-in-out turbo spin-echo imaging using con-  
2240 comitant gradient compensation and echo reordering at 0.55 t. Magnetic resonance  
2241 in medicine doi:10.1002/mrm.30380. pMID: 39567364.

2242 Wen, Y., Ma, H., Xiang, S., Feng, Z., Guan, C., Li, X., 2025. Accelerating brain t2-  
2243 weighted imaging using artificial intelligence-assisted compressed sensing combined  
2244 with deep learning-based reconstruction: a feasibility study at 5.0t MRI. BMC  
2245 medical imaging doi:10.1186/s12880-025-01763-5. pMID: 40596927.

2246 Whiting, P.F., Rutjes, A.W.S., Westwood, M.E., Mallett, S., Deeks, J.J., Reitsma,  
2247 J.B., Leeflang, M.M.G., Sterne, J.A.C., Bossuyt, P.M.M., 2011. QUADAS-2: A  
2248 revised tool for the quality assessment of diagnostic accuracy studies. Annals of  
2249 Internal Medicine 155, 529–536.

2250 Wichtmann, B.D., Katemann, C., Kadrija, M., Layer, Y.C., Bischoff, L.M., Scheuver,  
2251 Y., Mezger, M., Weber, O.M., Luetkens, J.A., Attenberger, U.I., Radbruch, A.,  
2252 Paech, D., 2025. Rapid and motion-robust pediatric brain imaging: T2-weighted  
2253 turbo-spin-echo propeller acquisition with compressed sensing. Pediatric radiology  
2254 doi:10.1007/s00247-024-06088-z. pMID: 39589491.

2255 Wismüller, A., Vosoughi, M.A., 2022. Self-supervised learning in medical imaging:  
2256 anomaly detection in MRI using autoencoders, in: Real-Time Image Processing  
2257 and Deep Learning 2022, p. 9. doi:10.1117/12.2622233.

2258 Wu, F., Luo, H., Wang, X., Yang, Q., Zhuang, Y., Lin, L., Dong, Y., Tulupov, A.,  
2259 Zhang, Y., Cai, S., Chen, Z., Cai, C., Bao, J., Cheng, J., 2025a. Application of  
2260 anti-motion ultra-fast quantitative MRI in neurological disorder imaging: Insights  
2261 from huntington's disease. Journal of magnetic resonance imaging : JMRI doi:10.  
2262 1002/jmri.29682.

2263 Wu, J., Chen, L., Li, Z., Li, X., Sun, T., Wang, L., Wang, R., Wei, H., Zhang,  
2264 Y., 2025b. 3d isotropic high-resolution fetal brain MRI reconstruction from motion  
2265 corrupted thick data based on physical-informed unsupervised learning. IEEE Jour-  
2266 nal of Biomedical and Health Informatics 29, 8256–8267. doi:10.1109/jbhi.2025.  
2267 3586049.

2268 Xia, L., Qi, X., Liu, J., Yuan, Y., Wang, Y., Xu, N., Zhang, J., 2026. Accurate multi-b-  
2269 value DWI generation using two-stage deep learning: multicenter study. European  
2270 Journal of Radiology 194, 112497. doi:10.1016/j.ejrad.2025.112497.

2271 Xie, K., Gao, L., Zhang, H., Zhang, S., Xi, Q., Zhang, F., Sun, J., Lin, T., Sui, J., Ni,  
2272 X., 2024. GAN-based metal artifacts region inpainting in brain MRI imaging with  
2273 reflective registration. Medical physics doi:10.1002/mp.16724.

2274 Xiong, Z., Gao, Y., Liu, F., Emery, D., Butcher, K., Wilman, A.H., Sun, H., 2026.  
2275 Advancing quantitative susceptibility mapping with 2.5d diffusion models for rapid  
2276 intracranial hemorrhage quantification. *Magnetic resonance in medicine* doi:10.  
2277 1002/mrm.70358. pMID: 41873533.

2278 Xu, J., Zu, T., Hsu, Y.C., Wang, X., Chan, K.W.Y., Zhang, Y., 2025. DEISM: Deep  
2279 reconstruction framework with self-calibration mechanisms for accelerated chemical  
2280 exchange saturation transfer imaging. *IEEE Transactions on Biomedical Engineering*  
2281 72, 2413–2424. doi:10.1109/tbme.2025.3543403.

2282 Yang, G., Yu, S., Dong, H., Slabaugh, G., Dragotti, P.L., Ye, X., Liu, F., Arridge, S.,  
2283 Keegan, J., Guo, Y., Firmin, D., 2018. DAGAN: Deep de-aliasing generative adver-  
2284 sarial networks for fast compressed sensing MRI reconstruction. *IEEE Transactions*  
2285 *on Medical Imaging* 37, 1310–1321.

2286 Yang, Q., Lin, Y., Wang, J., Bao, J., Wang, X., Ma, L., Zhou, Z., Yang, Q., Cai,  
2287 S., He, H., Cai, C., Dong, J., Cheng, J., Chen, Z., Zhong, J., 2022. Model-based  
2288 synthetic data-driven learning (most-dl): Application in single-shot t2 mapping  
2289 with severe head motion using overlapping-echo acquisition. *IEEE transactions on*  
2290 *medical imaging* doi:10.1109/tmi.2022.3179981. pMID: 35657830.

2291 Yarach, U., Chatnuntawech, I., Liao, C., Teerapittayanon, S., Iyer, S.S., Kim, T.H.,  
2292 Haldar, J., Cho, J., Bilgic, B., Hu, Y., Hargreaves, B., Setsompop, K., 2025.  
2293 Blip-up blip-down circular epi (buda-cepi) for distortion-free dmri with rapid un-  
2294 rolled deep learning reconstruction. *Magnetic resonance imaging* doi:10.1016/j.mri.  
2295 2024.110277. pMID: 39566835.

2296 Yarach, U., Chatnuntawech, I., Setsompop, K., Suwannasak, A., Angkurawaranon,  
2297 S., Madla, C., Hanprasertpong, C., Sangpin, P., 2024. Improved reconstruction  
2298 for highly accelerated propeller diffusion 1.5t clinical MRI. *Magnetic Resonance*  
2299 *Materials in Physics, Biology and Medicine* 37, 283–294. doi:10.1007/s10334-023-  
2300 01142-7.

2301 Yasaka, K., Akai, H., Kato, S., Tajima, T., Yoshioka, N., Furuta, T., Kageyama, H.,  
2302 Toda, Y., Akahane, M., Ohtomo, K., Abe, O., Kiryu, S., 2024. Iterative motion  
2303 correction technique with deep learning reconstruction for brain MRI: A volunteer  
2304 and patient study. *Journal of imaging informatics in medicine* doi:10.1007/s10278-  
2305 024-01184-w. pMID: 38942939.

2306 Yasaka, K., Akai, H., Sugawara, H., Tajima, T., Akahane, M., Yoshioka, N., Kabasawa,  
2307 H., Miyo, R., Ohtomo, K., Abe, O., Kiryu, S., 2022. Impact of deep learning recon-  
2308 struction on intracranial 1.5 t magnetic resonance angiography. *Japanese journal of*  
2309 *radiology* doi:10.1007/s11604-021-01225-2. pMID: 34851499.

2310 Yilmaz, K., Çukur, T., Patel, V.M., 2023. Self-supervised MRI reconstruction with  
2311 unrolled diffusion models. *Medical Image Computing and Computer Assisted Inter-*  
2312 *vention – MICCAI 2023* doi:10.1007/978-3-031-43999-5\_47.

2313 Yoo, H., Moon, H.E., Kim, S., Kim, D.H., Choi, Y.H., Cheon, J.E., Lee, J.S., Lee,  
2314 S., 2025. Evaluation of image quality and scan time efficiency in accelerated 3D  
2315 t1-weighted pediatric brain MRI using deep learning-based reconstruction. *Korean*  
2316 *journal of radiology* doi:10.3348/kjr.2024.0701. pMID: 39898398.

2317 Yoshida, N., Kageyama, H., Akai, H., Sasaki, K., Sakurai, N., Koori, N., Yamamoto,  
2318 S., Kodama, N., 2026. Deep learning approach to super-resolution correction of  
2319 brain MRI motion artifacts for accurate hippocampal volumetry. Scientific reports  
2320 doi:10.1038/s41598-026-44834-5. pMID: 41866419.

2321 Yoshida, N., Kageyama, H., Akai, H., Yasaka, K., Sugawara, H., Okada, Y., Kuni-  
2322 matsu, A., 2022. Motion correction in MR image for analysis of vsrad using gener-  
2323 ative adversarial network. PloS one doi:10.1371/journal.pone.0274576. pMID:  
2324 36103561.

2325 You, S., Gondova, A., Izaguirre, C.S.A., Milo, G.T., Jeong, S., Lee, H.J., Tarui, T.,  
2326 Rollins, C.K., Yun, H.J., Grant, P.E., Im, K., 2025. Conditional deep generative  
2327 normative modeling for structural and developmental anomaly detection in the fetal  
2328 brain. NeuroImage doi:10.1016/j.neuroimage.2025.121442. pMID: 40912664.

2329 Yu, S., Dai, G., Wang, Z., Li, L., Wei, X., Xie, Y., 2018. A consistency evaluation of  
2330 signal-to-noise ratio in the quality assessment of human brain magnetic resonance  
2331 images. BMC medical imaging doi:10.1186/s12880-018-0256-6. pMID: 29769079.

2332 Yuan, S., Liu, Y., Wei, R., Zhu, J., Men, K., Dai, J., 2024. A novel loss function to  
2333 reproduce texture features for deep learning-based MRI-to-CT synthesis. Medical  
2334 physics doi:10.1002/mp.16850.

2335 Zahneisen, B., Keating, B., Singh, A., Herbst, M., Ernst, T., 2016. Reverse retrospec-  
2336 tive motion correction. Magnetic resonance in medicine doi:10.1002/mrm.25830.  
2337 pMID: 26140504.

2338 Zaitsev, M., Maclaren, J., Herbst, M., 2015. Motion artifacts in MRI: A review. NMR  
2339 in Biomedicine 28, 911–935.

2340 Zhang, C., Karkalousos, D., Bazin, P.L., Coolen, B.F., Vrenken, H., Sonke, J.J.,  
2341 Forstmann, B.U., Poot, D.H.J., Caan, M.W.A., 2022. A unified model for recon-  
2342 struction and r2\* mapping of accelerated 7t data using the quantitative recurrent  
2343 inference machine. NeuroImage doi:10.1016/j.neuroimage.2022.119680. pMID:  
2344 36240989.

2345 Zhang, C., Klein, S., Cristobal-Huerta, A., Hernandez-Tamames, J.A., Poot, D.H.J.,  
2346 2021. Apir4emc: Autocalibrated parallel imaging reconstruction for extended multi-  
2347 contrast imaging. Magnetic resonance imaging doi:10.1016/j.mri.2021.02.002.  
2348 pMID: 33592248.

2349 Zhang, J., Wu, J., Chen, S., Zhang, Z., Cai, S., Cai, C., Chen, Z., 2019a. Robust  
2350 single-shot T2 mapping via multiple overlapping-echo acquisition and deep neural  
2351 network. IEEE Transactions on Medical Imaging 38, 1801–1811. doi:10.1109/tmi.  
2352 2019.2896085.

2353 Zhang, L., Wu, M., Wang, L., Steffens, D.C., Potter, G.G., Liu, M., 2026. Iterative  
2354 learning for joint image denoising and motion artifact correction of 3D brain MRI.  
2355 Medical image analysis doi:10.1016/j.media.2025.103872. pMID: 41274083.

2356 Zhang, Q., Ruan, G., Yang, W., Liu, Y., Zhao, K., Feng, Q., Chen, W., Wu, E.X.,  
2357 Feng, Y., 2019b. MRI gibbs?ring artifact reduction by means of machine learning  
2358 using convolutional neural networks. Magnetic Resonance in Medicine 82, 2133–  
2359 2145. doi:10.1002/mrm.27894.



2360 Zhao, C., Shao, M., Carass, A., Li, H., Dewey, B.E., Ellingsen, L.M., Woo, J.,  
2361 Guttman, M.A., Blitz, A.M., Stone, M., Calabresi, P.A., Halperin, H., Prince,  
2362 J.L., 2019. Applications of a deep learning method for anti-aliasing and super-  
2363 resolution in MRI. *Magnetic resonance imaging* doi:10.1016/j.mri.2019.05.038.  
2364 pMID: 31247254.

2365 Zhao, R., Yaman, B., Zhang, Y., Stewart, R., Dixon, A., Knoll, F., Huang, Z., Lui,  
2366 Y.W., Hansen, M.S., Lungren, M.P., 2022. fastMRI+, clinical pathology annota-  
2367 tions for knee and brain fully sampled magnetic resonance imaging data. *Scientific*  
2368 *Data* 9, 152. doi:10.1038/s41597-022-01255-z.

2369 Zhou, L., Ni, X., Kong, Y., Zeng, H., Xu, M., Zhou, J., Wang, Q., Liu, C., 2023. Miti-  
2370 gating misalignment in MRI-to-CT synthesis for improved synthetic CT generation:  
2371 an iterative refinement and knowledge distillation approach. *Physics in medicine*  
2372 *and biology* doi:10.1088/1361-6560/ad0ddc.

2373 Zhou, X., Balachandra, A.R., Romano, M.F., Chin, S.P., Au, R., Kolachalama, V.B.,  
2374 2024. Adversarial learning for MRI reconstruction and classification of cognitively  
2375 impaired individuals. *IEEE Access* doi:10.1109/access.2024.3408840.

2376 Zhu, D., Xu, F., Liu, D., Qin, Q., 2025a. A straightforward approach for 3D single-  
2377 shot arterial spin labeling-based brain perfusion imaging: Preventing artifacts due to  
2378 signal fluctuations. *Magnetic resonance in medicine* doi:10.1002/mrm.30439. pMID:  
2379 39887515.

2380 Zhu, P., Liu, C., Fu, Y., Chen, N., Qiu, A., 2025b. Cycle-conditional diffusion model  
2381 for noise correction of diffusion-weighted images using unpaired data. *Medical image*  
2382 *analysis* doi:10.1016/j.media.2025.103579. pMID: 40273728.

2383 Zhu, S., Jin, Y., Spears, T., Zawar, I., Fletcher, P.T., 2026. Realdeal: Enhancing  
2384 realism and details in brain image generation via image-to-image diffusion models.  
2385 *Deep Generative Models* doi:10.1007/978-3-032-05472-2\_15.

2386 Zibetti, M.V.W., Knoll, F., Regatte, R.R., 2022. Alternating learning approach for  
2387 variational networks and undersampling pattern in parallel MRI applications. *IEEE*  
2388 *Transactions on Computational Imaging* doi:10.1109/tci.2022.3176129.